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AG Piping: Hot Tapping

BY

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Introduction

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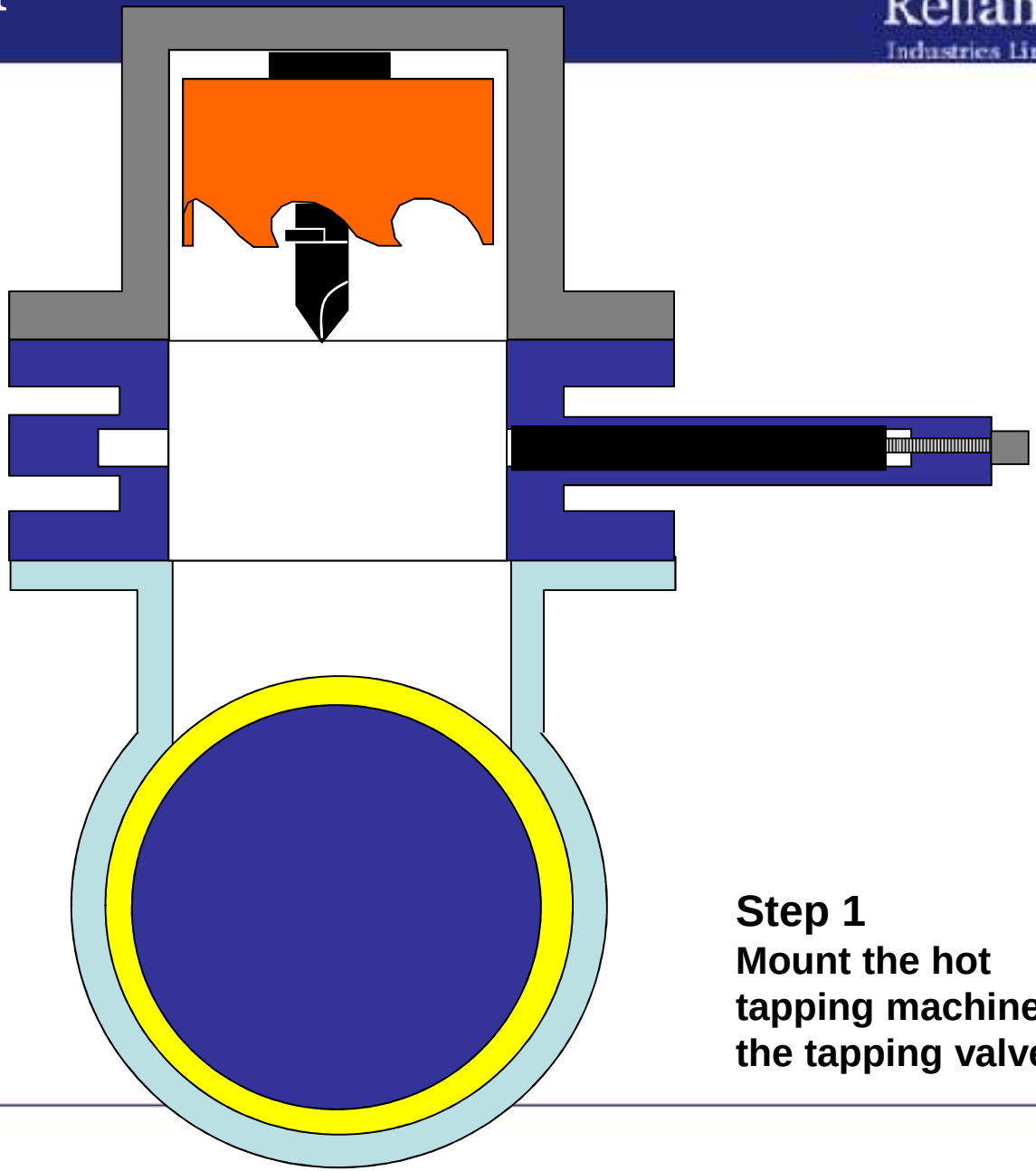
Definition:

Hot tapping is the technique of attaching a welded branch fitting to piping or equipment in service, and then creating an opening in that piping or equipment by drilling or cutting a portion of the piping or equipment within the attached fitting.

(In certain specific situations a bolted or mechanical fitting may be used)

Hot tapping is usually performed when it is not feasible, or is impractical, to take the equipment or piping out of service, or to purge or clean it by conventional methods.

Introduction



Step 1

Mount the hot tapping machine to the tapping valve.

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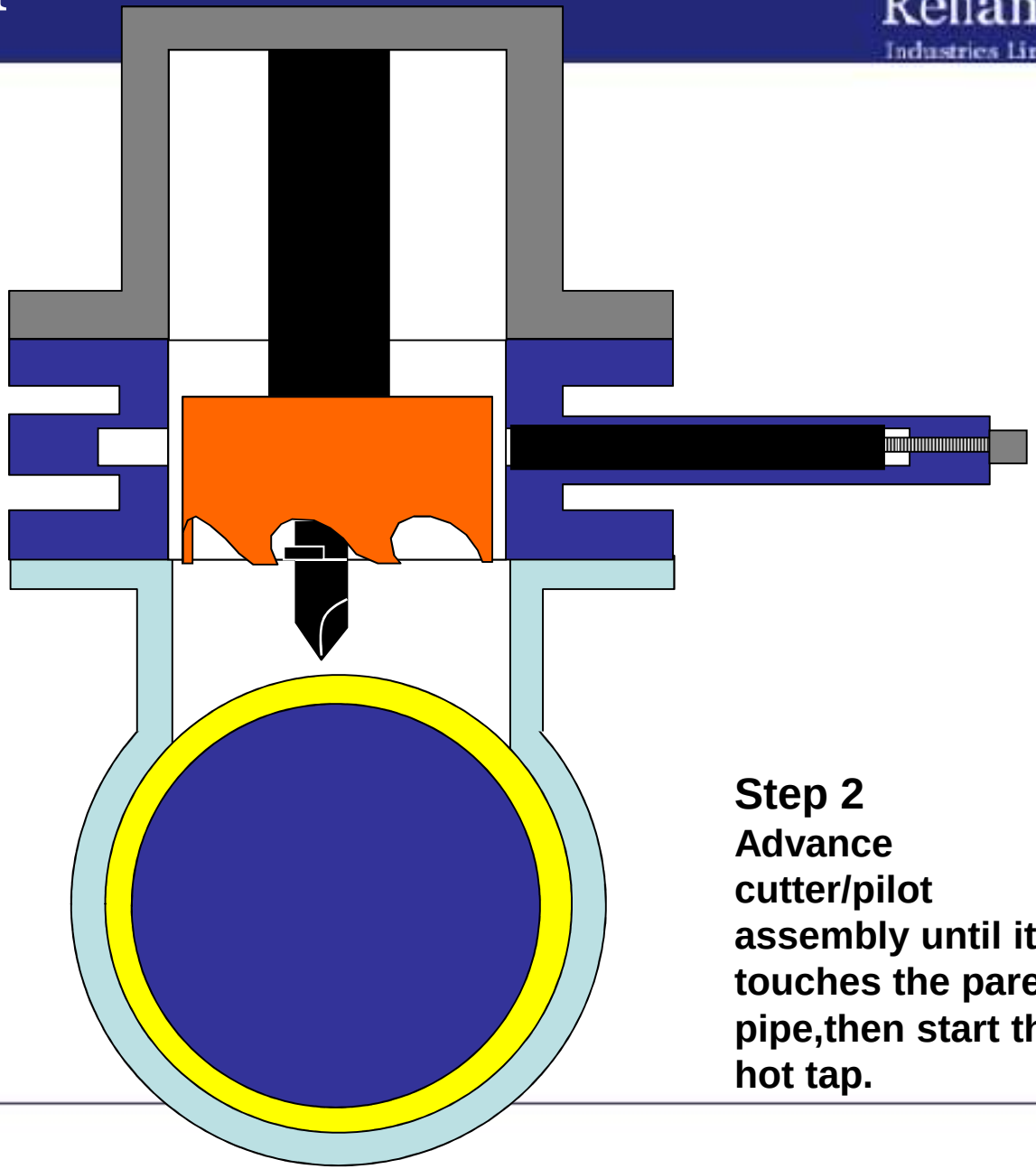

Step 2

Advance cutter/pilot assembly until it touches the parent pipe,then start the hot tap.

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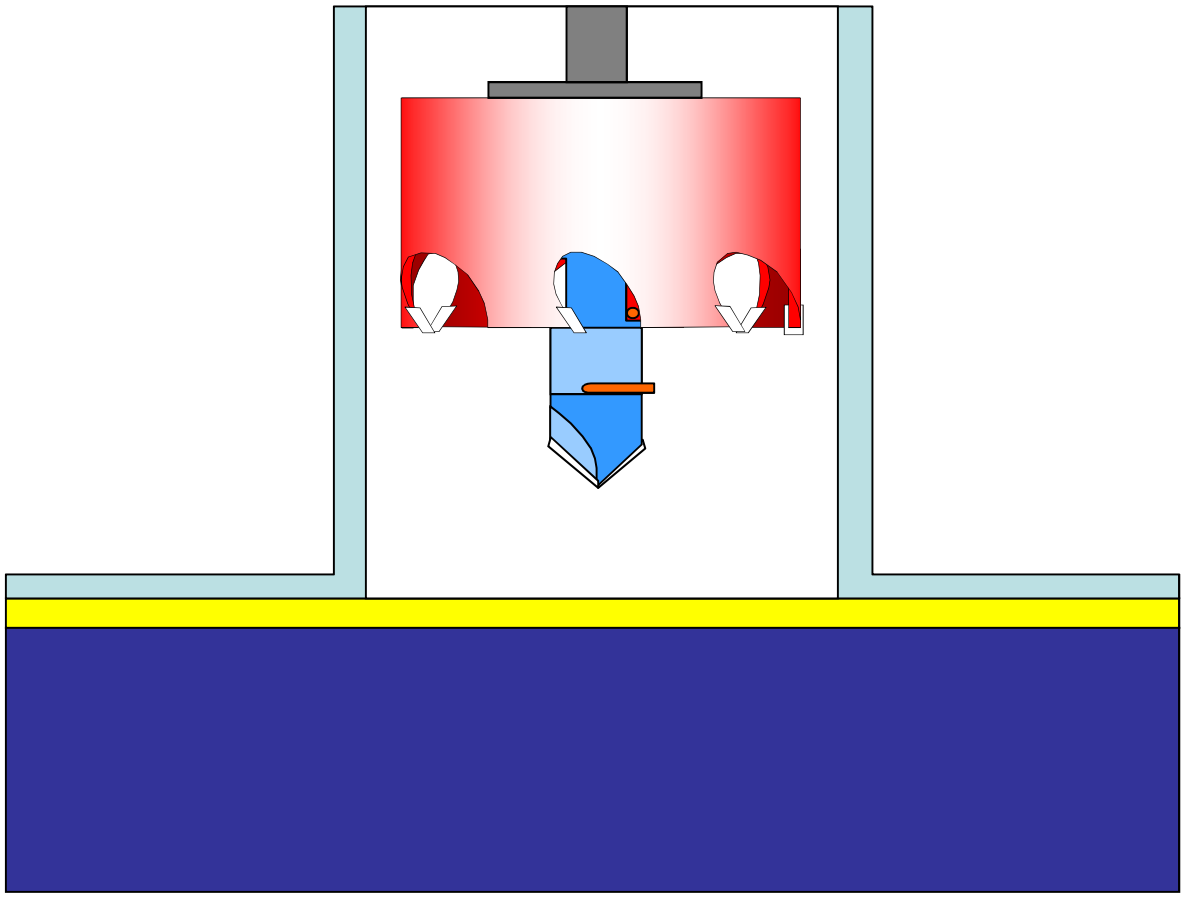

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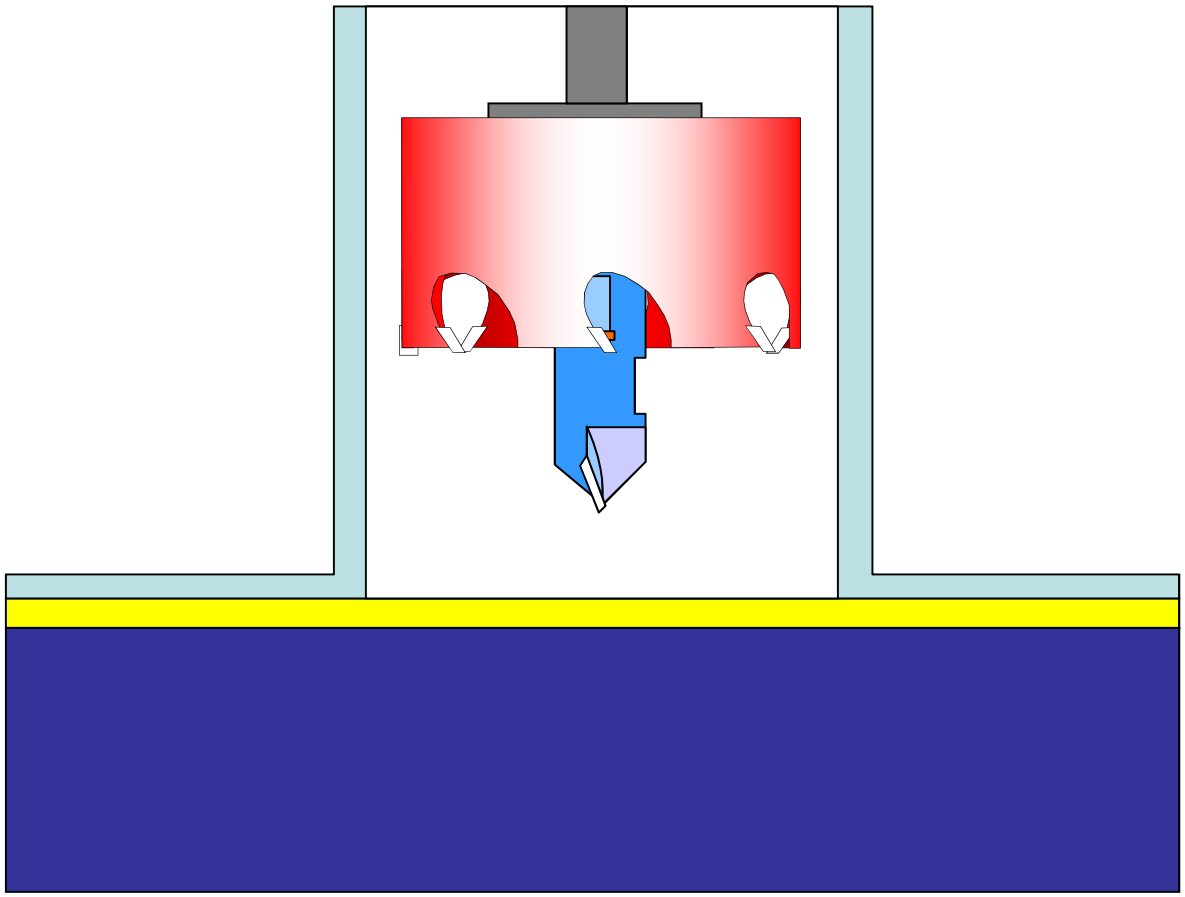

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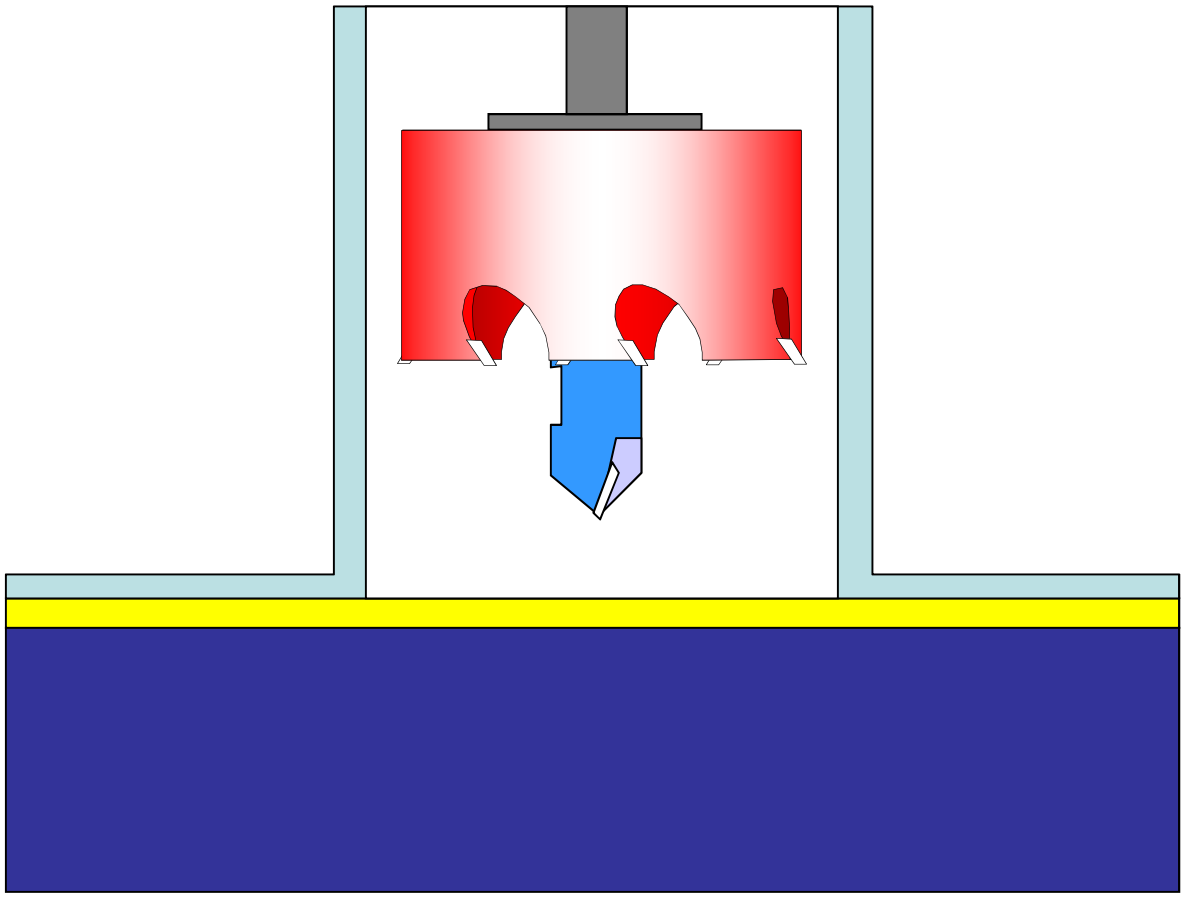

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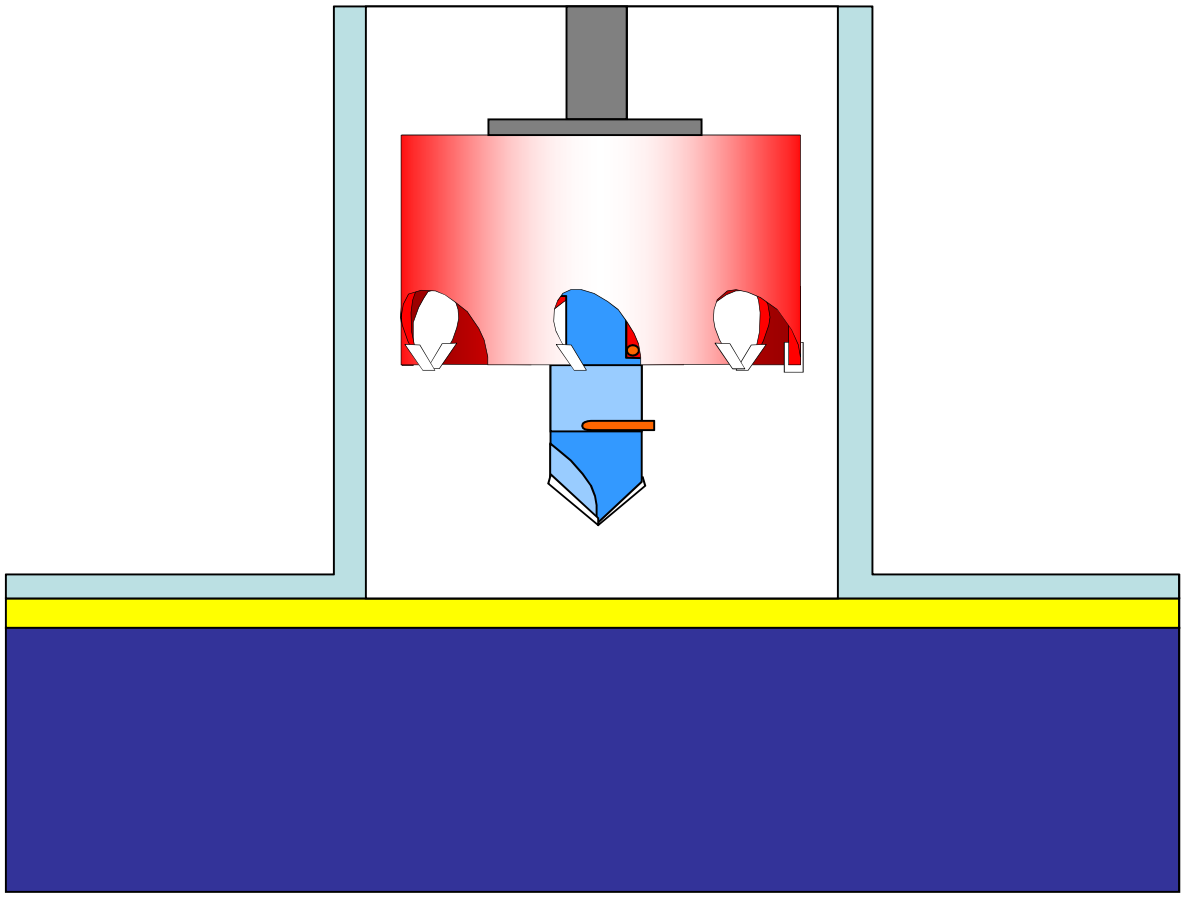

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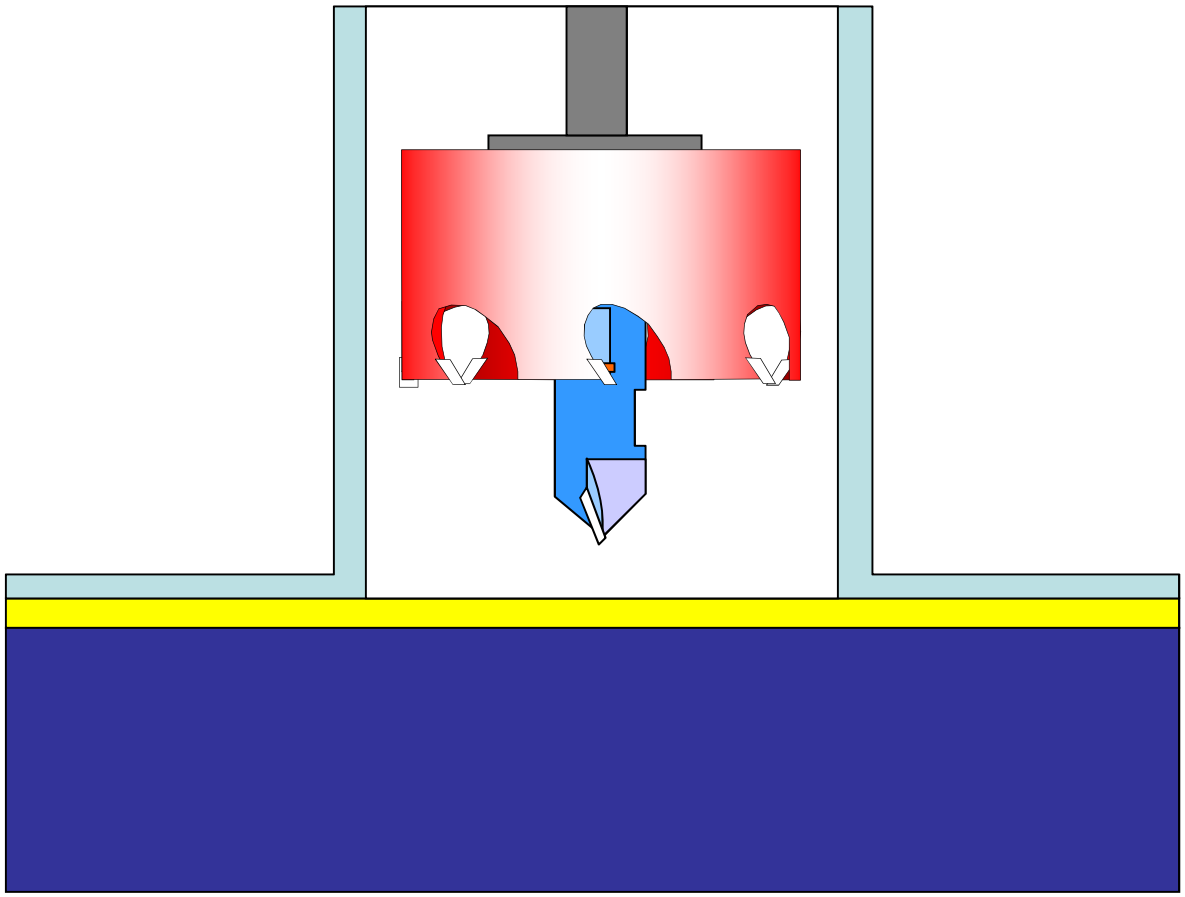

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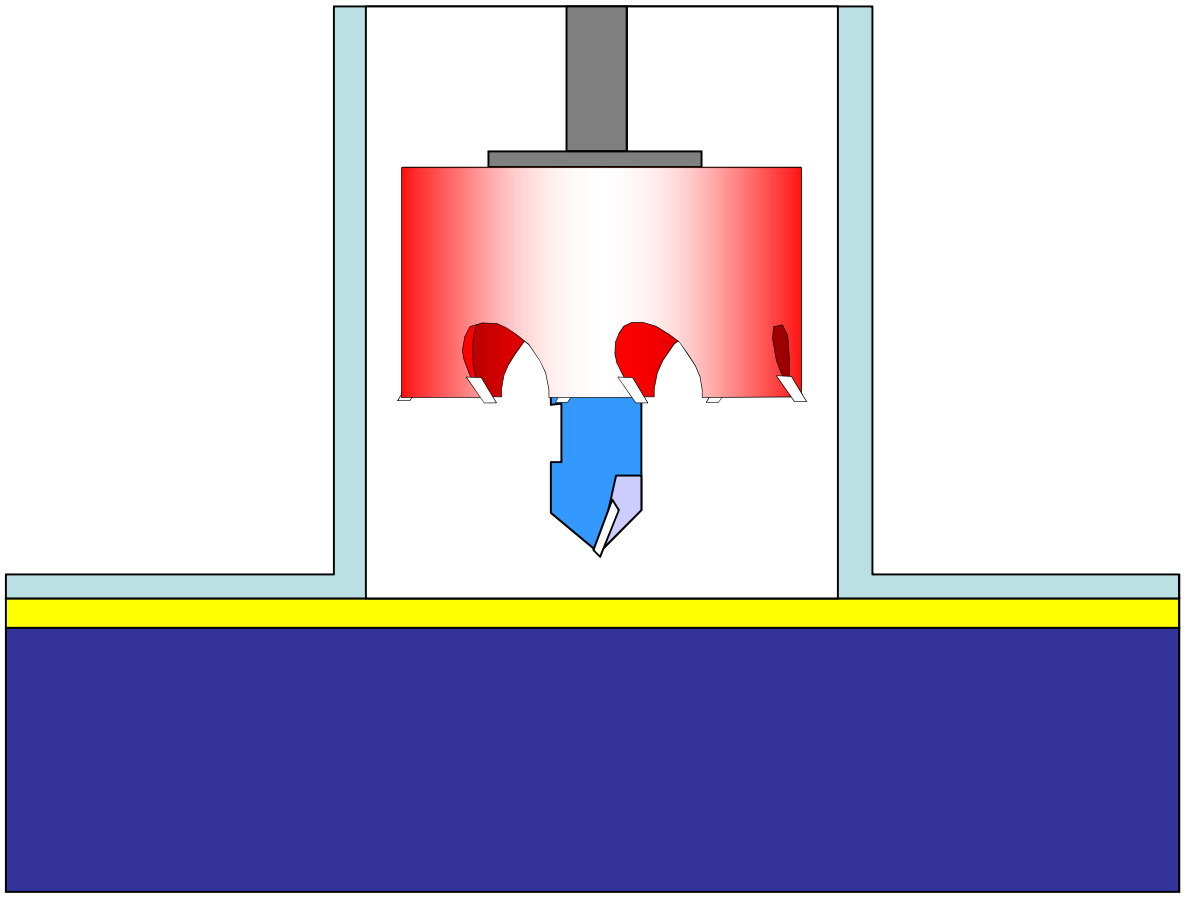

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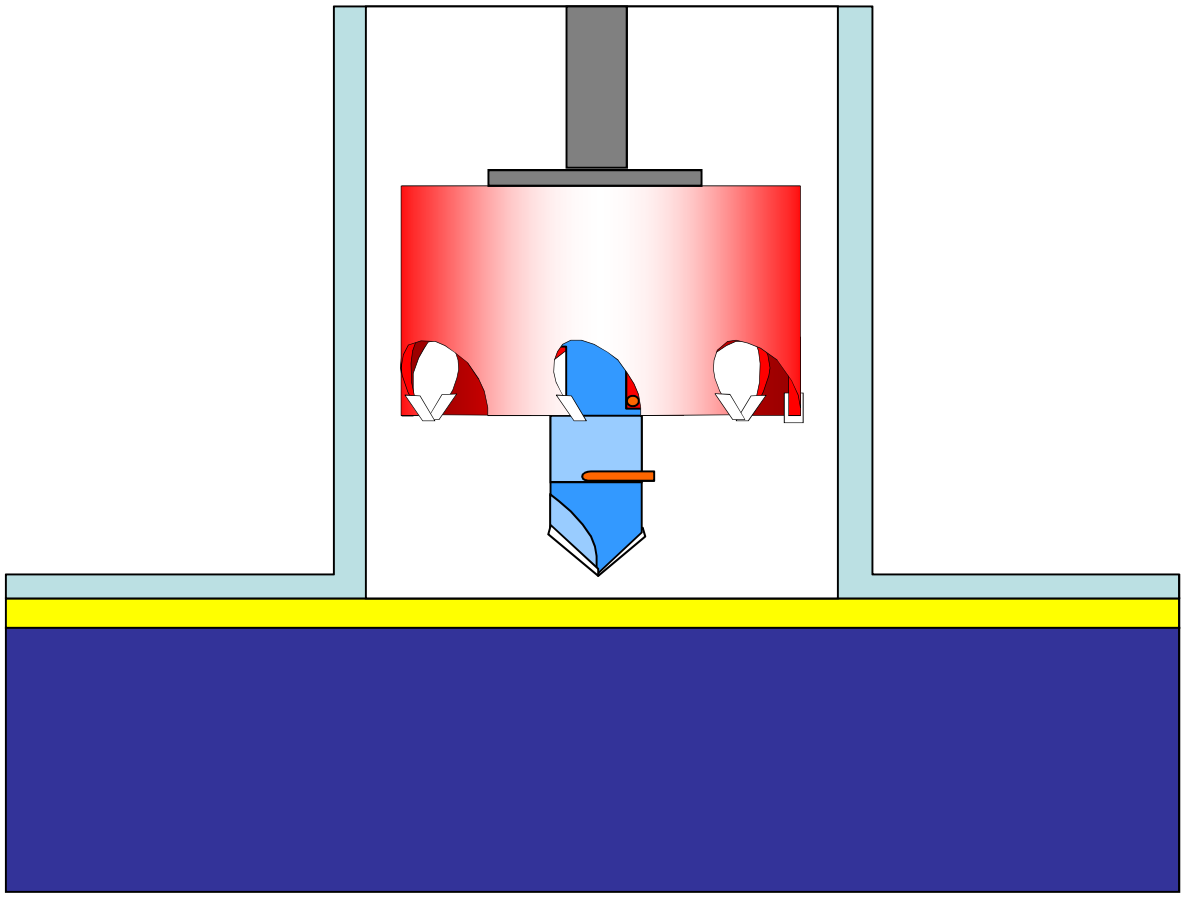

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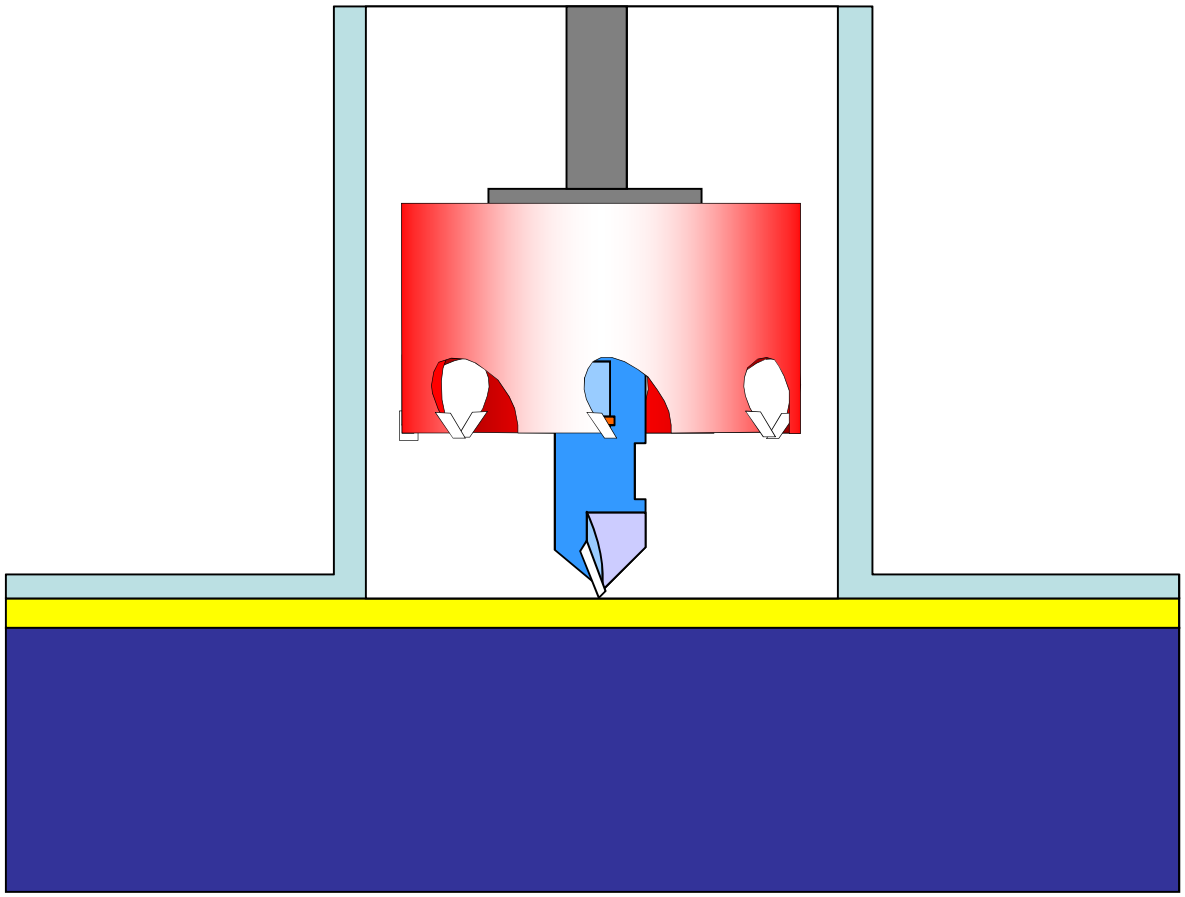

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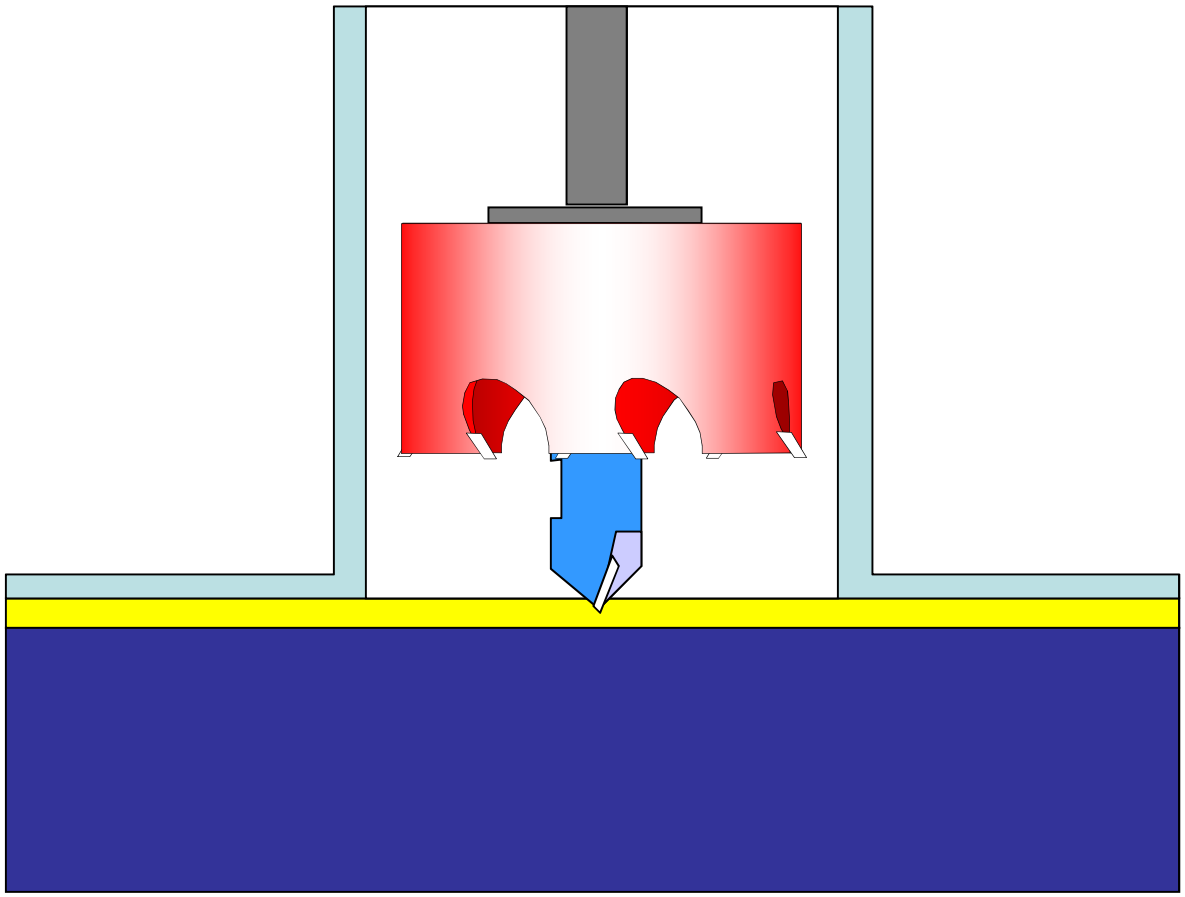

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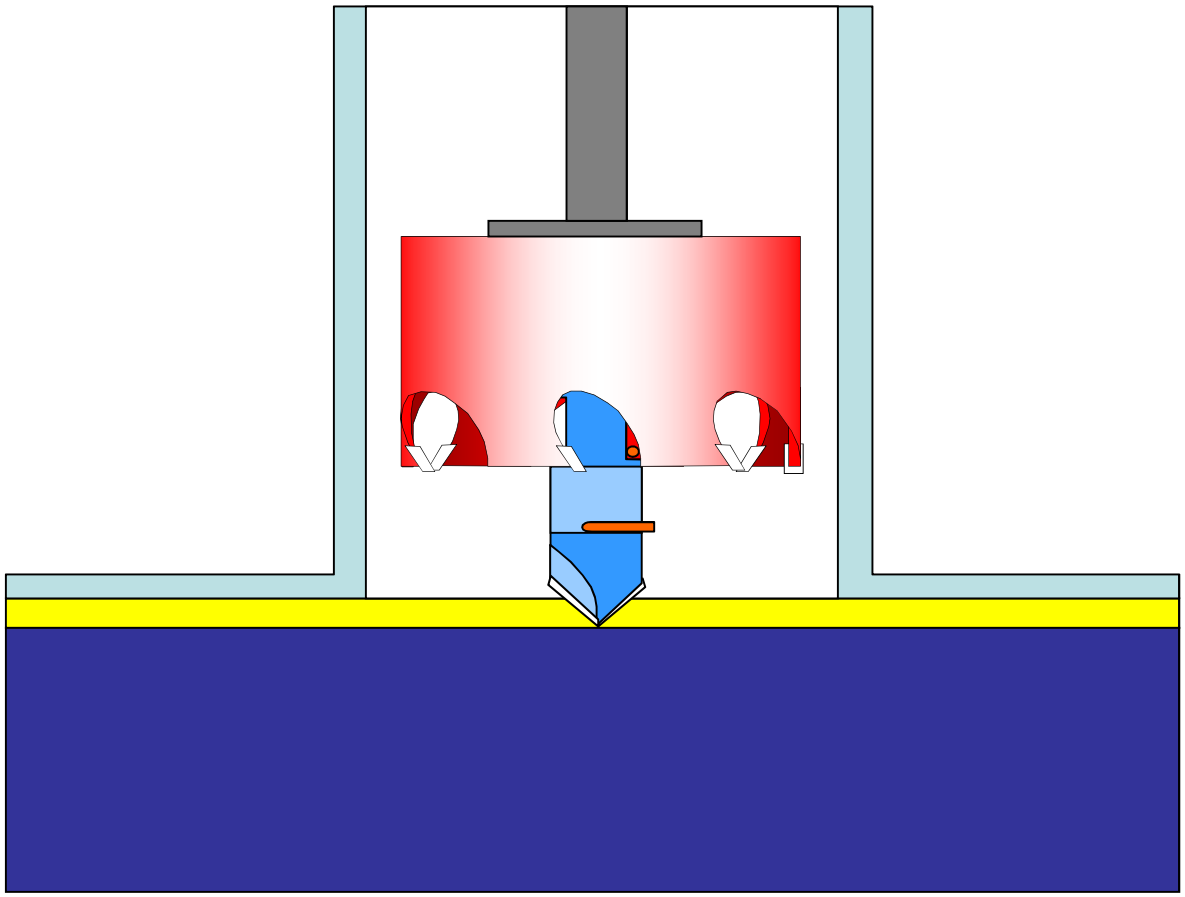

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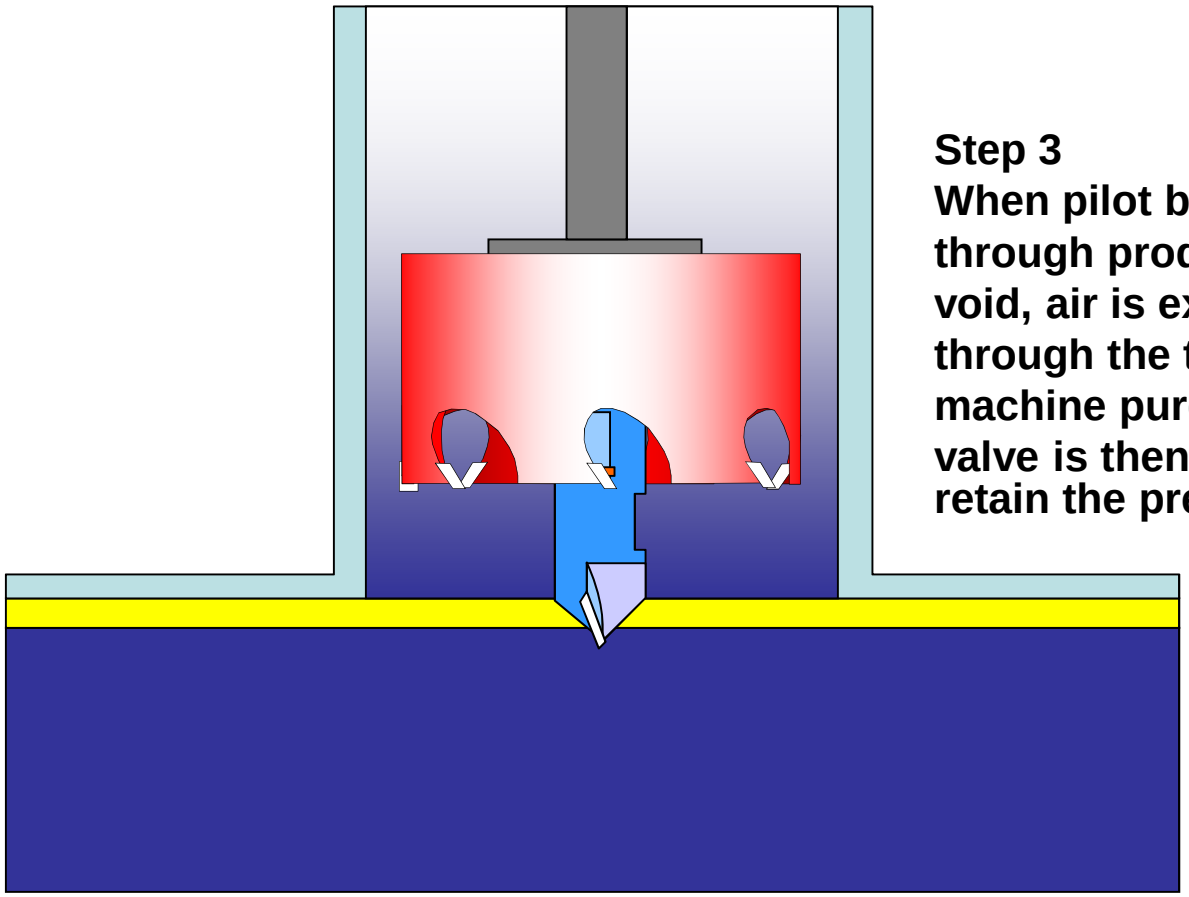

Step 3
When pilot breaks through product fills the void, air is expelled through the tapping machine purge valve and valve is then closed to retain the pressure.

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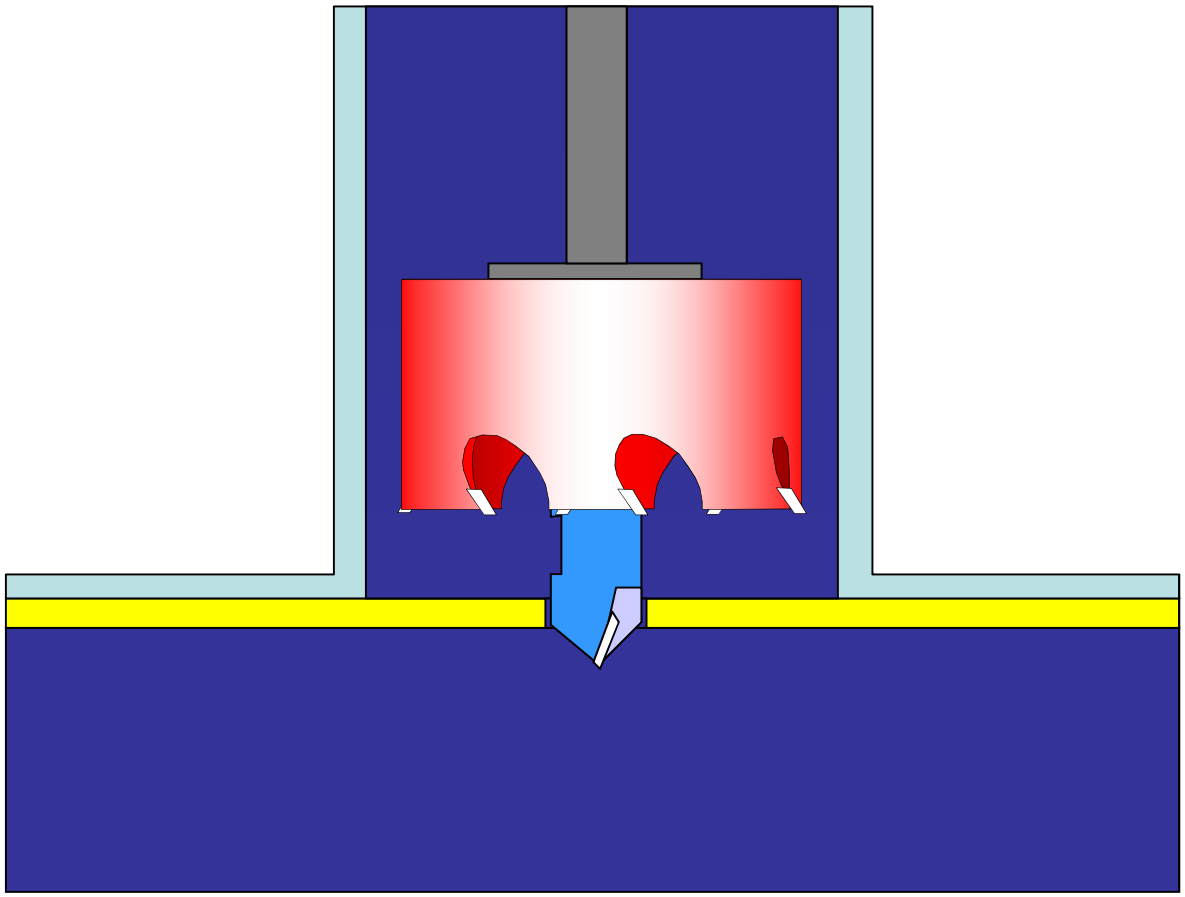

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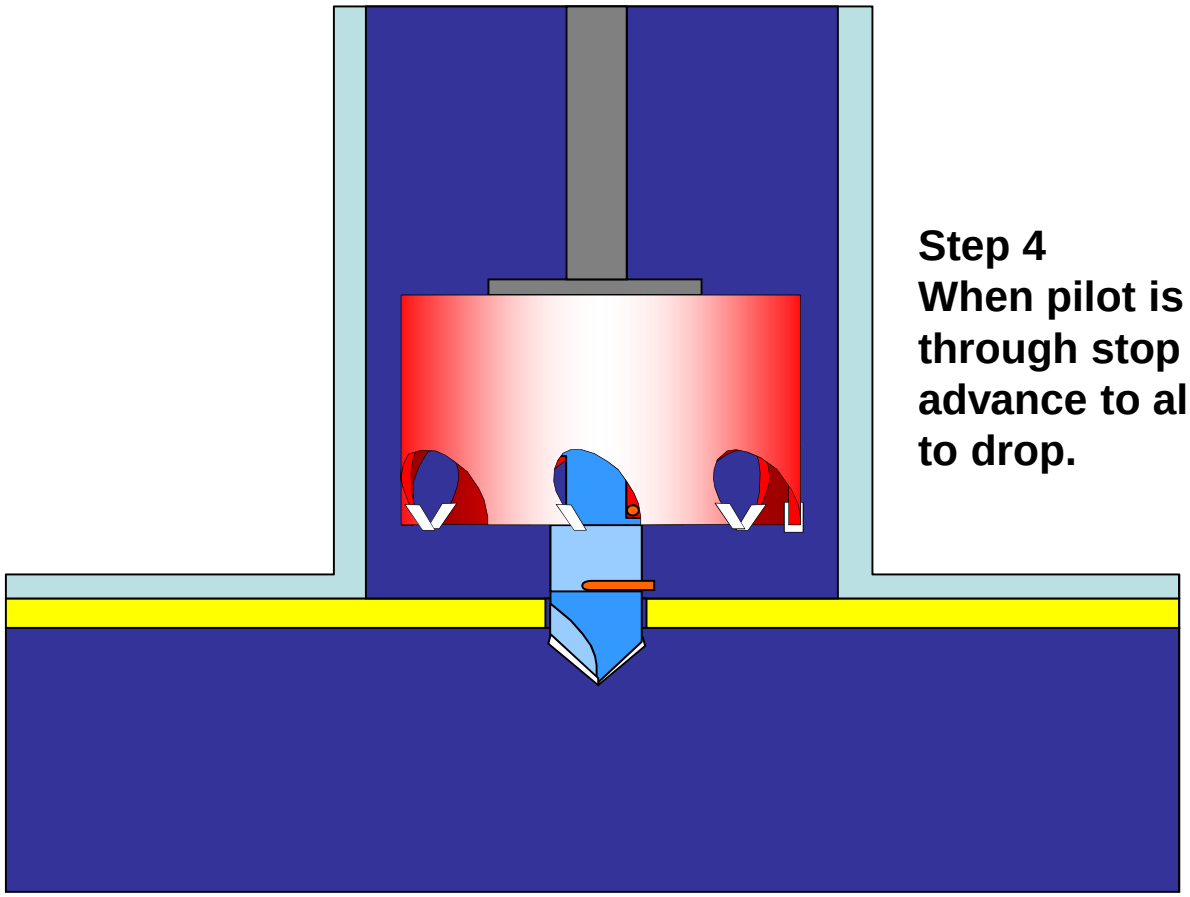

Step 4
When pilot is completely through stop rotation & advance to allow u-wires to drop.

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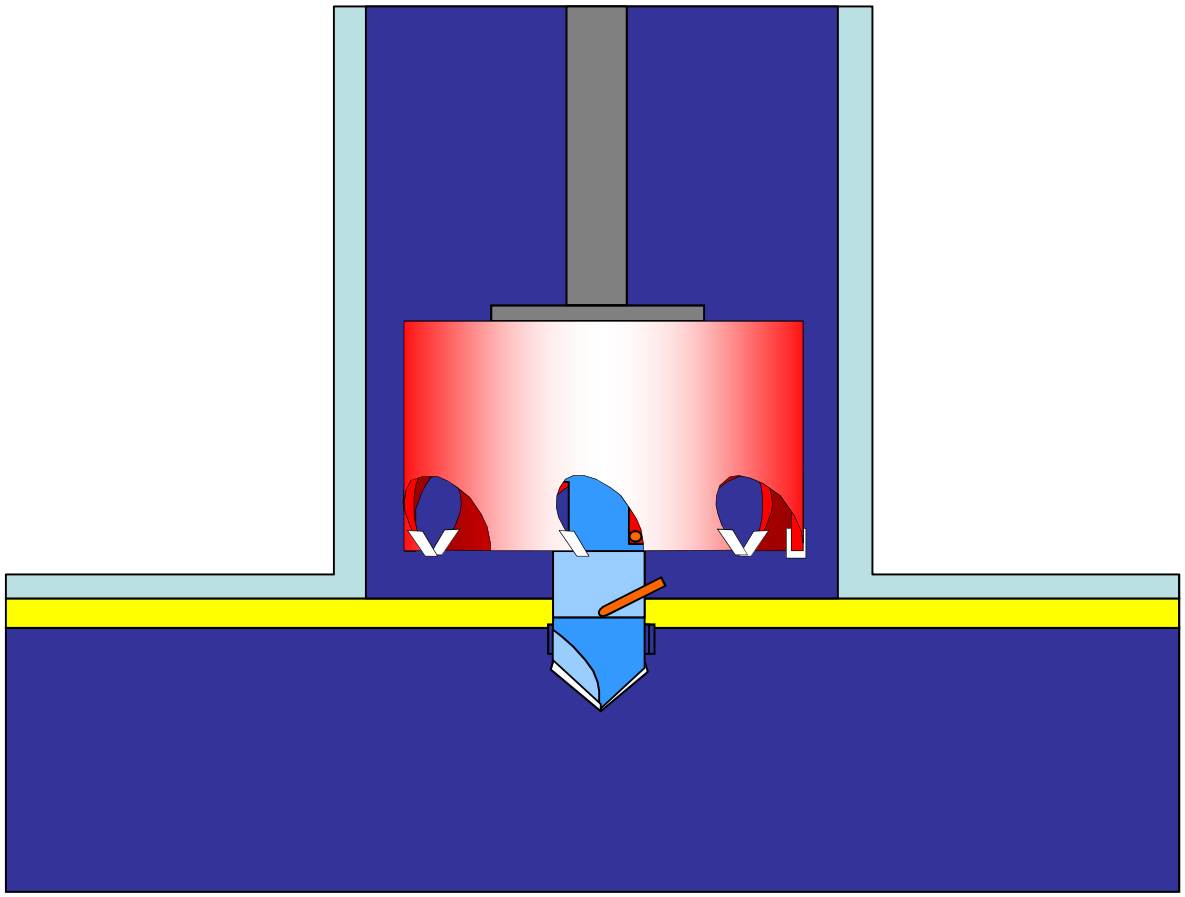

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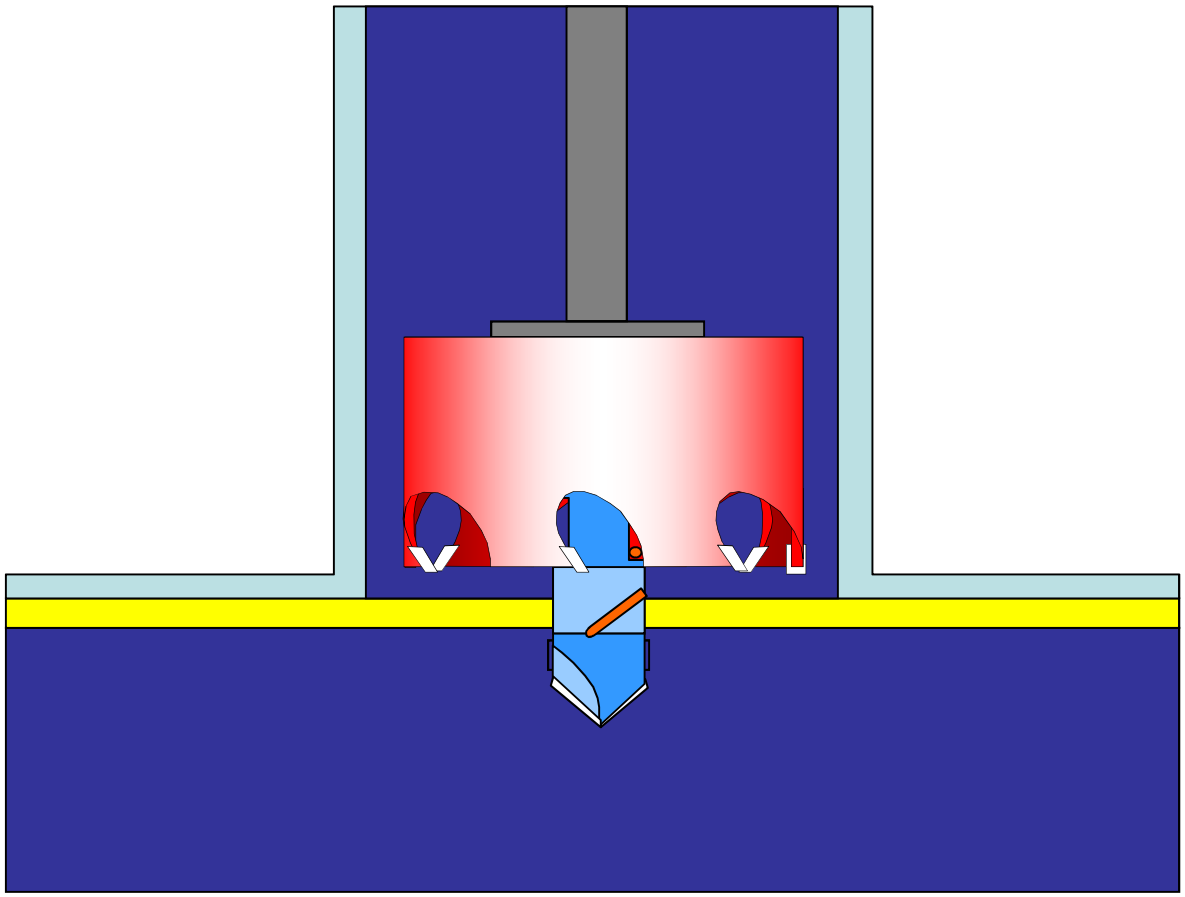

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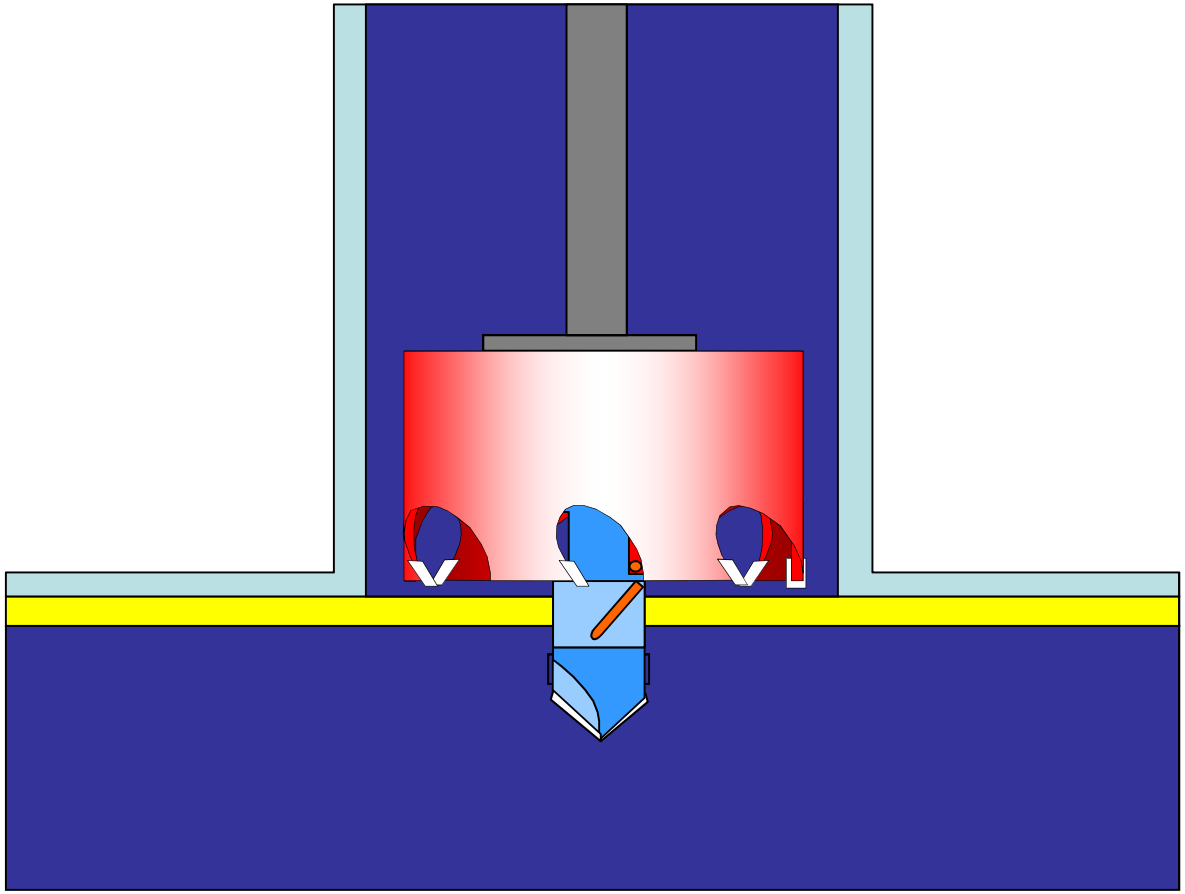

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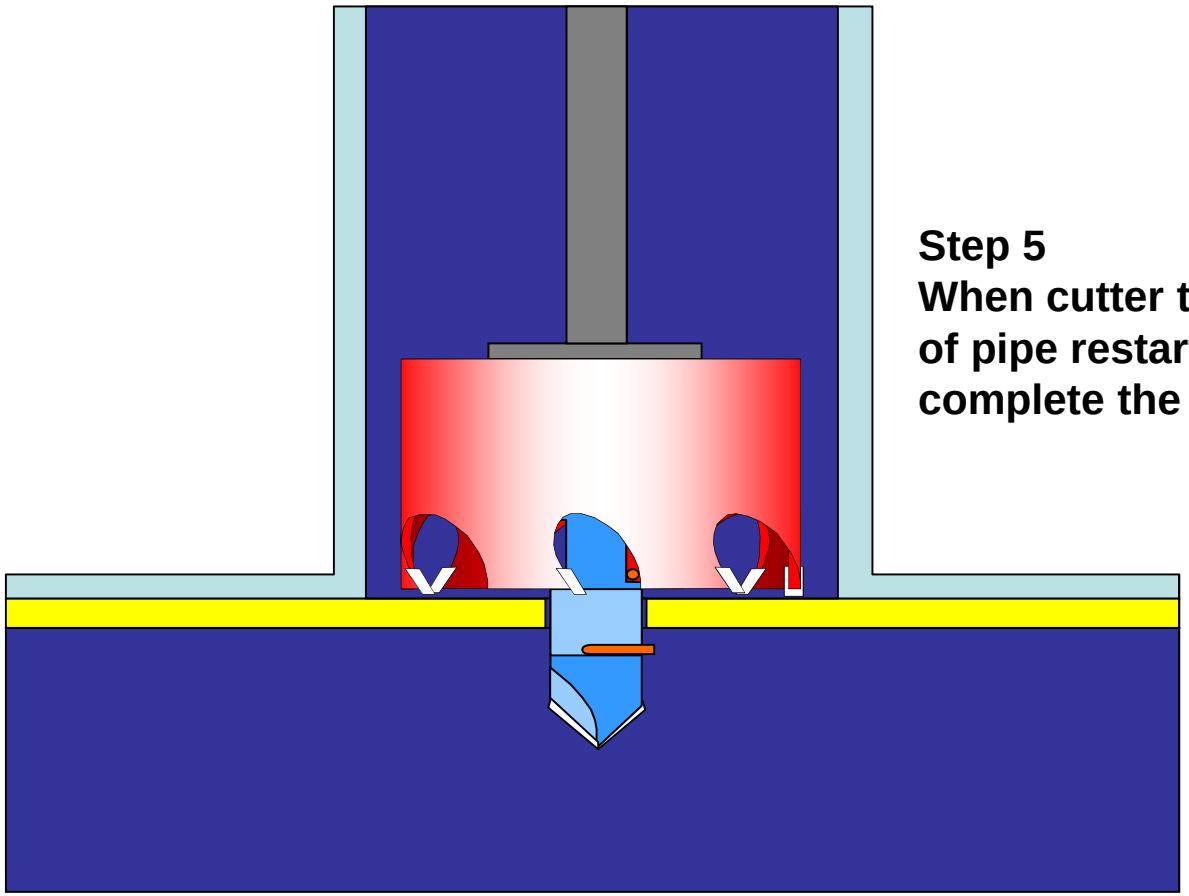

Step 5
When cutter touches top of pipe restart machine to complete the cut.

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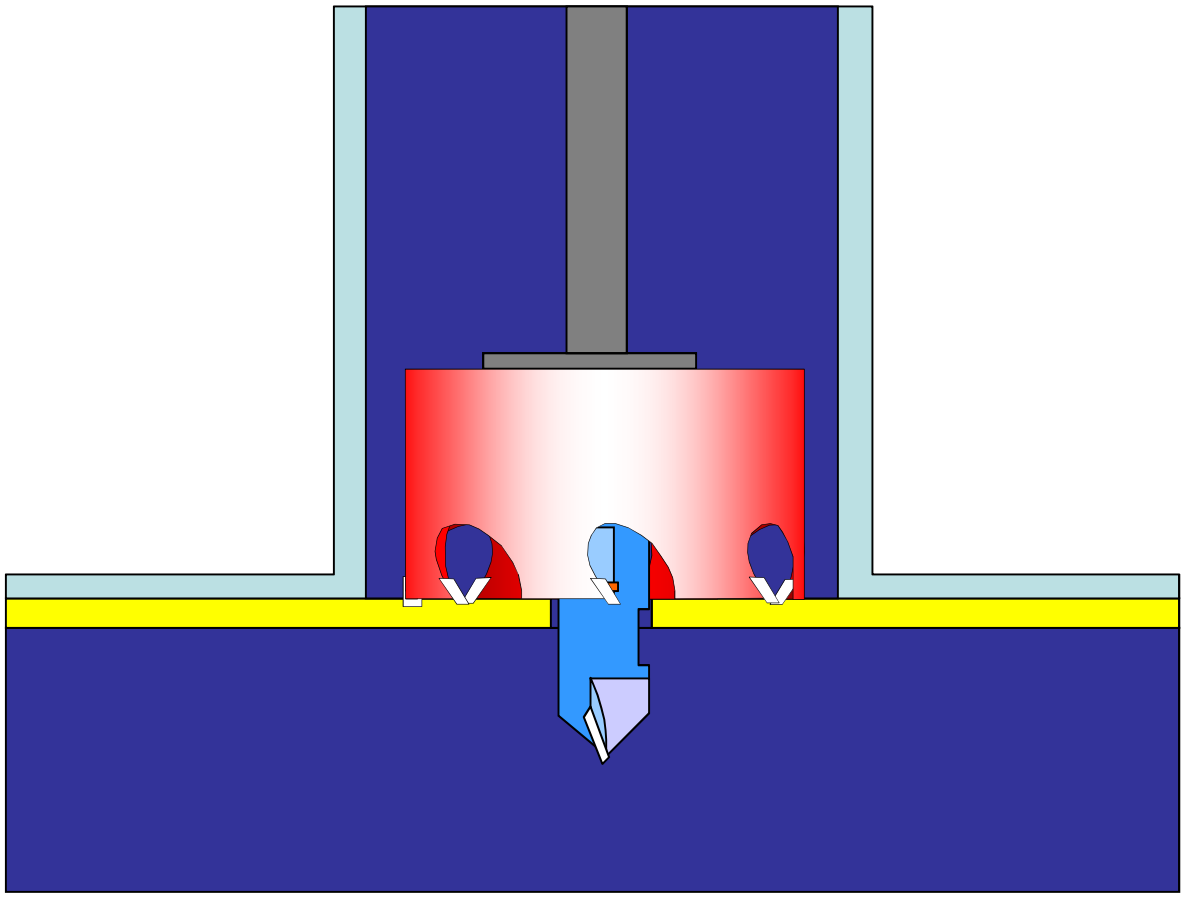

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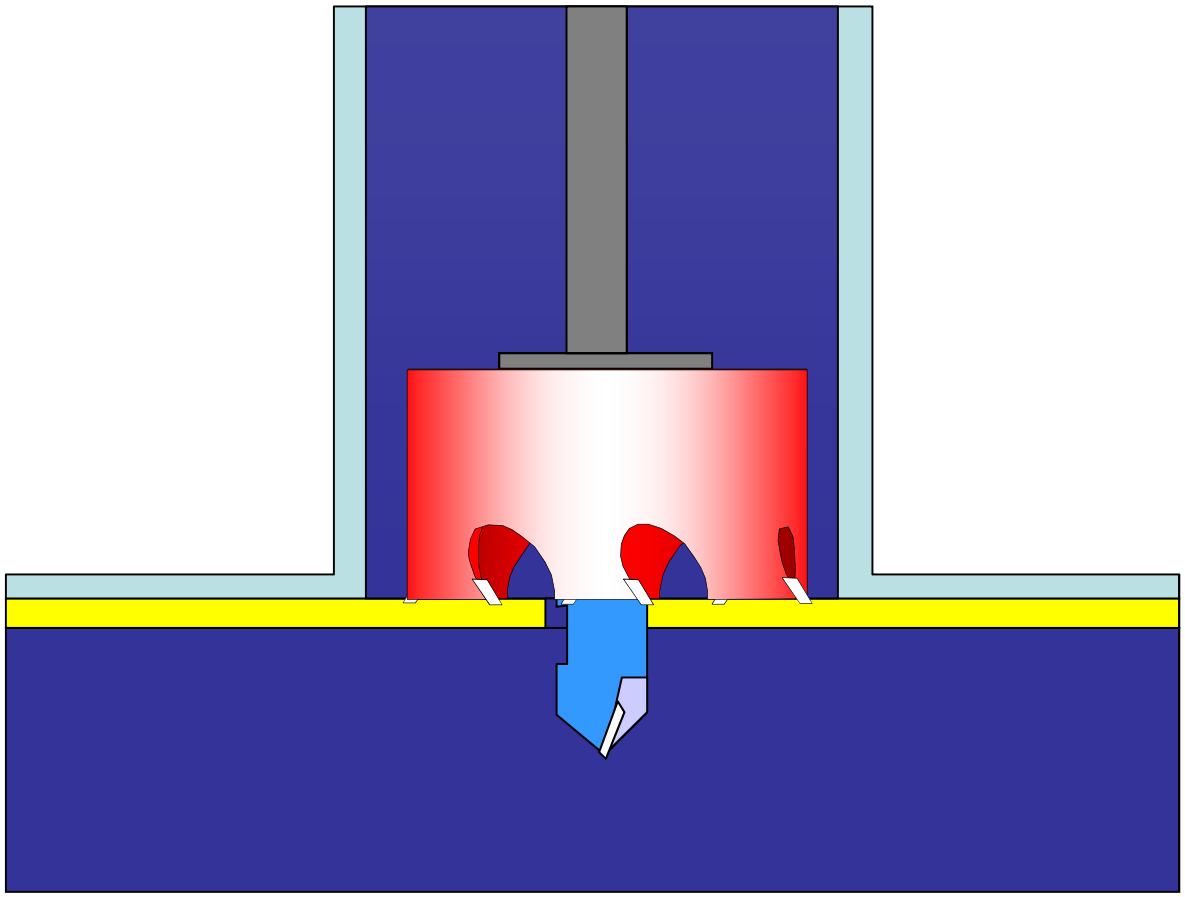

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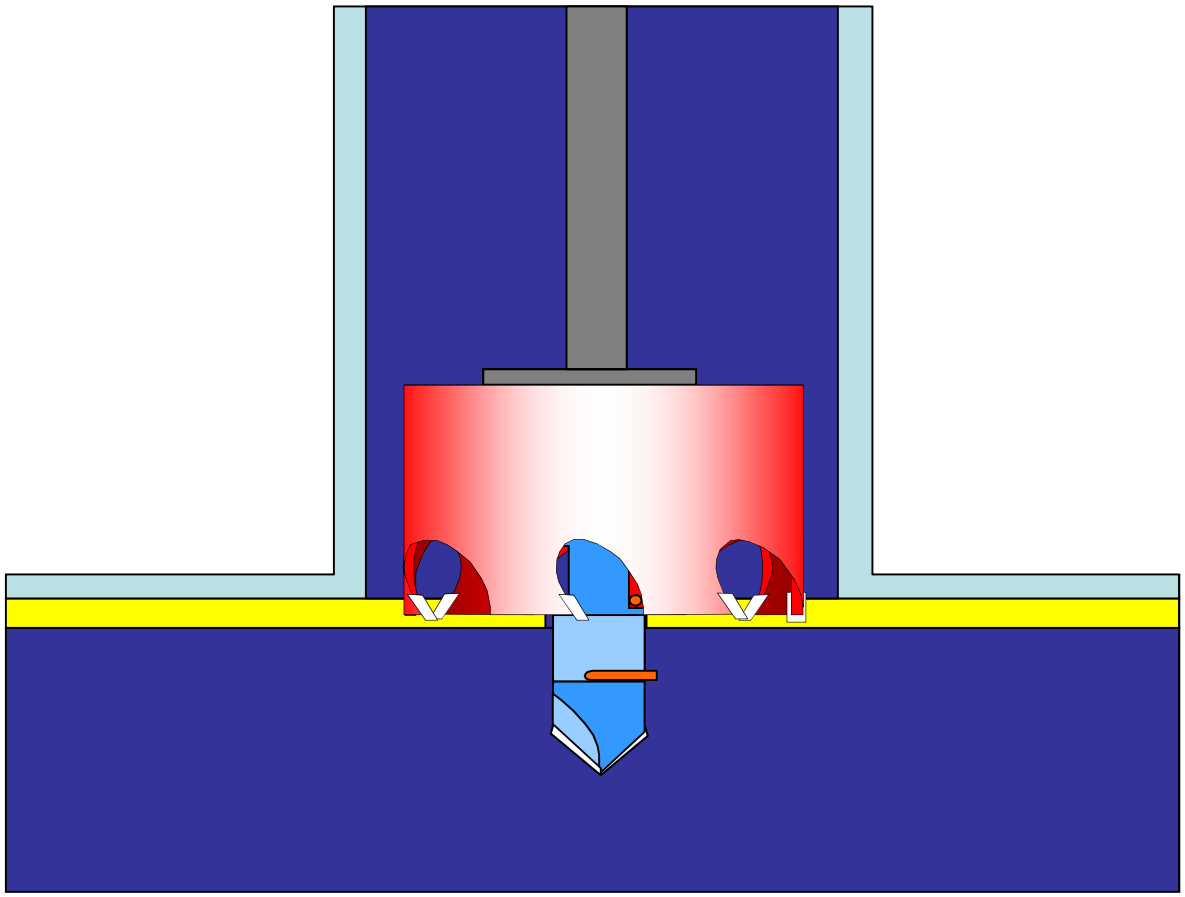

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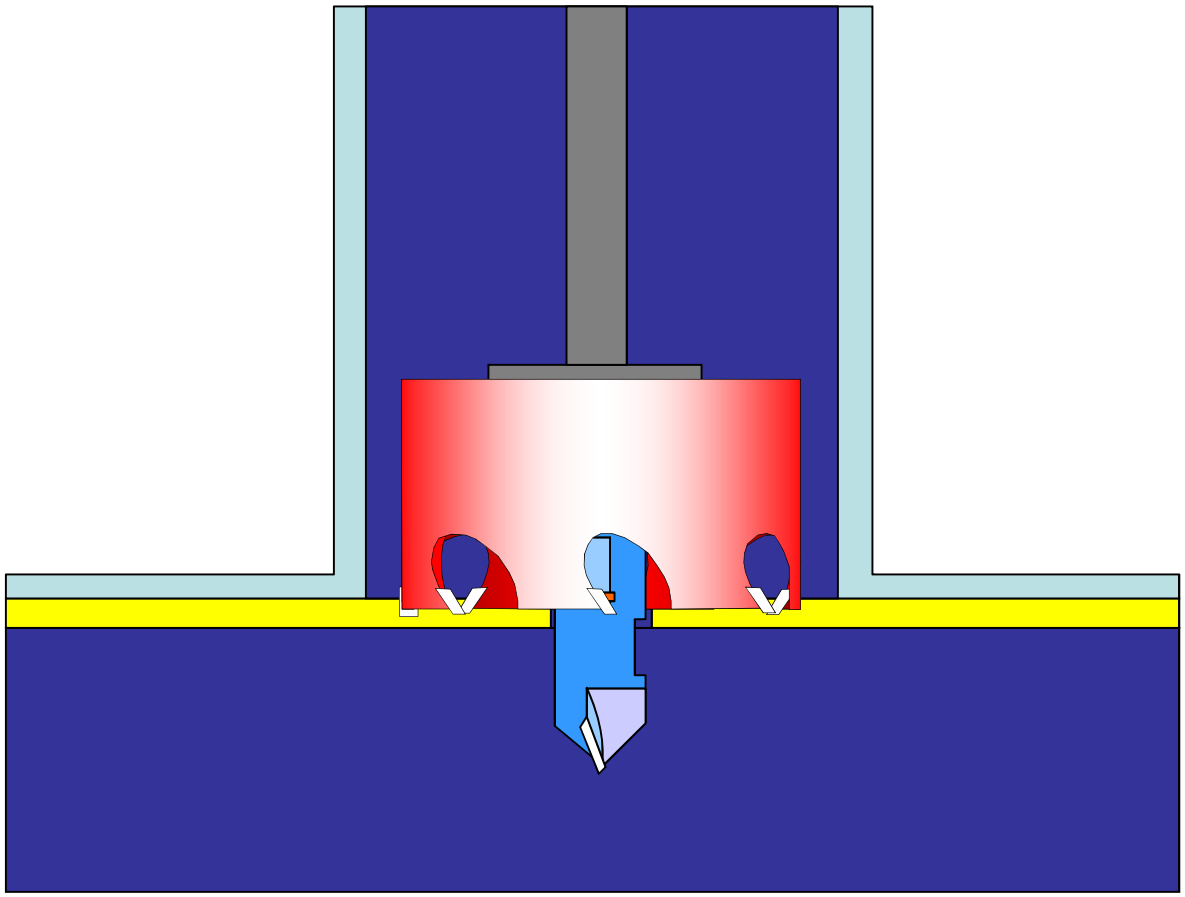

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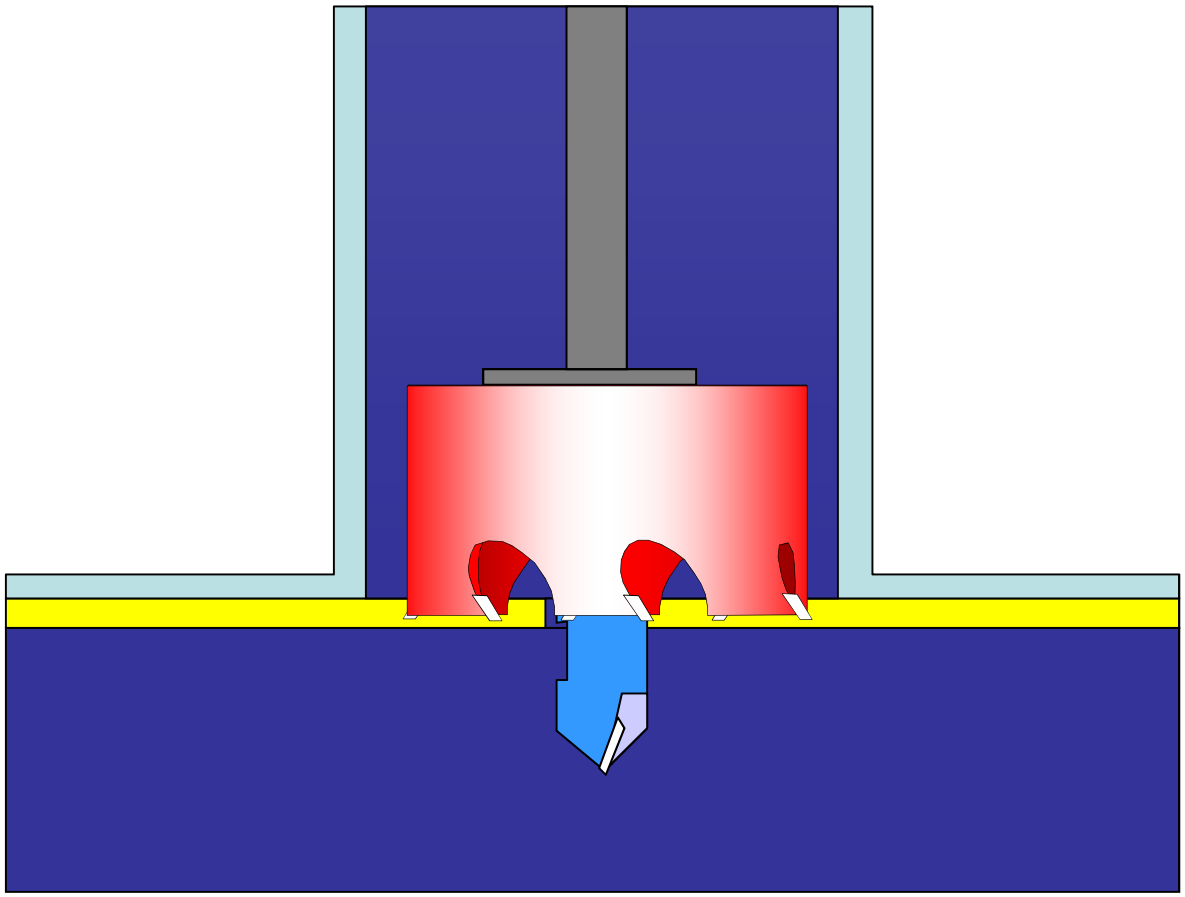

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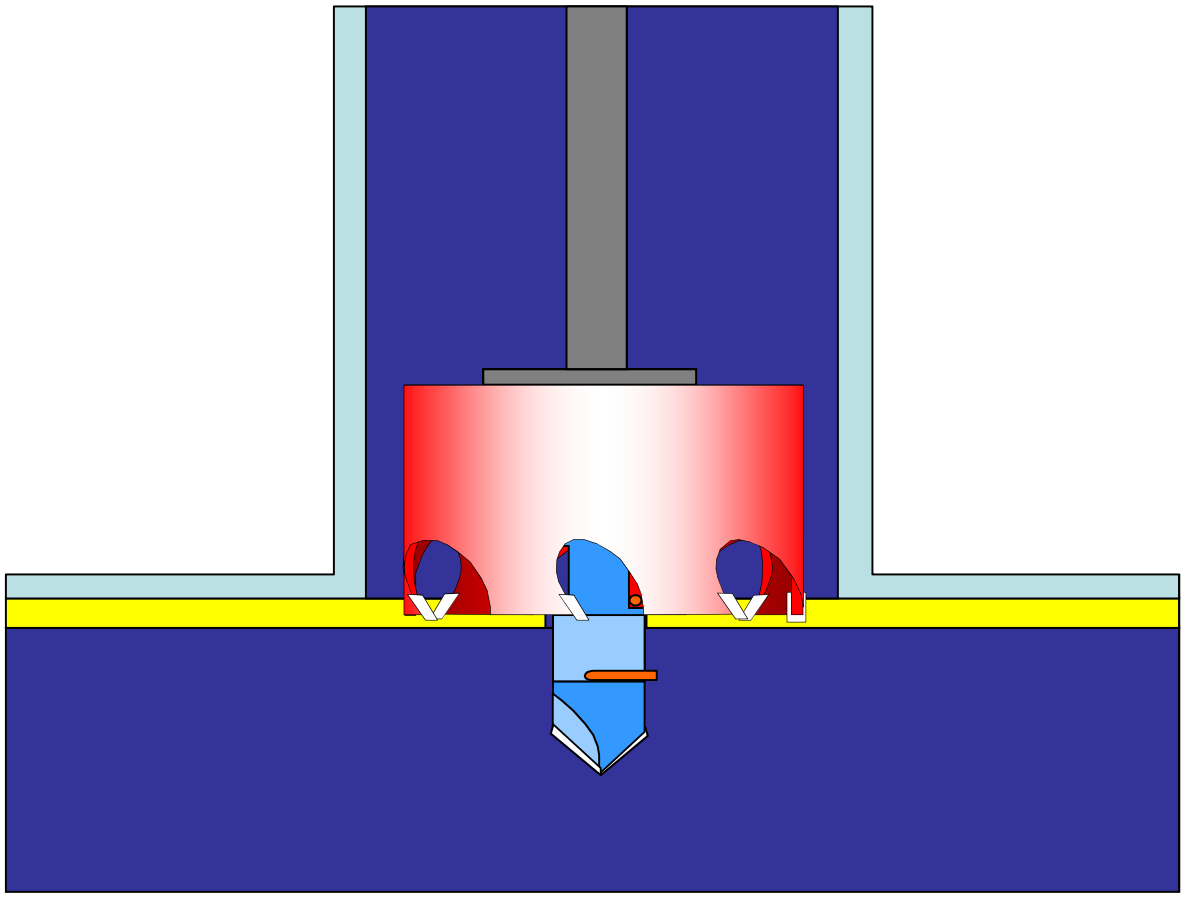

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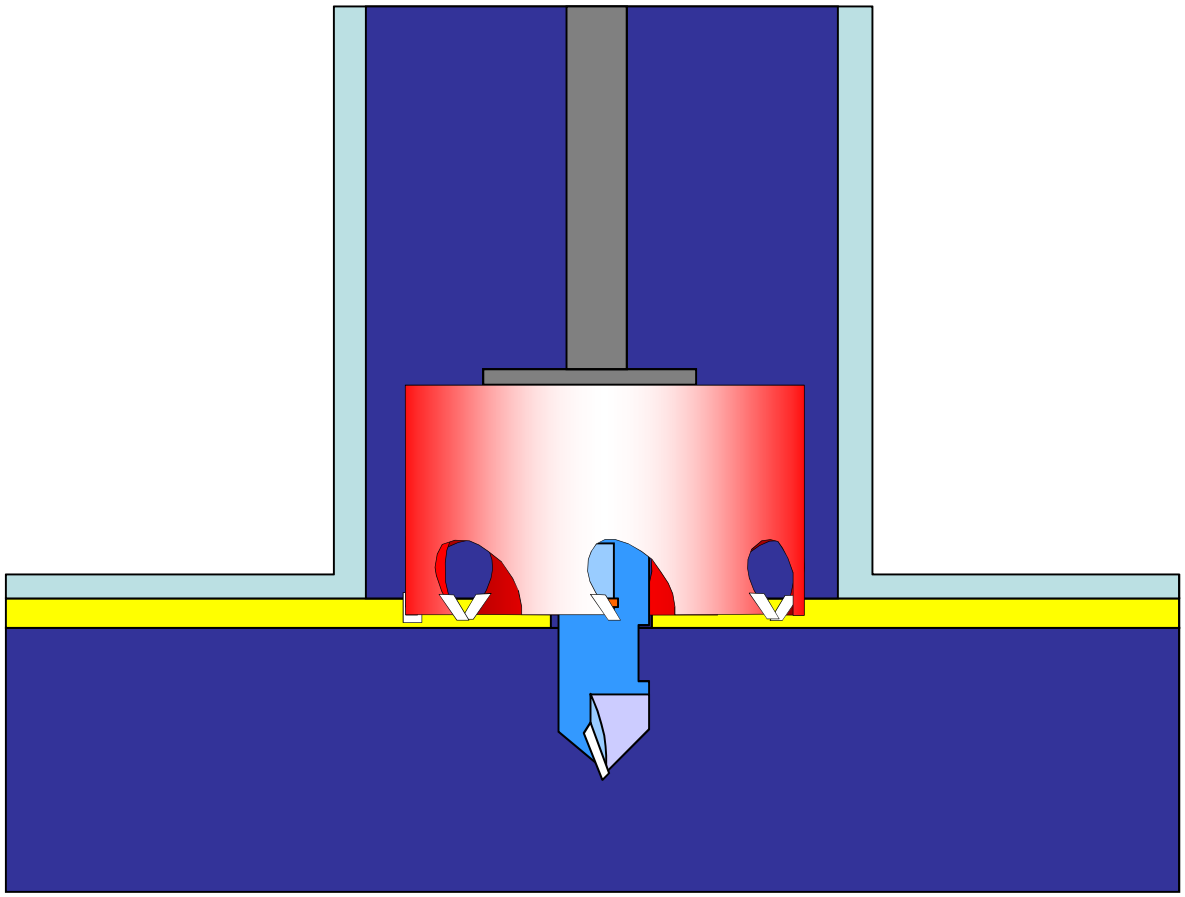

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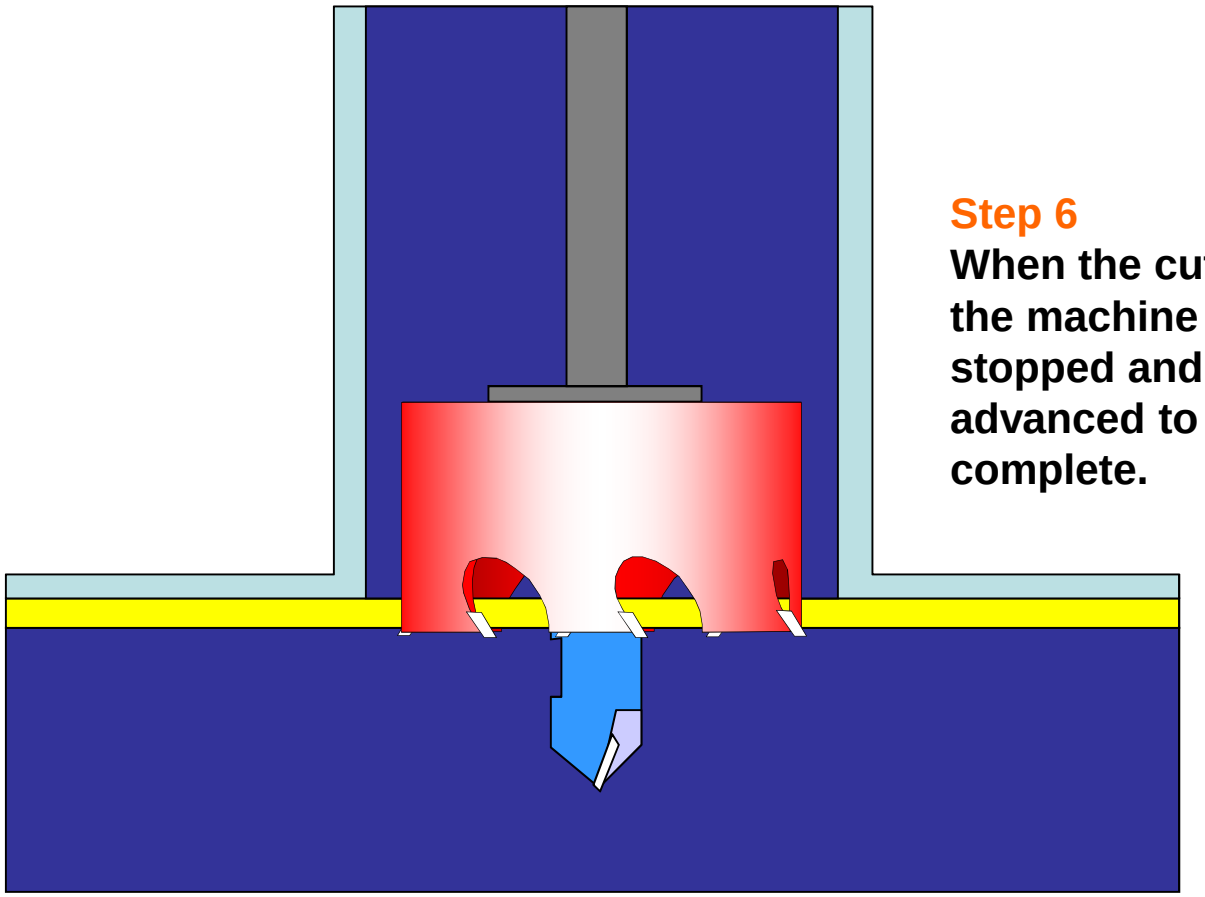

Step 6

When the cut is complete the machine should be stopped and cutter advanced to confirm tap complete.

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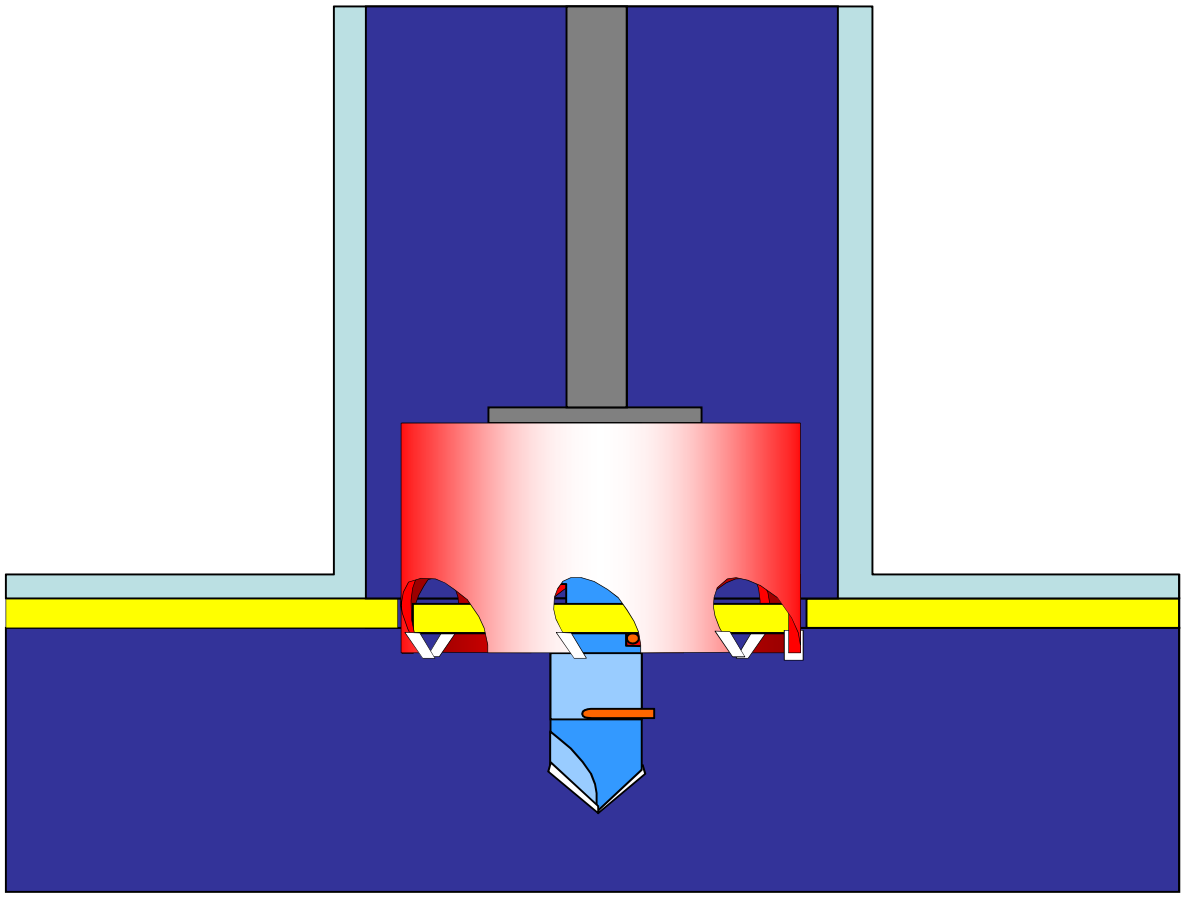

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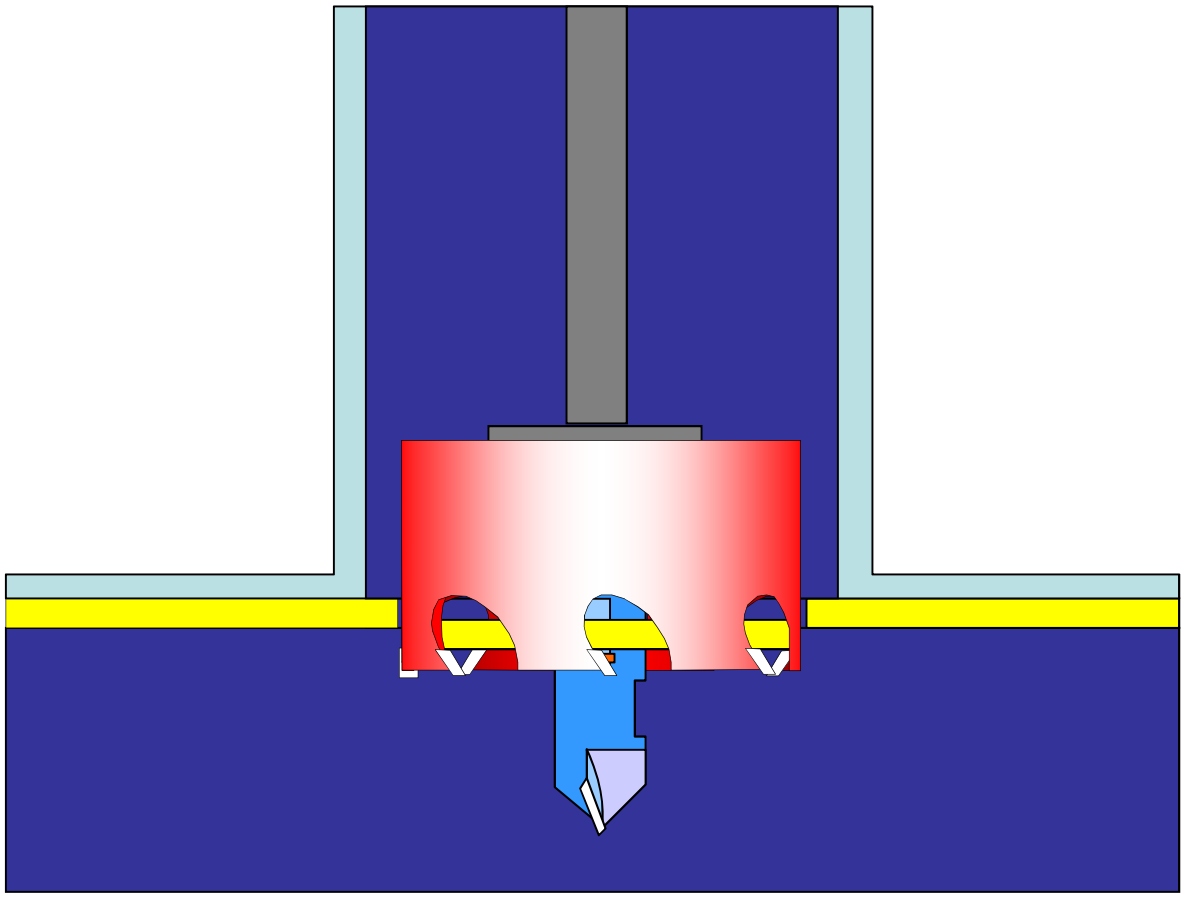

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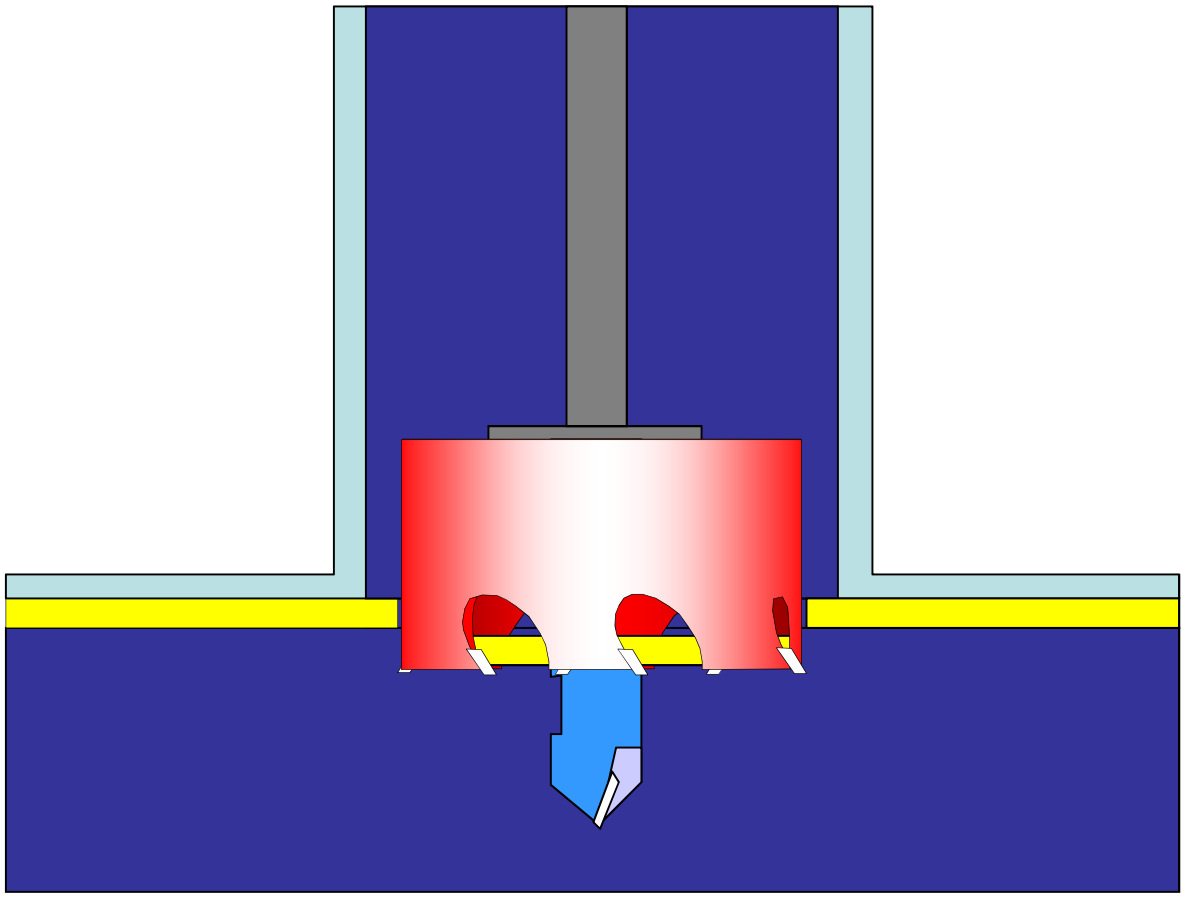

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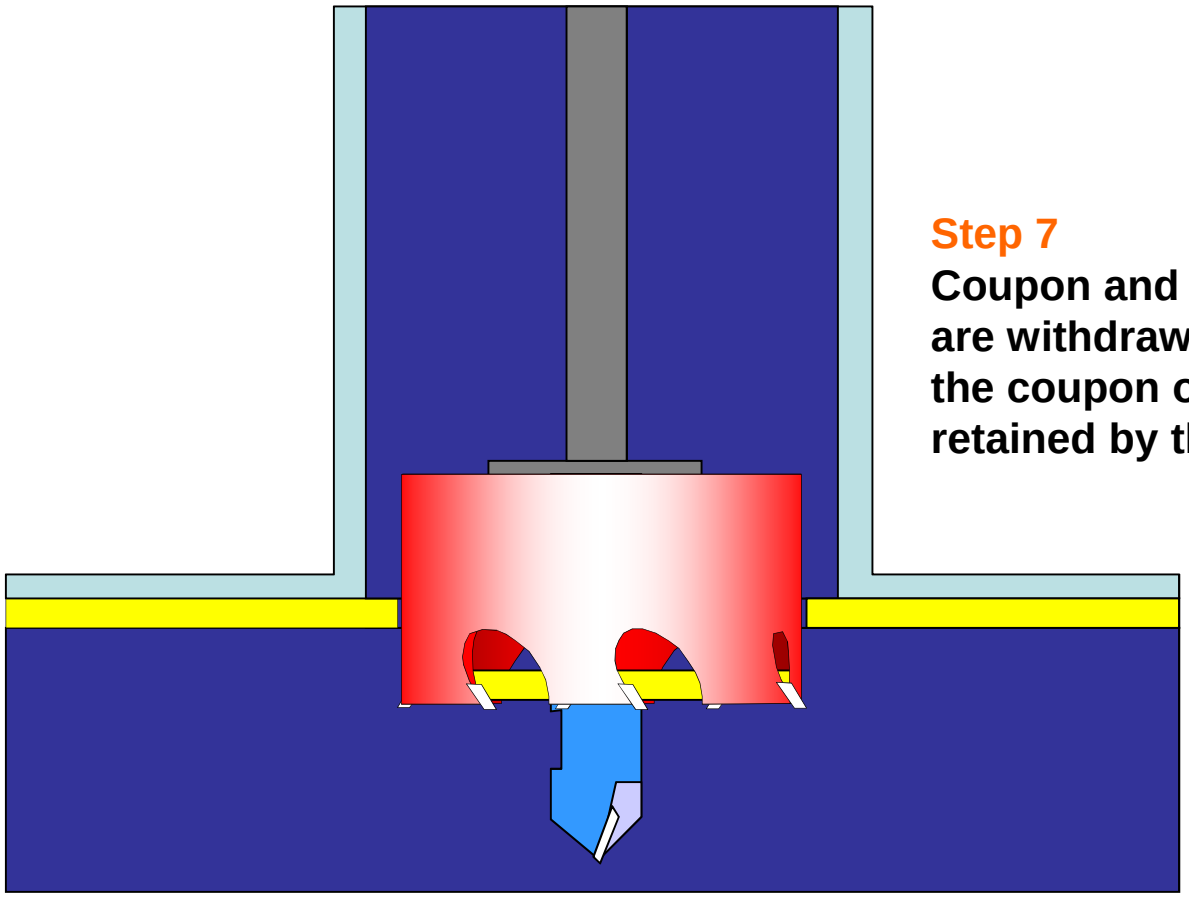

Step 7

Coupon and cutter are withdrawn with the coupon or cut out retained by the u-wires.

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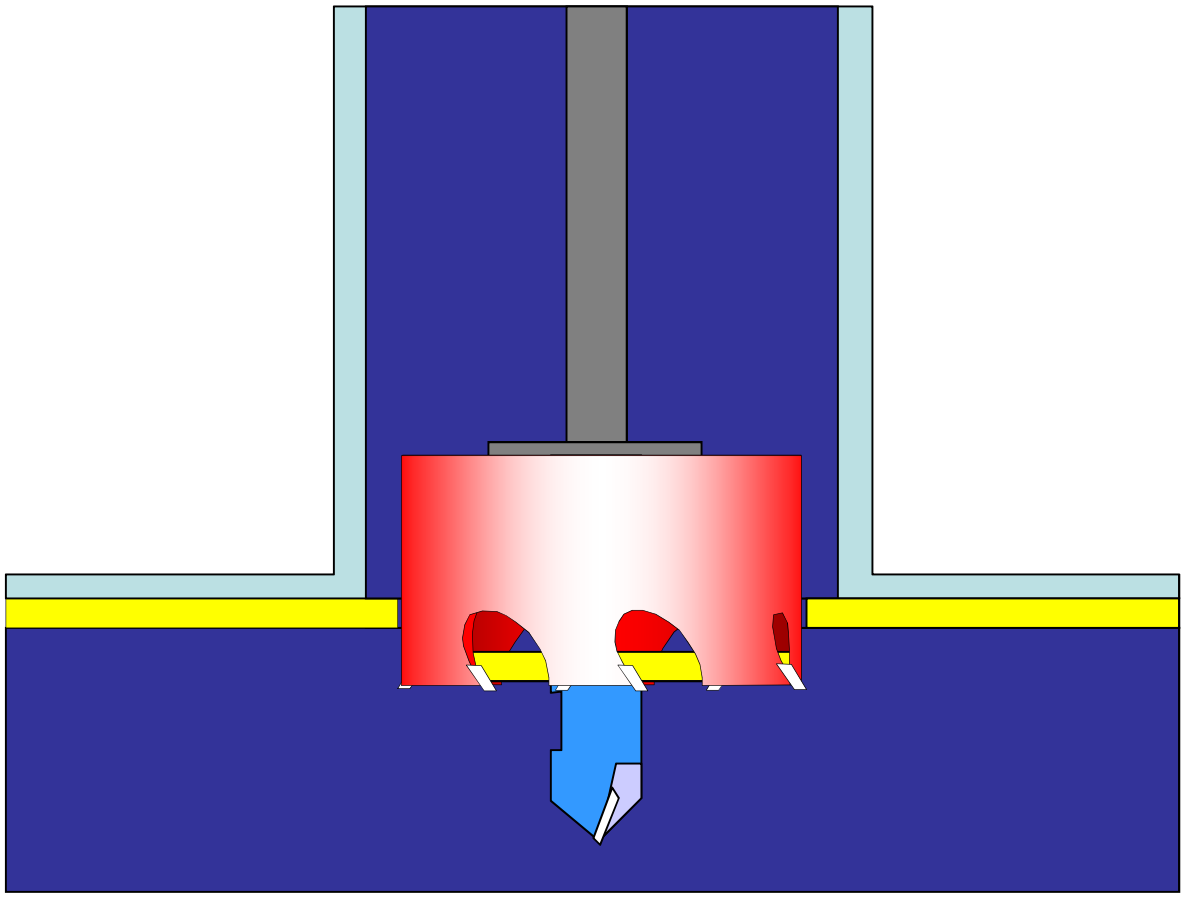

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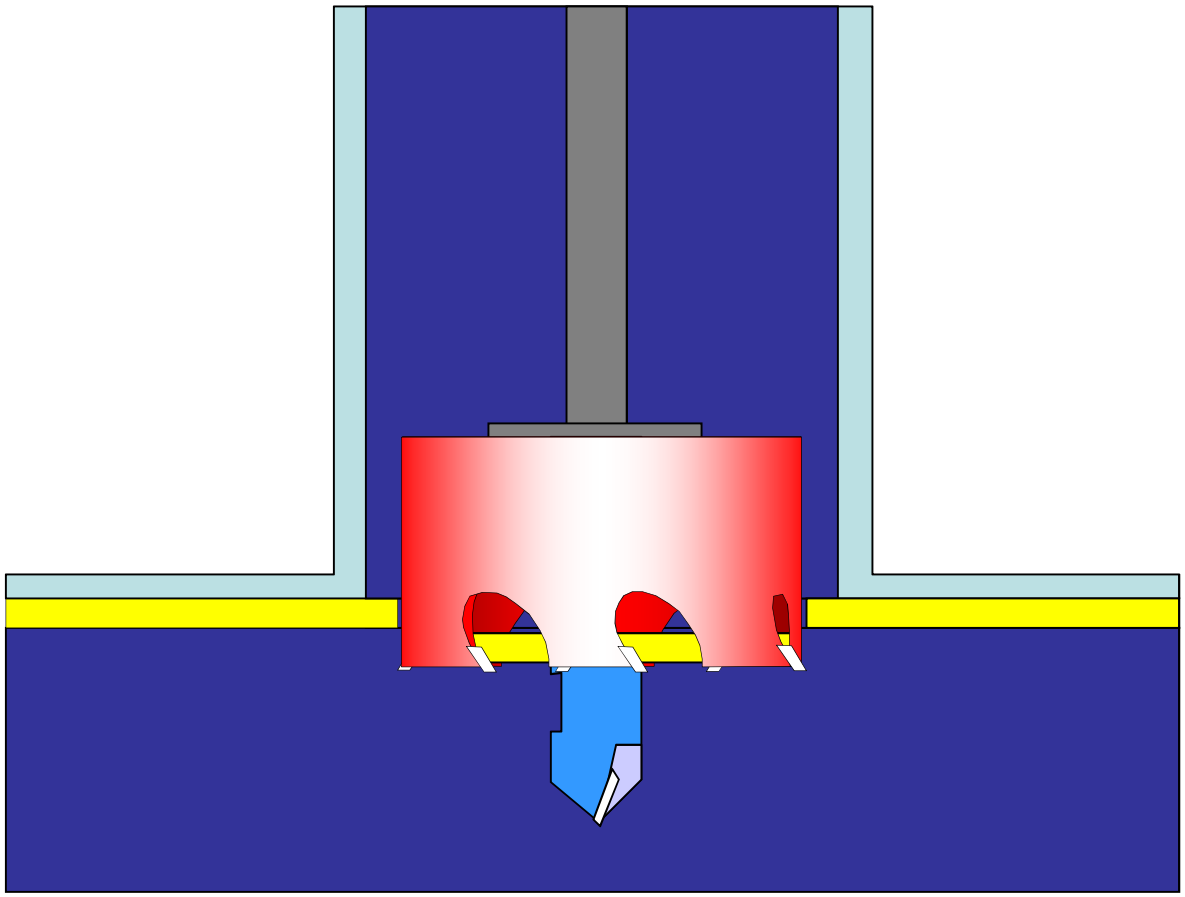

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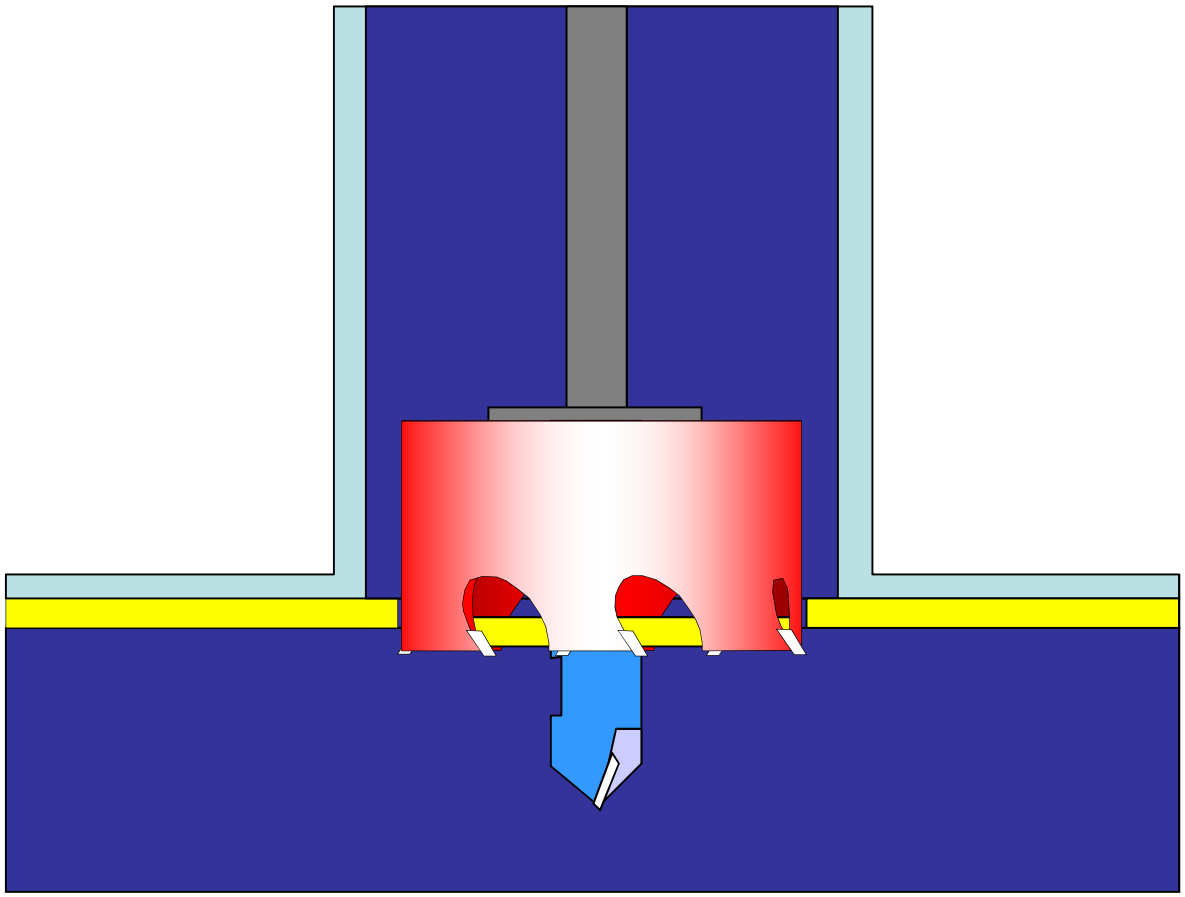

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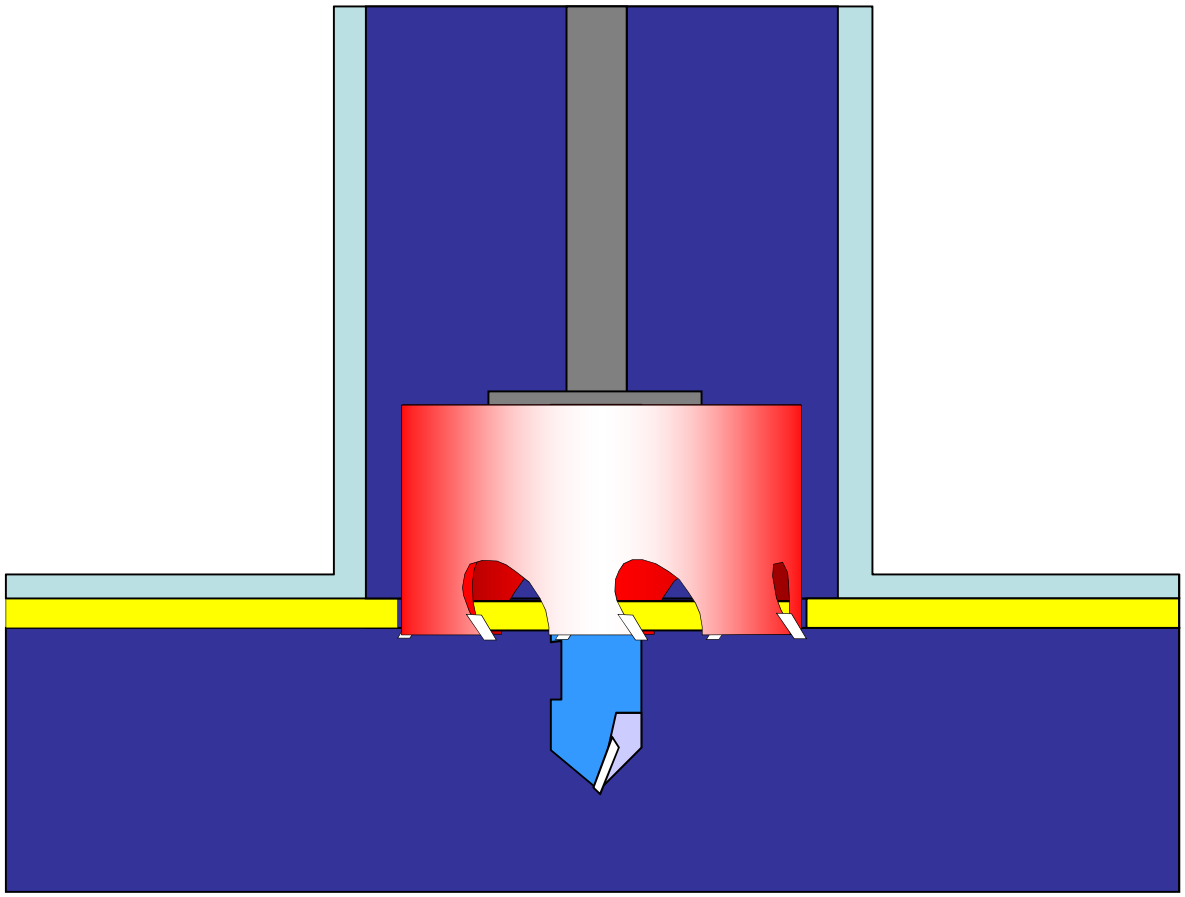

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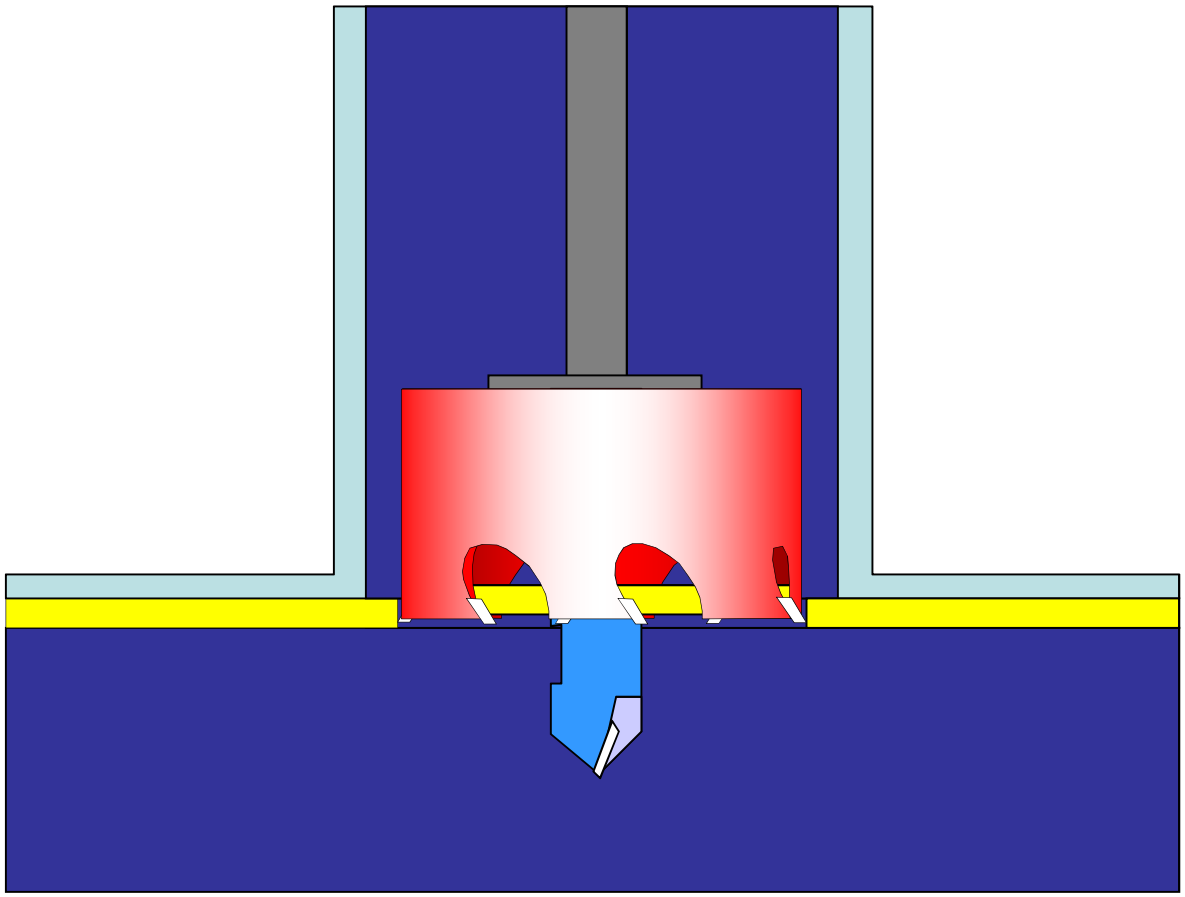

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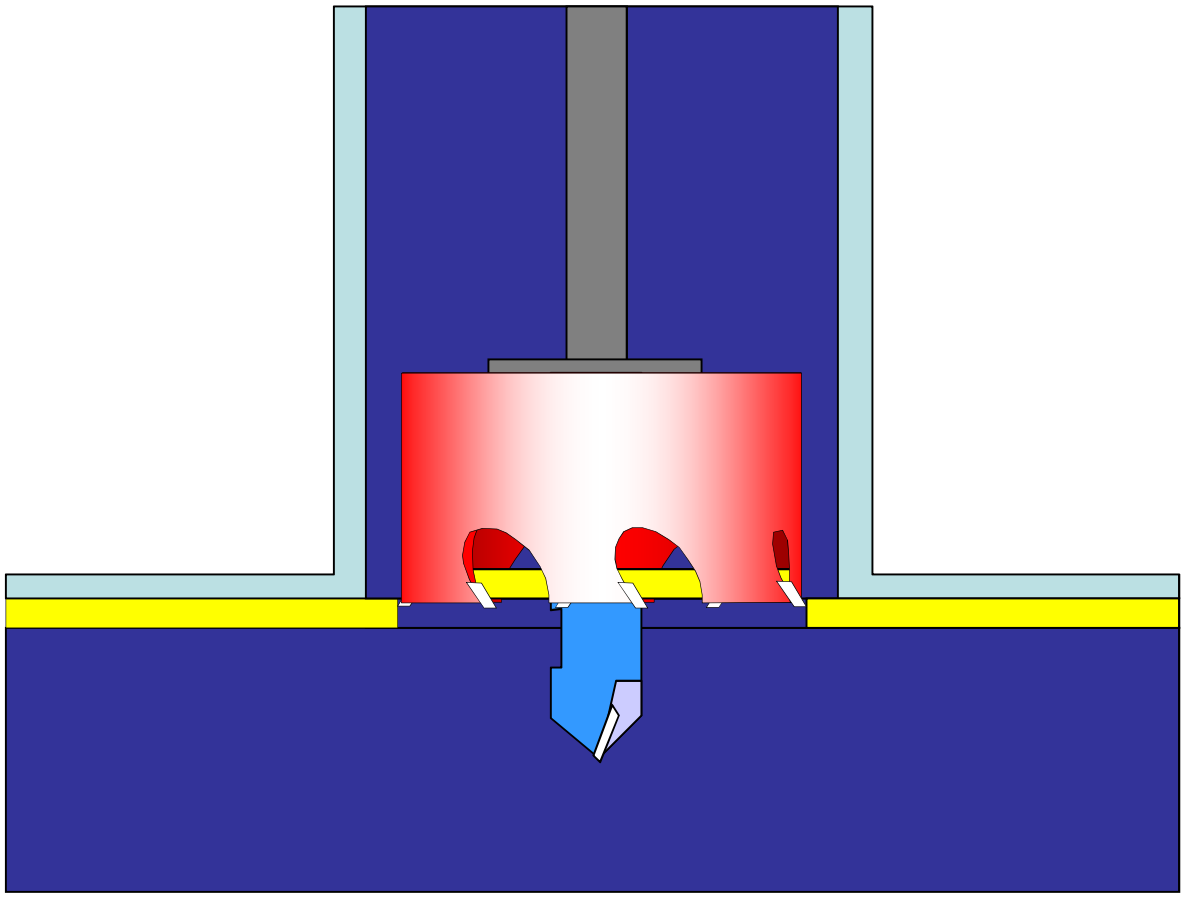

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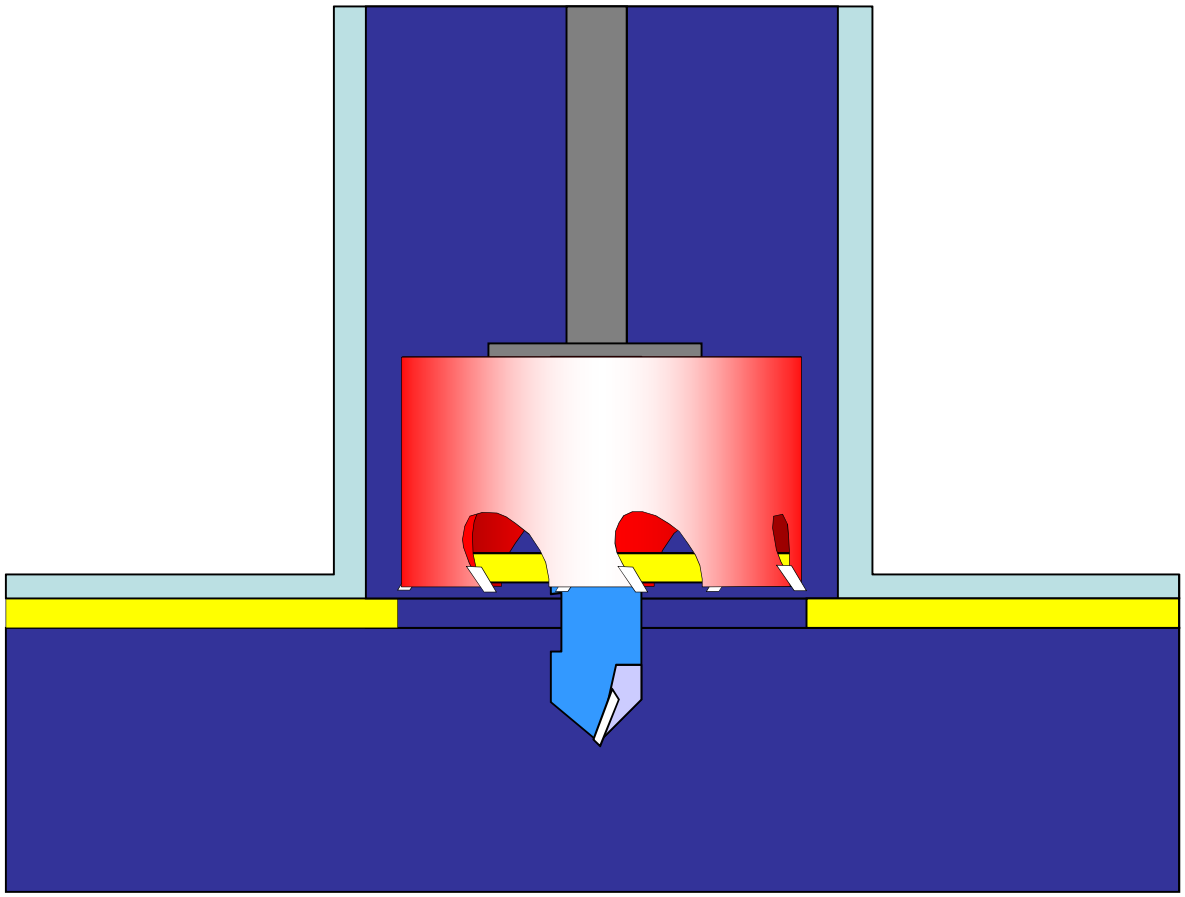

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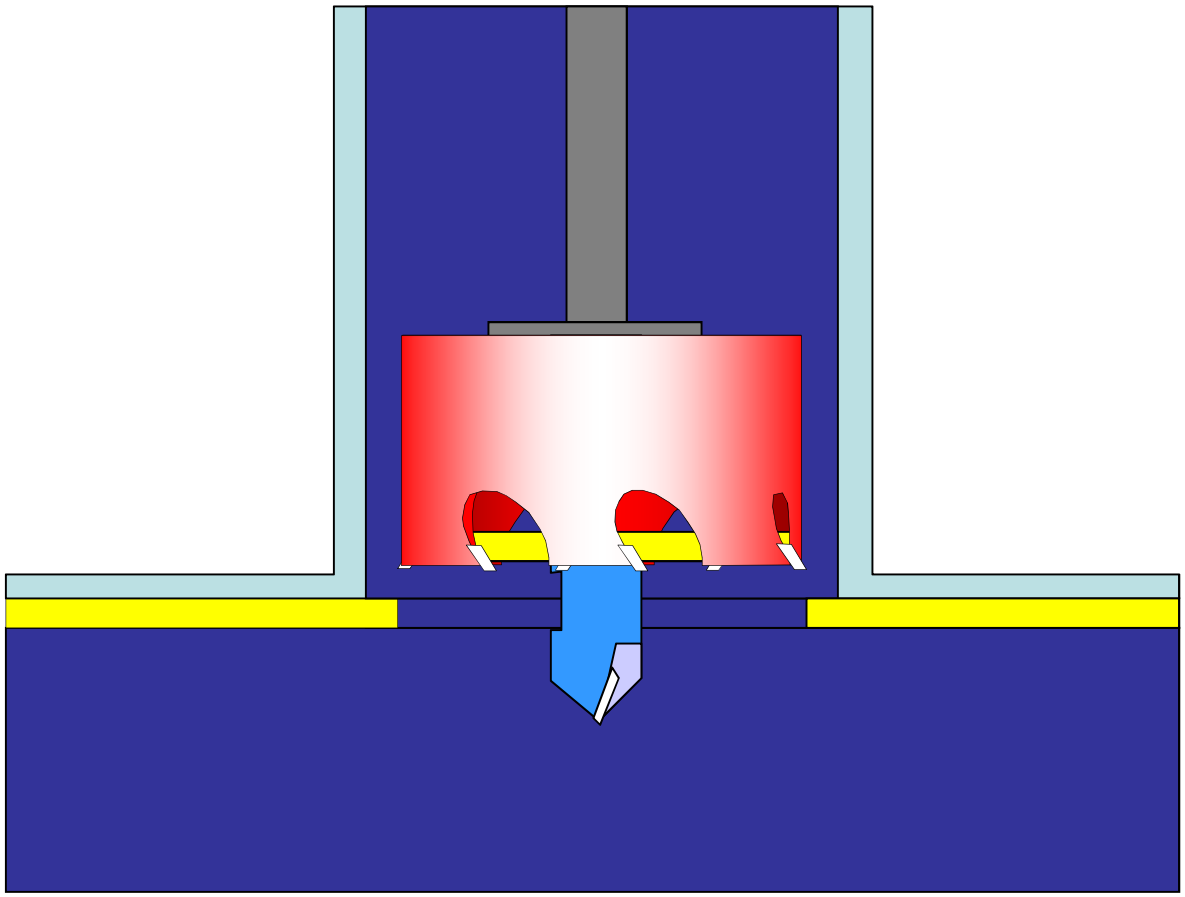

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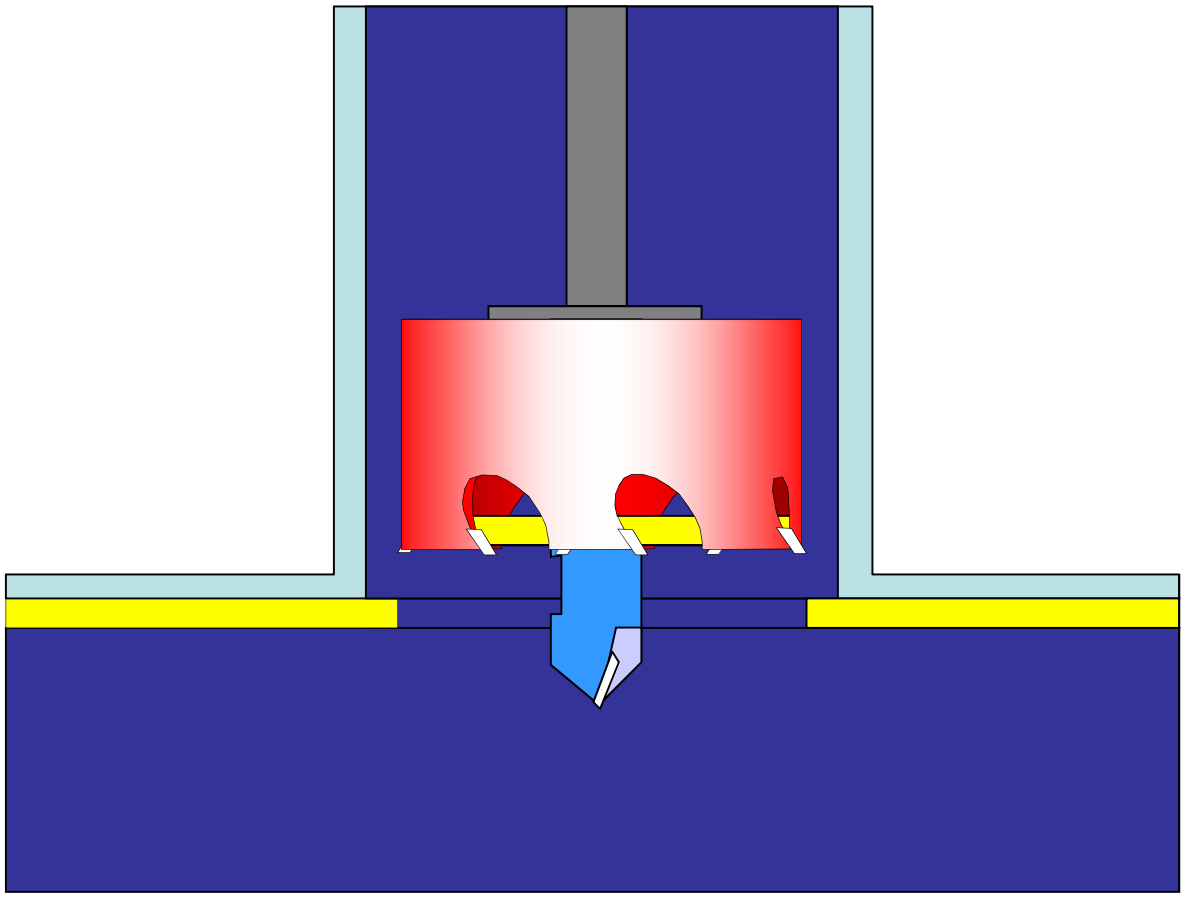

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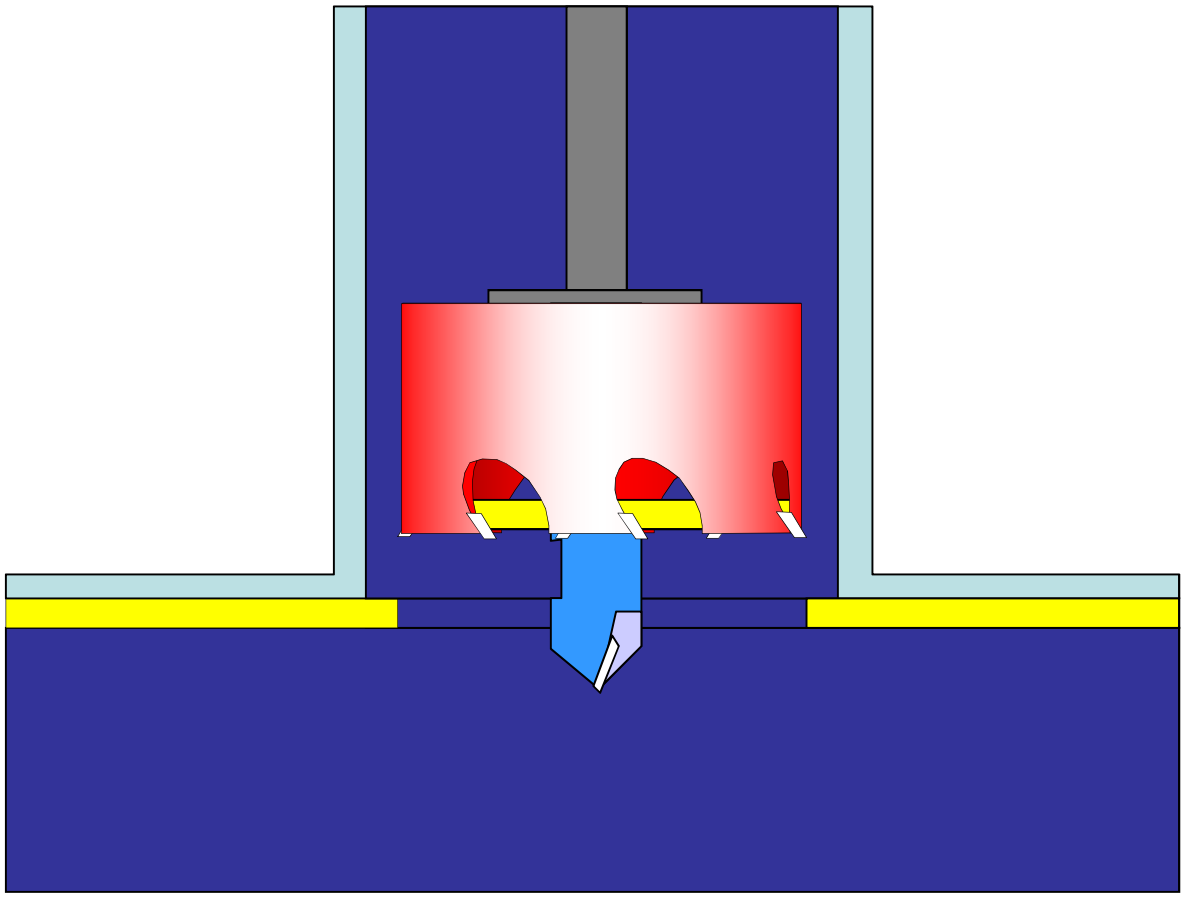

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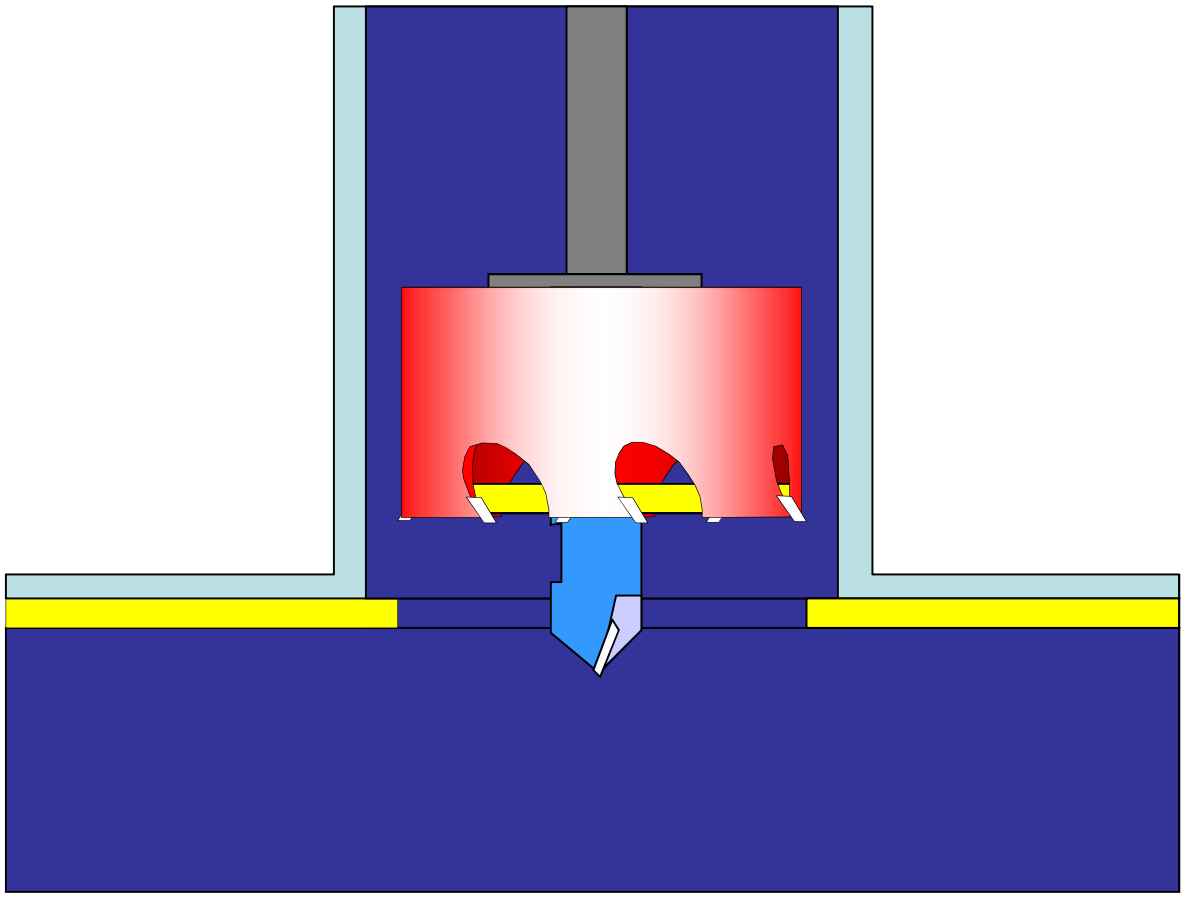

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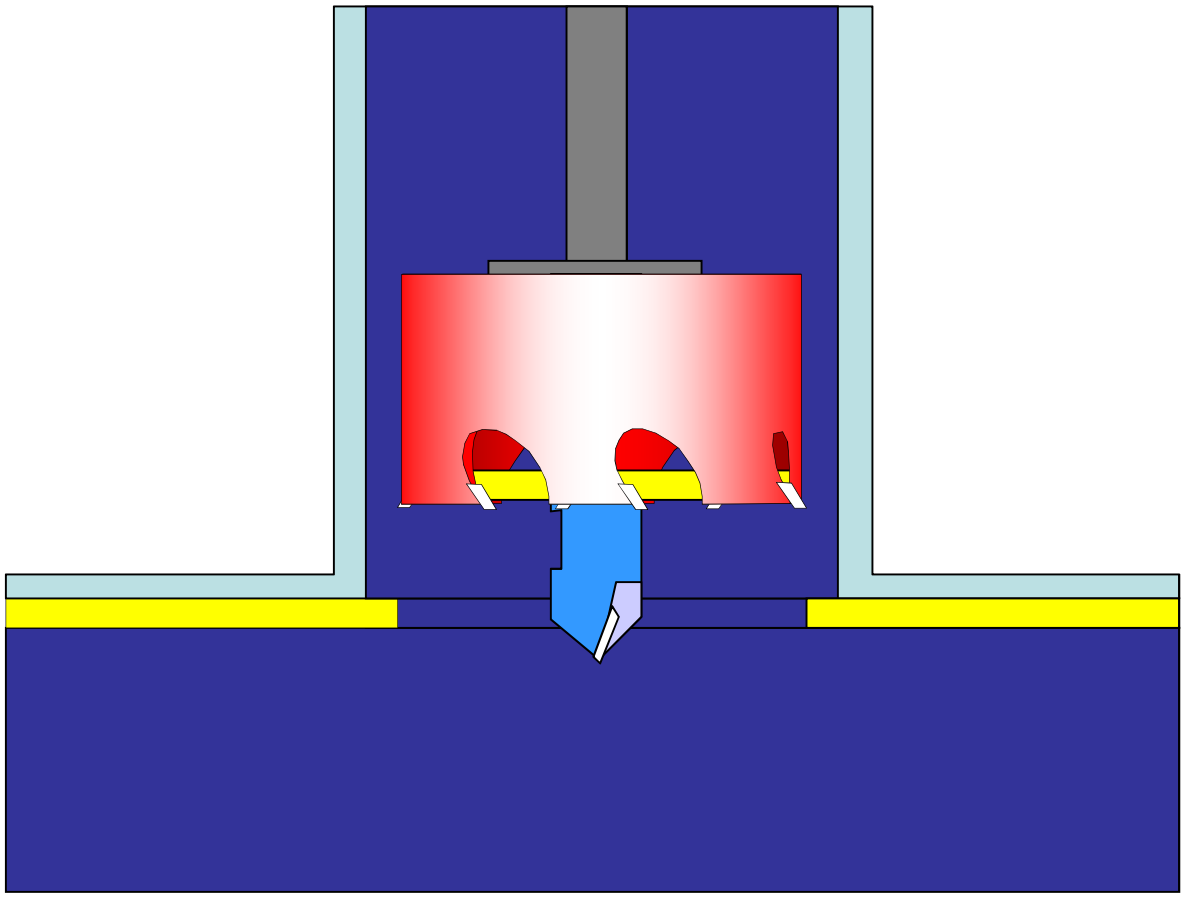

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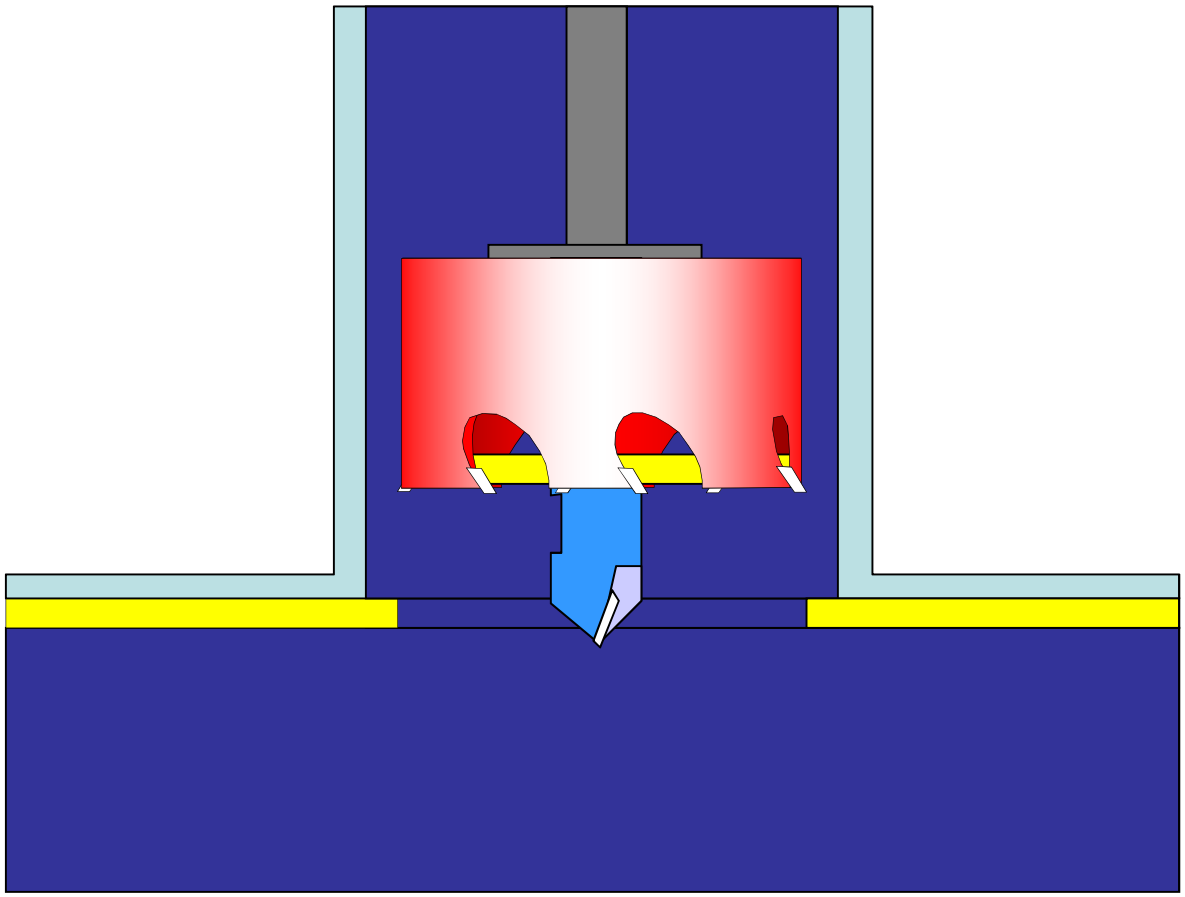

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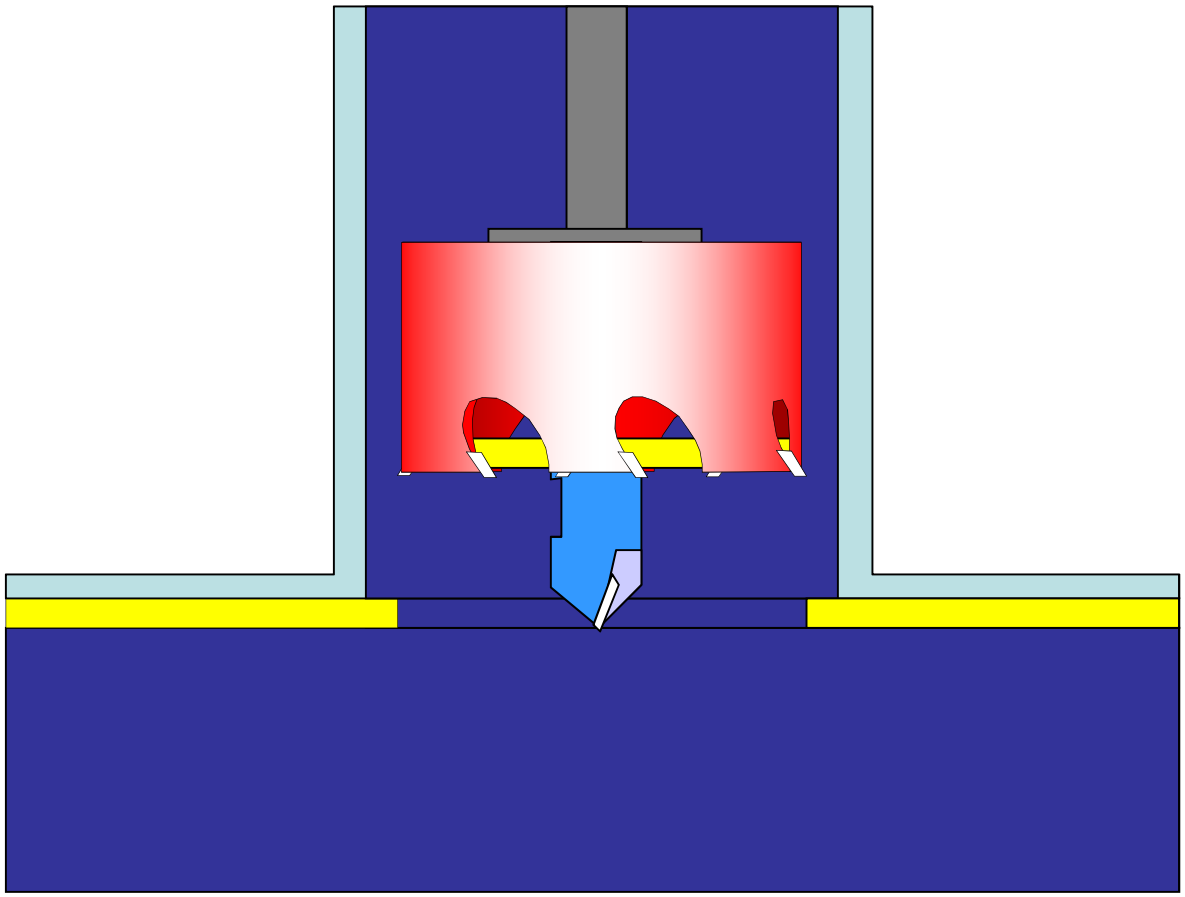

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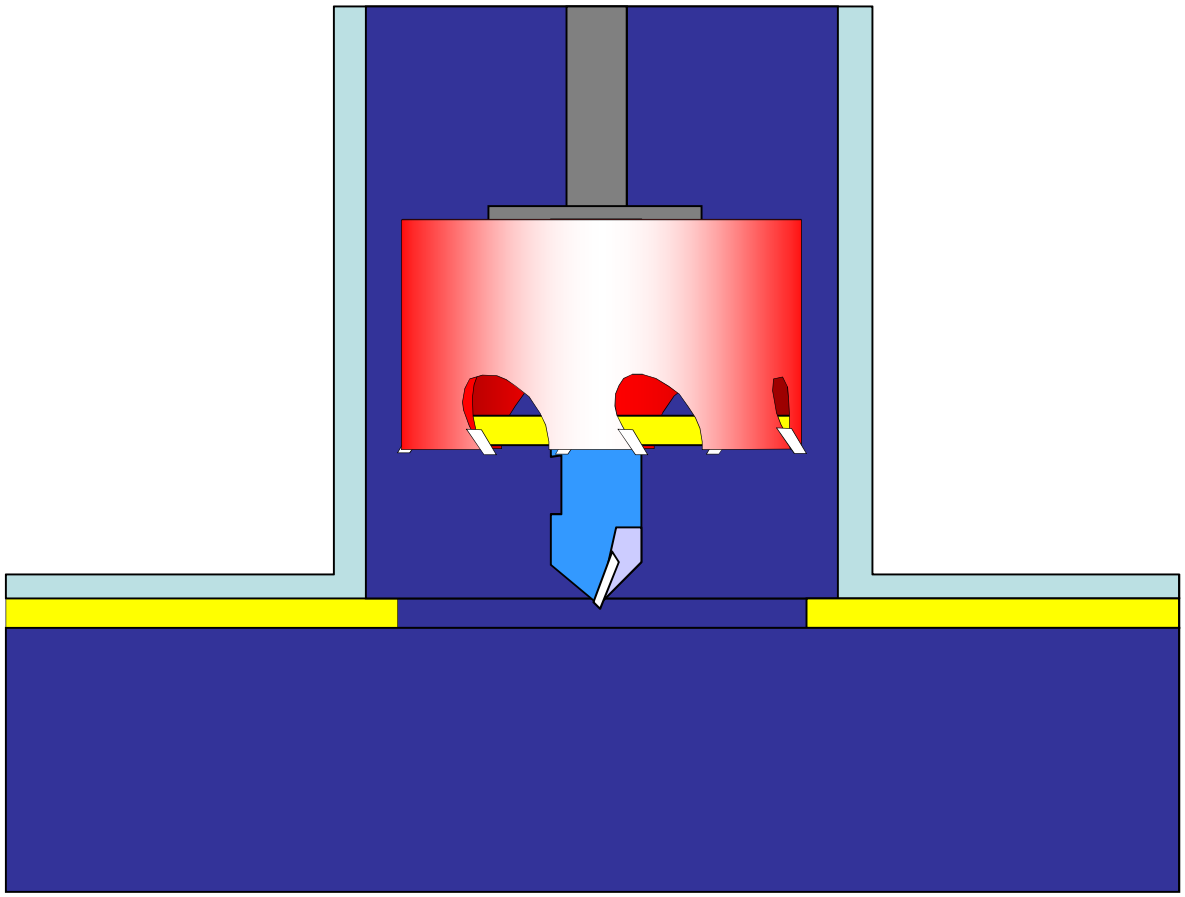

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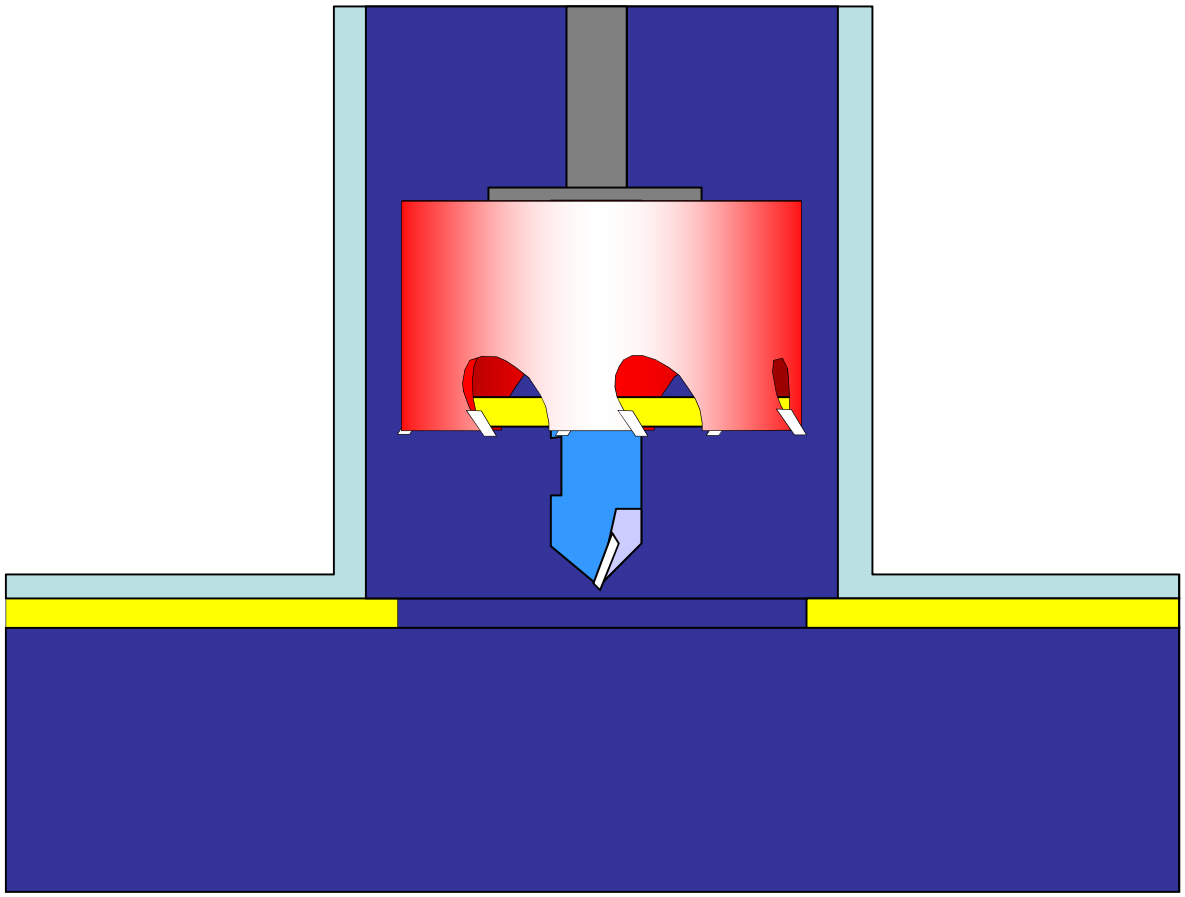

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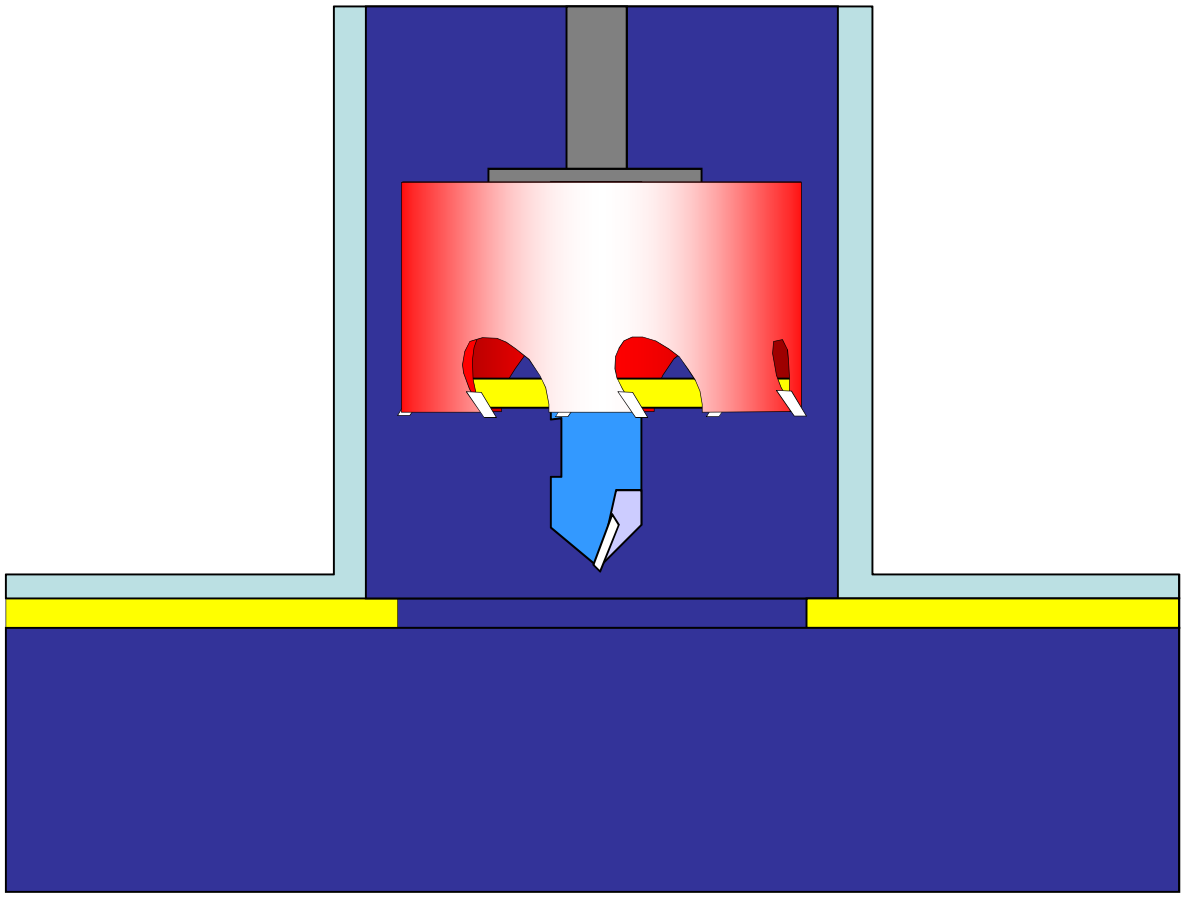

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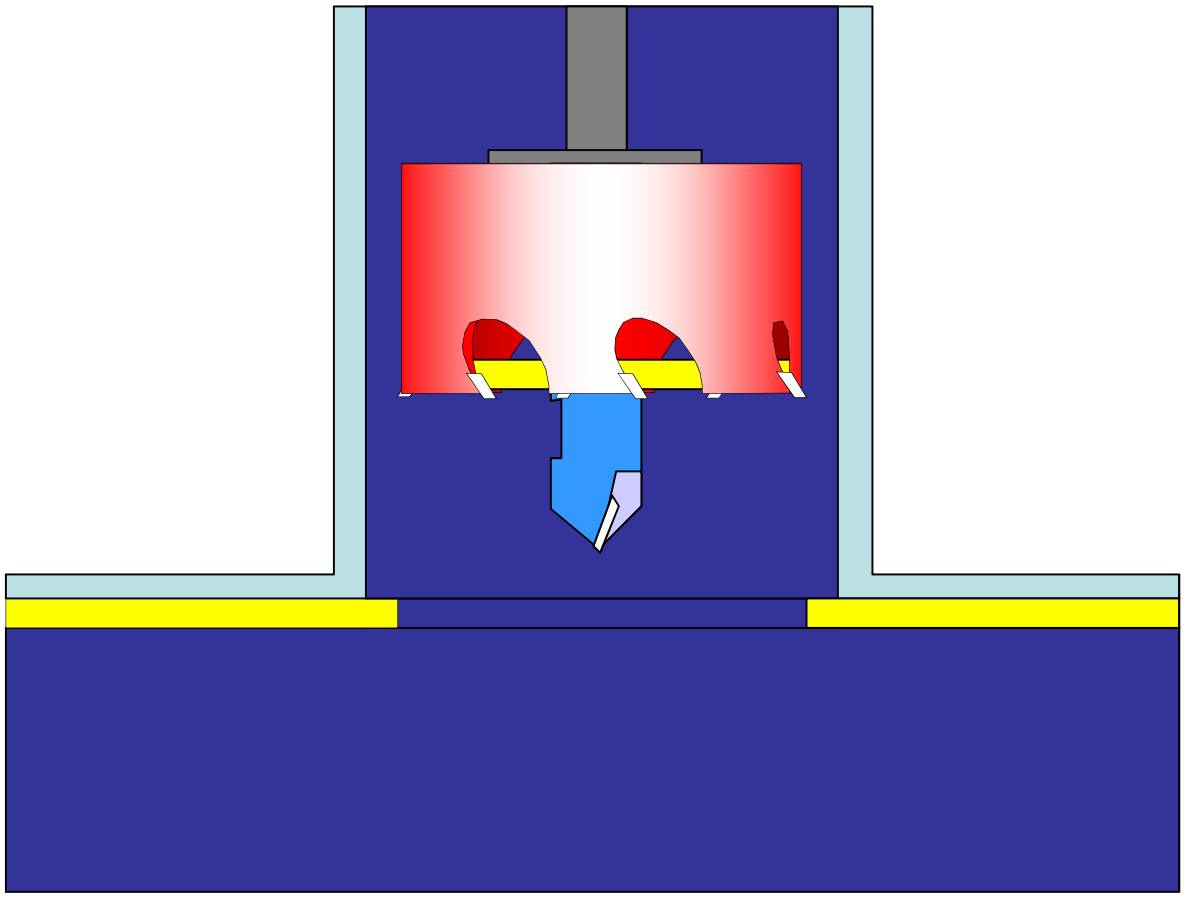

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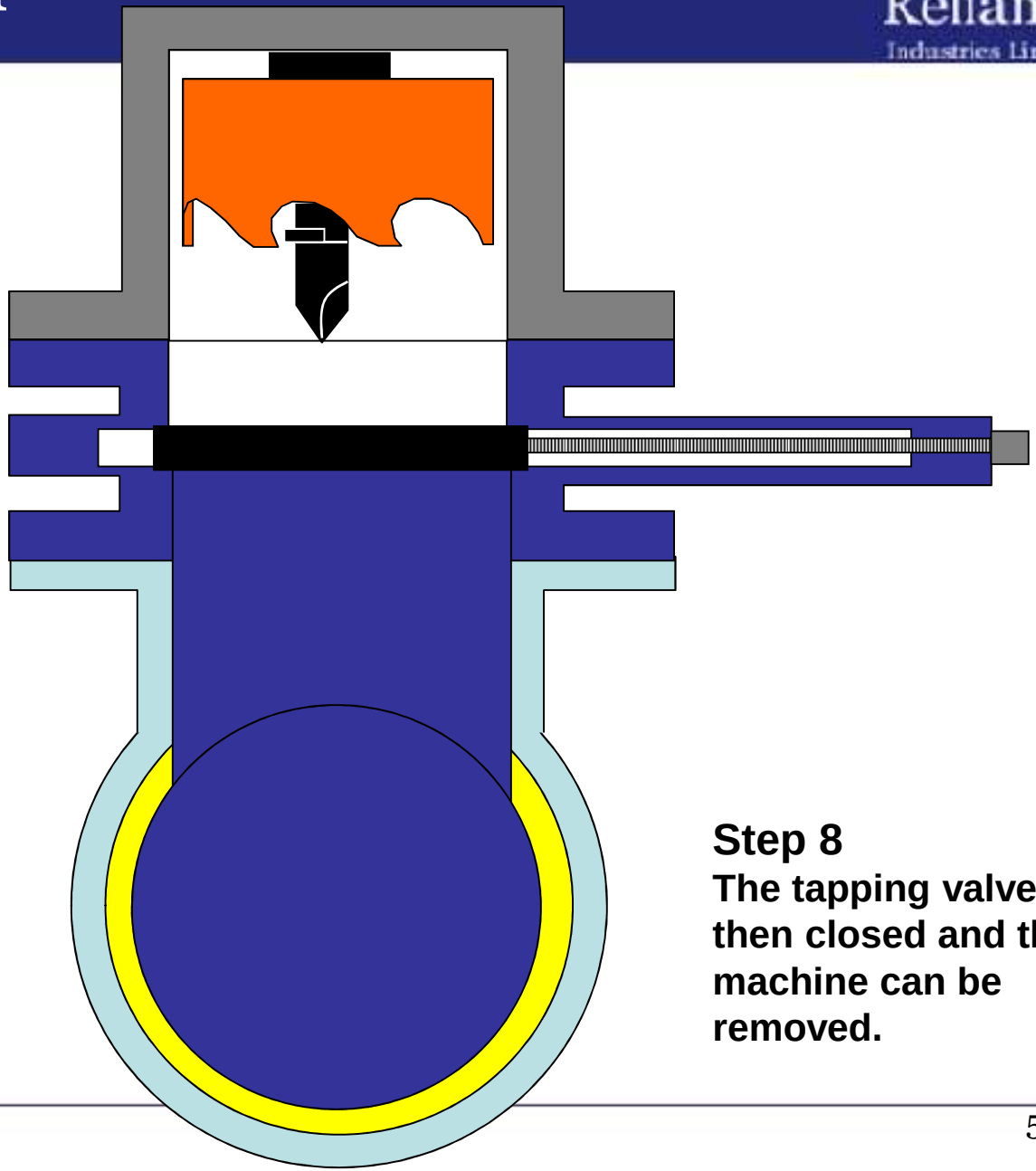

Step 8

The tapping valve is then closed and the machine can be removed.

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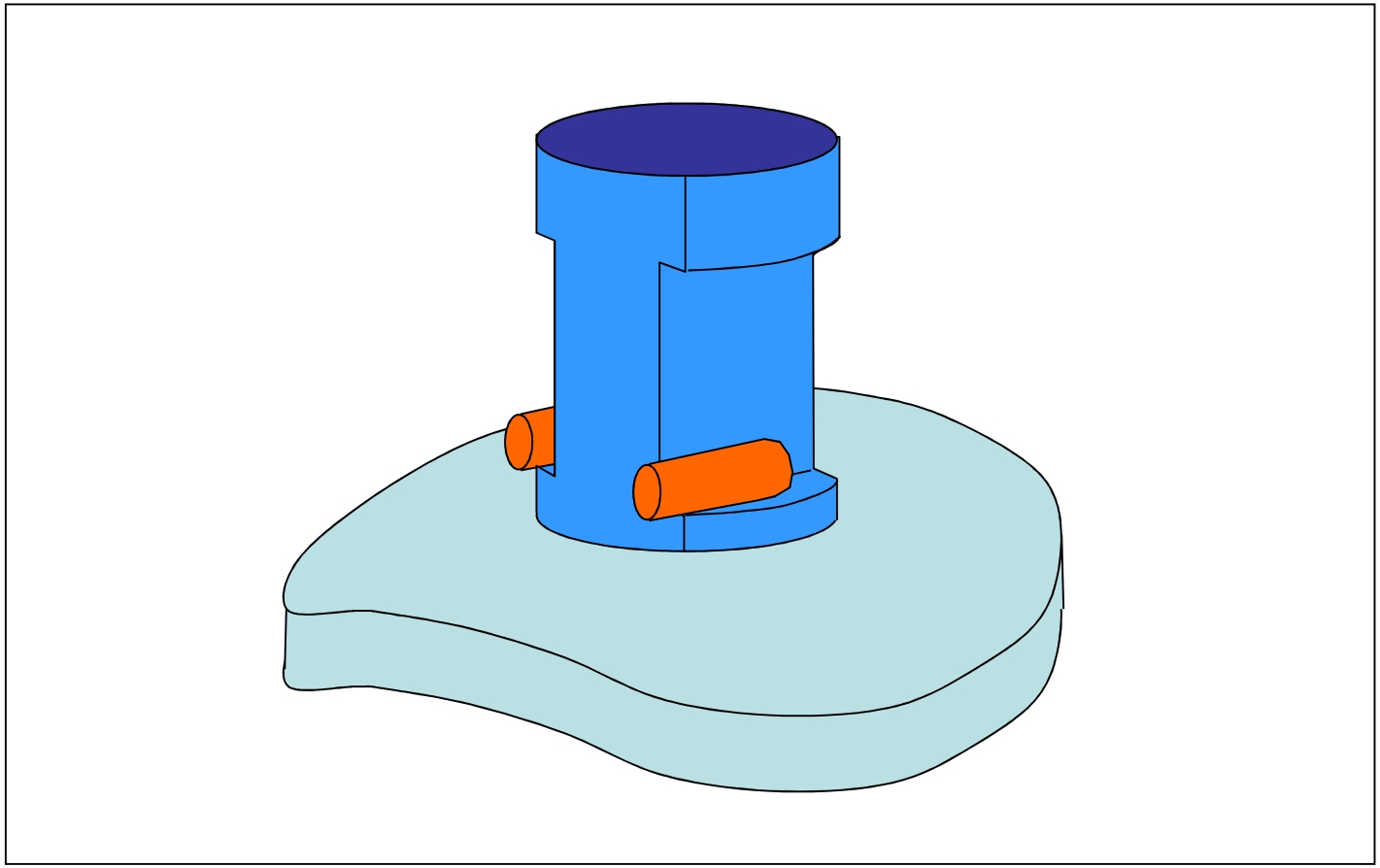

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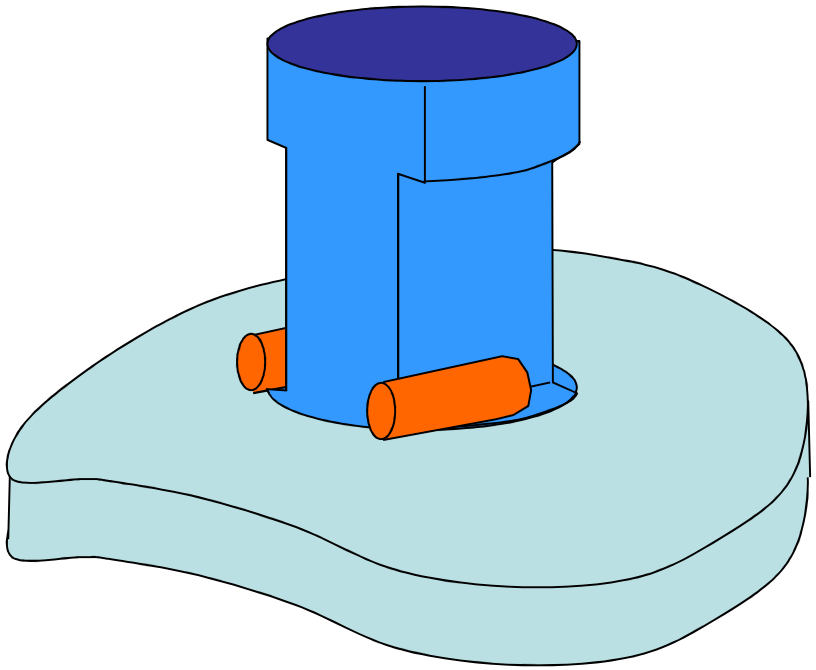

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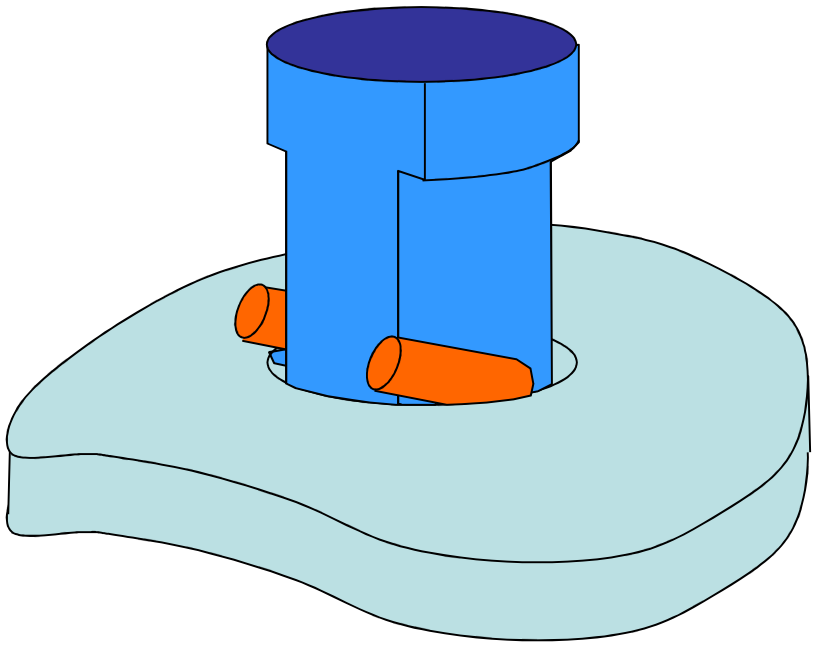

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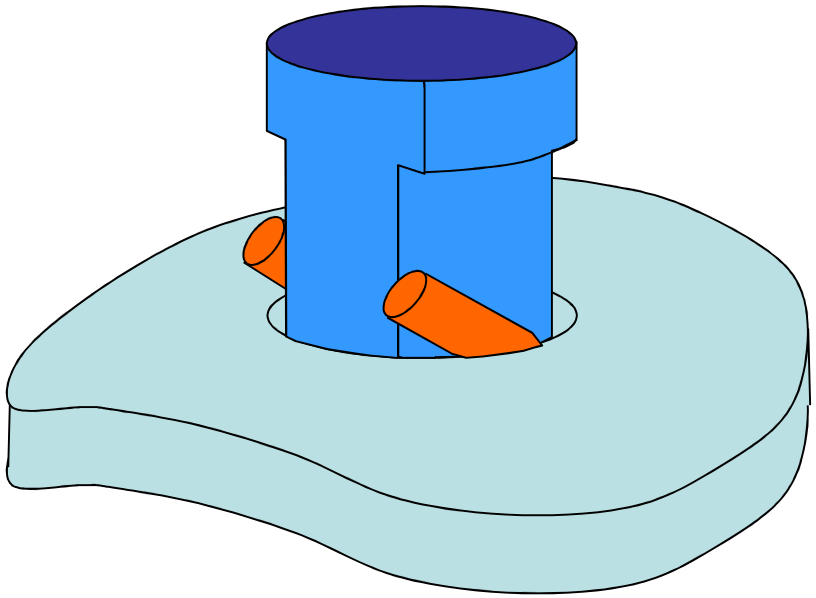

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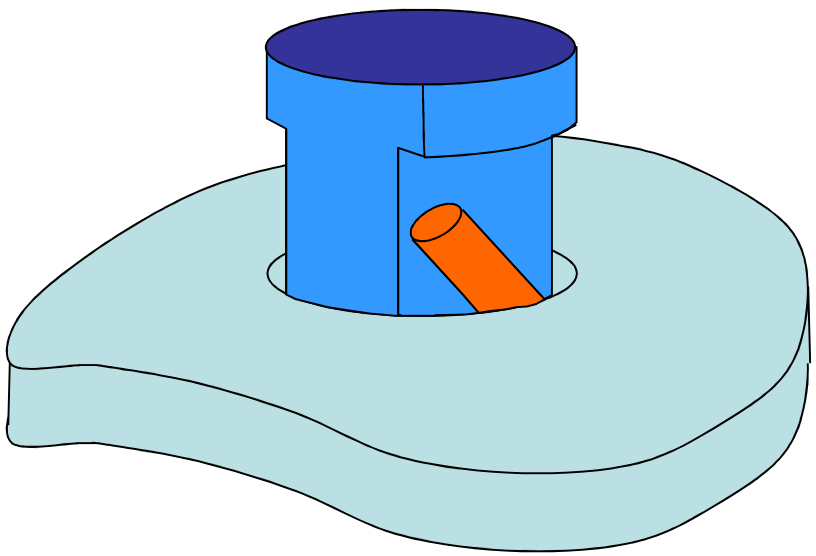

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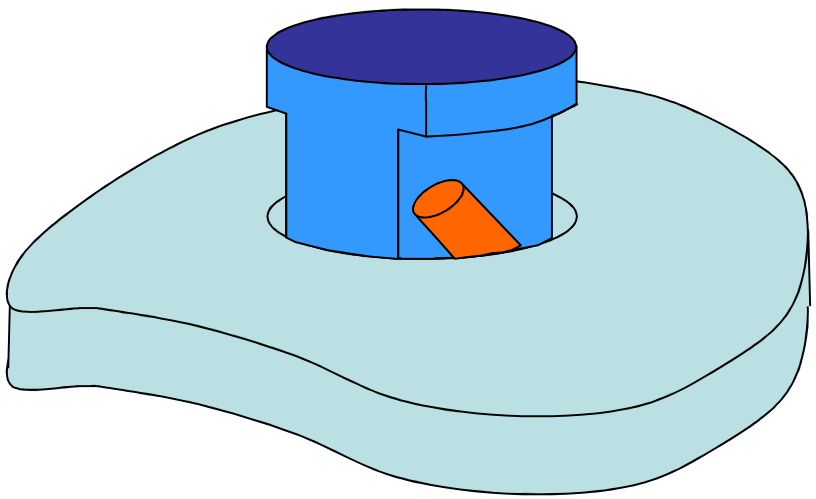

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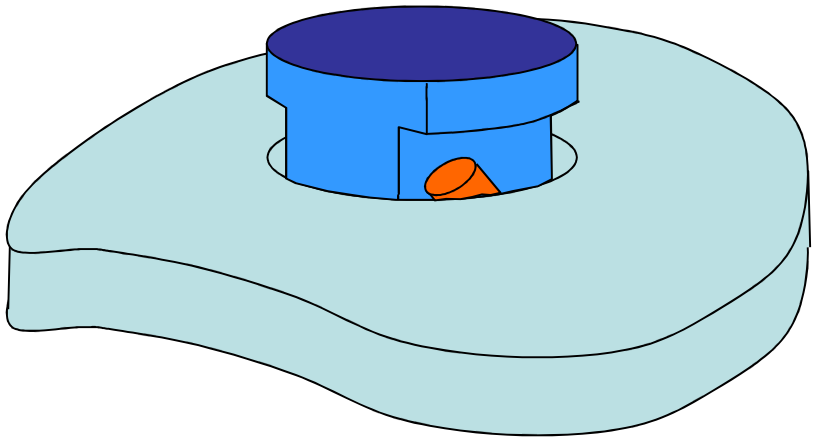

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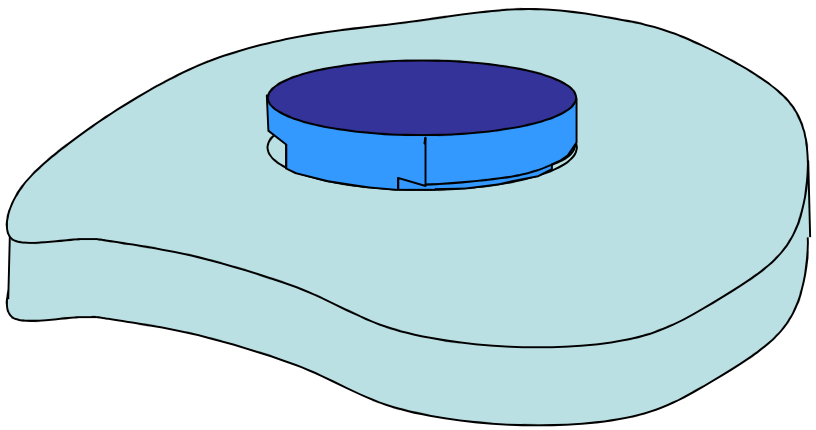

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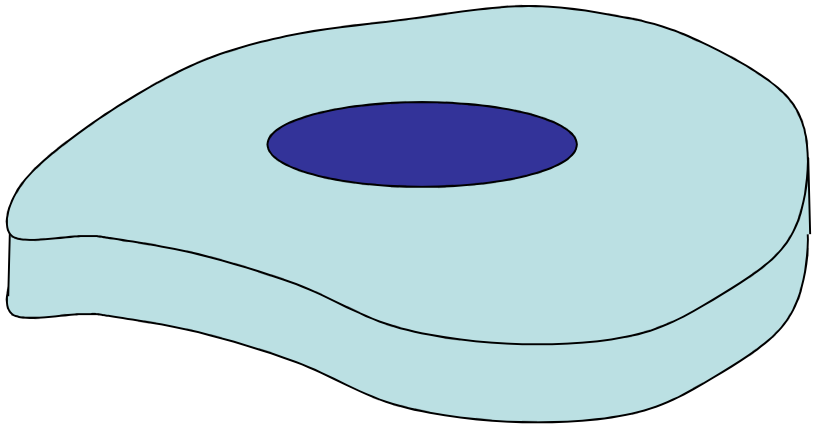

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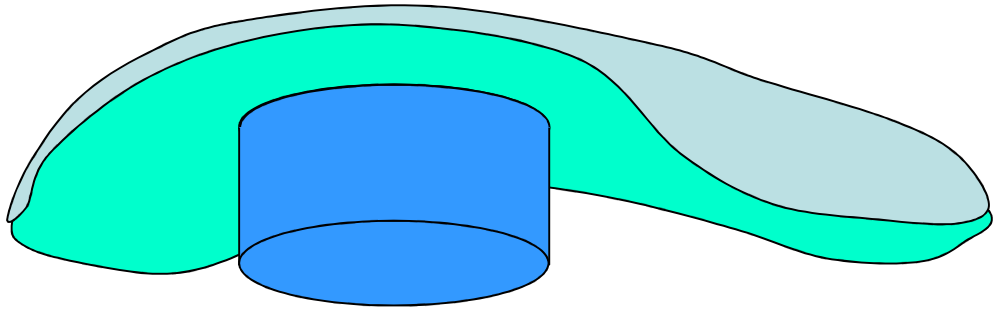

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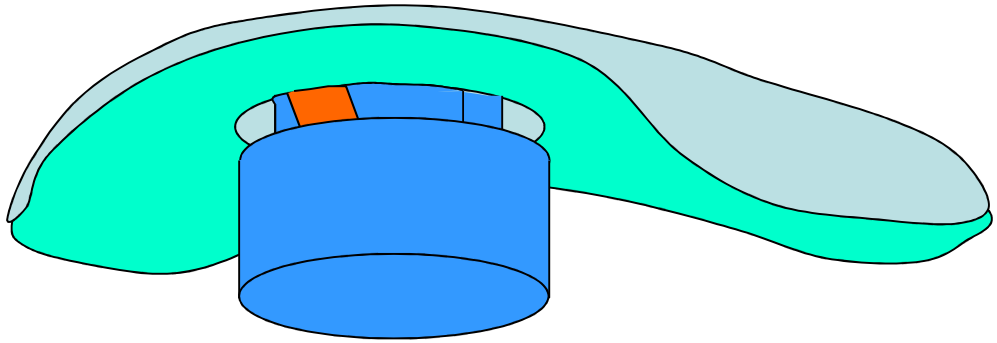

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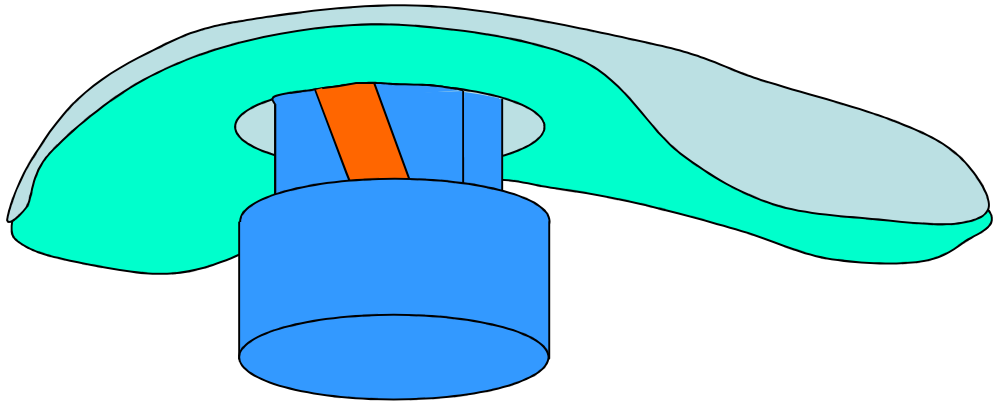

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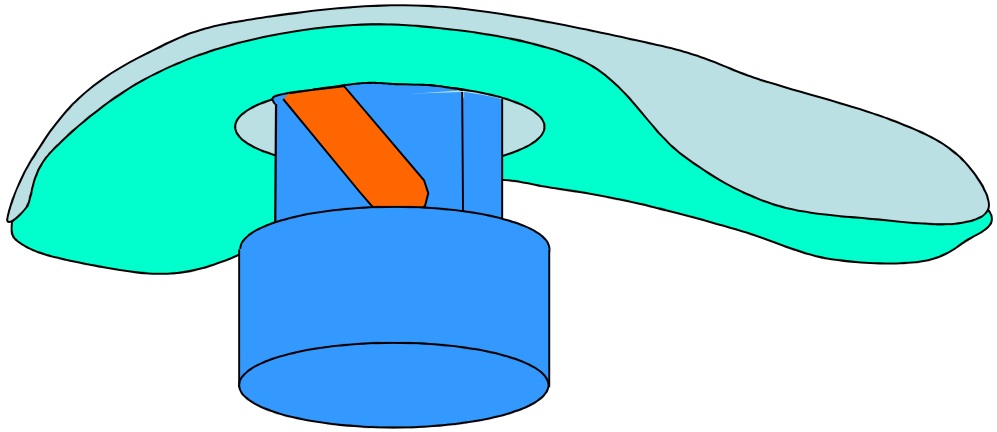

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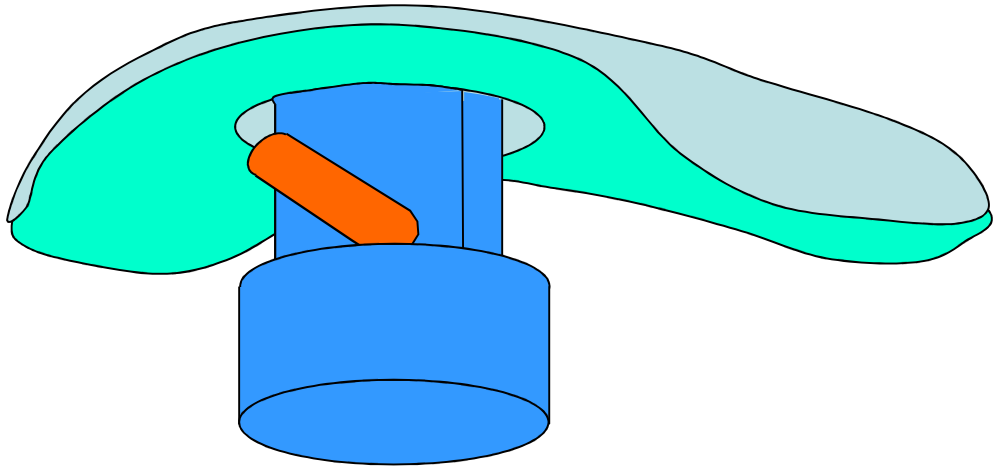

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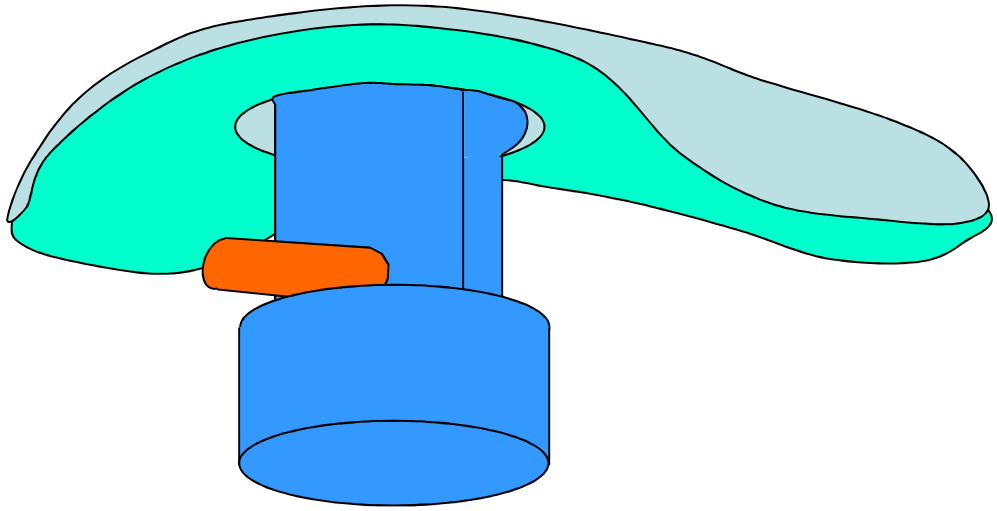

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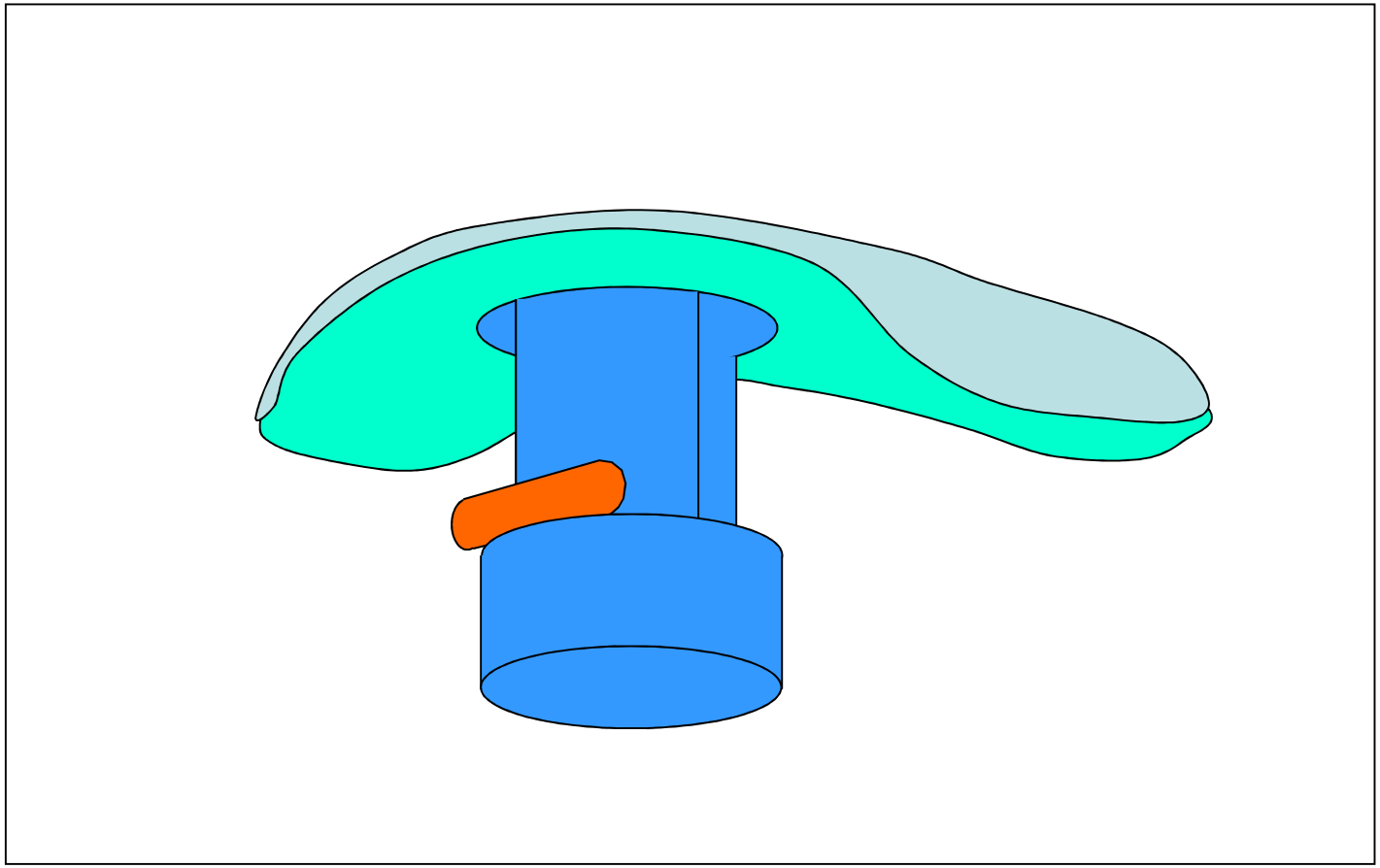

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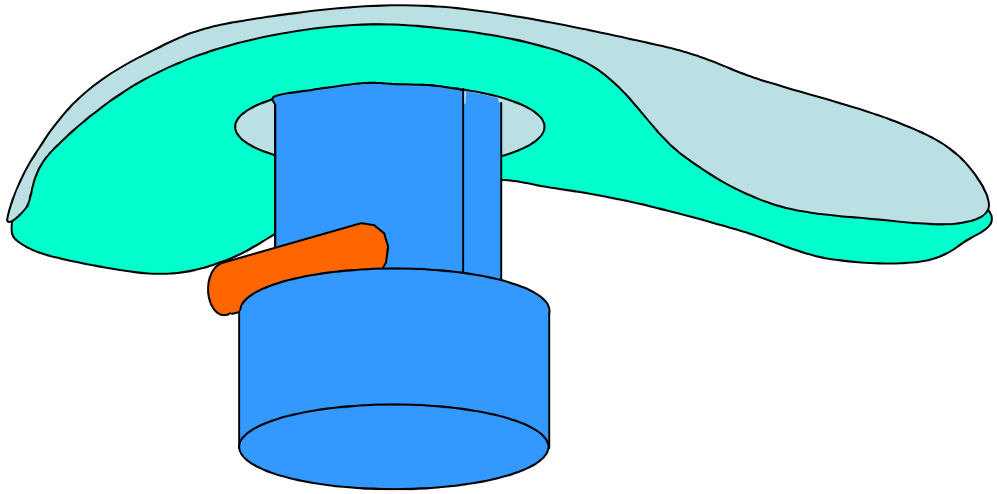

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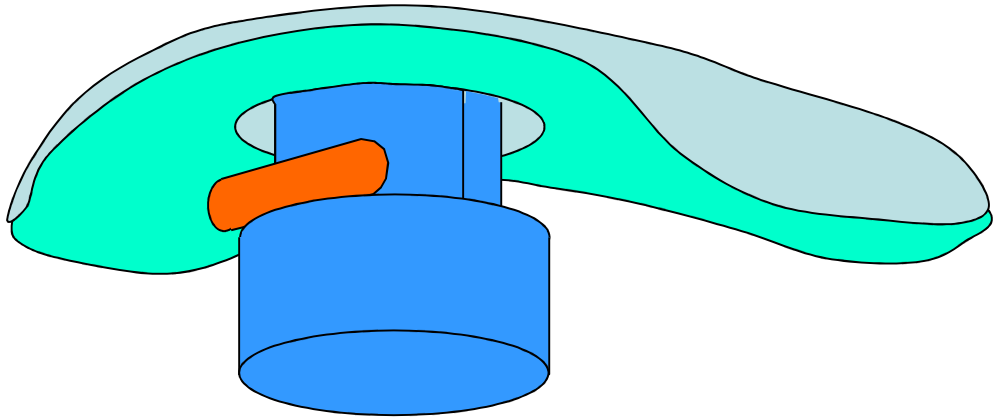

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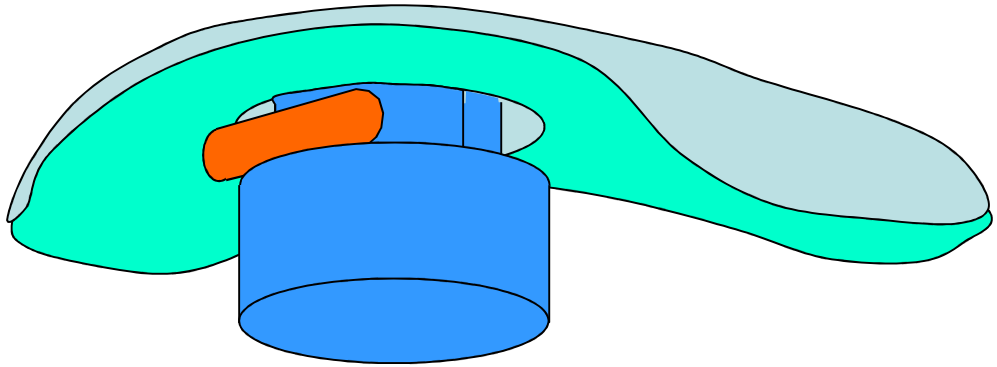

Introduction

Hot tapping

Codes and standards

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ASME B31.3
Process piping

ASME PCC-2
Repair of pressure equipment and piping

API RP 2201
Safe hot tapping practices in the petroleum and petrochemical industries

DEP 31.38.60.10-Gen
Hot tapping on pipelines, Piping and Equipment

RIL GROUP STANDARD
Hot tapping on Pipes and Equipment's in service

DG-PP-72012
Hot-tapping on piping and equipment in service

General considerations

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Justification for hot tapping

Hot-tapping is potentially hazardous and therefore should only be undertaken when shown to be technically feasible and to offer a clear advantage over alternatives.

The advantage may be economic and, in some situations, the safety and environmental risks may be less than those associated with more conventional methods.

A decision on whether hot-tapping is to be applied shall be based on careful considerations including at least the following aspects:

- safety;
- condition of the pipe/equipment under consideration;
- configuration of the connection;
- code/statutory requirements;
- operating conditions;
- technical capabilities of the drilling equipment under the operating conditions (pressure, temperature, nature of product);
- related welding problems;
- economic aspects;
- environmental/pollution aspects.

Restrictions on applications and operational conditions

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Materials:

Hot tapping shall be done only on materials with yield strength not greater than 450 N/mm2, operating at temperatures of not greater than 350 C and with a hoop stress of not greater than 72% of the specified minimum yield stress to prevent blow-out during welding.

Hot-tap welding should not normally be performed on materials which require post-weld heat treatment.

Welding shall not be performed on lined, clad or internally coated pipe.

Hot-tap welding shall not be performed on the following materials:
- Any material with a maximum allowable operating temperature greater than 370 °C;
- pipelines where the minimum allowable operating temperature is less than -20 °C.

For minimum allowable operating temperatures between -20 °C and 0 °C, additional attention shall be given to ensuring adequate material toughness.

Hardness of the run pipe should not be less than 200 BHN.

Restrictions on applications and operational conditions

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Process fluids:

Hot tapping shall only be performed under the following conditions:

- The contained process fluids remain stable when subjected to the high metal temperatures during welding;
- The process fluids remain stable when in contact with air during drilling and perforation operations;
- There shall be no decomposition or chemical reaction between components of the contained fluid (explosion, ignition, exothermic reaction);
- There shall be no chemical reaction between the fluid and the hot containment wall (burning, (stress) corrosion, embrittlement).

Restrictions on applications and operational conditions

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Hot tapping shall not be performed under the following conditions:

- Mixtures of gases or vapors within their flammable range or which may become flammable as a result of heat input in welding operations.

- Substances which may undergo reaction or decomposition leading to a dangerous increase in pressure, explosion or attack on metal.

- Oxygen-enriched atmospheres in the presence of hydrocarbons which may be present either in the atmosphere or deposited on the inside surface of the equipment or pipe.

- Compressed air in the presence of hydrocarbons which may be present either in the air or deposited on the inside surfaces of the equipment or pipe.

- Gaseous mixtures in which the partial pressure of hydrogen exceeds 700 kPa gauge, except where evidence from tests has demonstrated that hot-tapping can be done safely.

Restrictions on applications and operational conditions

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Substances:

- Acetylene;
- Acetonitrile;
- Butadiene;
- Caustic soda;*
- Chlorine;
- Compressed air at a pressure in excess of 3 000 kPa gauge;
- Ethylene;
- Ethylene oxide;
- Fat/lean DEA/MEA;
- HP steam (pressure in excess of 5 000 kPa (ga));
- Hydrogen (partial pressure in excess of 700 kPa (ga));
- Hydrogen sulphide;*
- Hydrofluoric acid;
- Oxygen;
- Propene;
- Propene oxide;
- Sulphuric acid;
- Toxic substances.*

*Constraint based on general hazard in the event of line puncturing during welding, not the welding process.

Restrictions on applications and operational conditions

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Process condition

Pressure

For general safety reasons, ***at any time*** during the hot-tap operation the pressure of fluid inside the run-pipe shall not exceed 7 000 kPa (ga) nor shall it exceed the maximum working pressure of the hot-tap equipment if this is lower.

Velocity

The velocity of the operating fluid shall be minimum of 0.4 meters/sec for gases and liquid service for liquids the maximum velocity shall be 1.75 meters/sec.

For metal thickness greater than 12.8 mm, the effect of flow on both weld cooling rates and the risk of burn through shall be considered as negligible.

Restrictions on applications and operational conditions

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Hot tap location:

The separation of the attachment fillet welds from existing circumferential welds should be at least one run pipe outside diameter and shall not be less than 150 mm.

Typical dimensions & weight of hot-tapping equipment and head-room clearance requirement

Tap size (inch)	Mass (kg)	Max. height (mm)	Head room clearance required from face of isolation valve (In meter)
1" to 2"	50	-	2
3"	75	1 270	4
4"	308	2 390	4
6"	310	2 440	4
8"	312	2 440	4
10"	320	2 745	4
12"	330	2 800	4
14"	1 000	3 180	6
16"	1 004	3 180	6
18"	1 008	3 300	* 7
20"	1 018	3 300	* 7
24"	1 090	3 400	* 7
30"	1 182	3 550	* 7
36"	1 220	3 800	* 7

Note: The above table applies to convential hot-tapping machines

* Head room clearance requirement to be consulted with hot-tap vendor to have exact detail

Restrictions on applications and operational conditions

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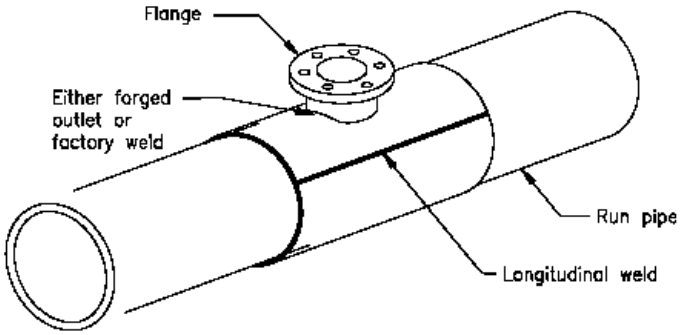
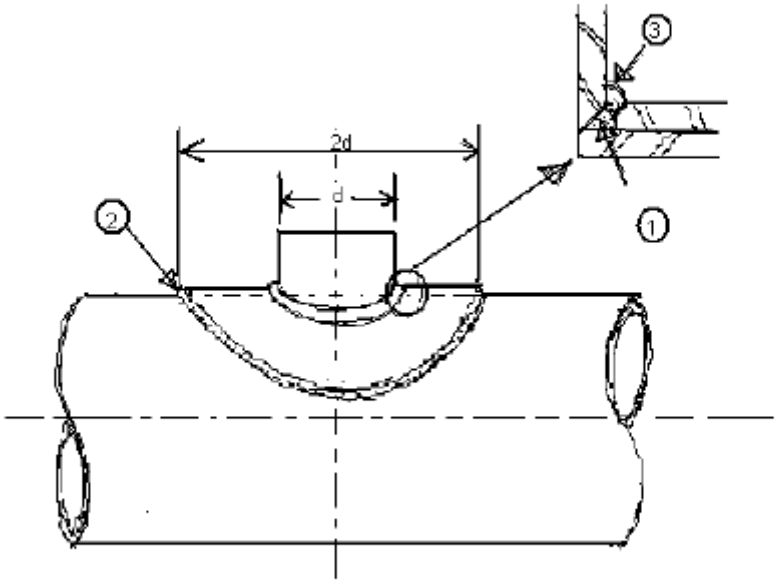
Branch connections:





NOTE: This is suggested welding sequence

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- **Maximum Allowable Operating Pressure**

$$P = \frac{2\,S\,t\,F\,E\,T}{D}$$

Where,

P =Maximum allowable operating pressure welding operation, in *MPa*;

S = Specified minimum yield stress in N/mm2 (for off-plot pipeline), or Basic allowable stress, in N/mm2 (for on-plot piping)

D =Nominal outside diameter of run pipe, in *mm* ;

T =Reduced wall thickness (where *t=ta-u*), in *mm* ;

U =Reduction in wall thickness during welding, representing the weld pool penetration, in *mm*;

ta=Minimum actual wall thickness in mm. This value shall be determined in advance by ultrasonic measurement;

F = safety factor for off-plot pipe (recommended maximum value 0.8) equivalent to the design factor. (In ASME B31.3 this factor is already included in the basic allowable stress);

E = Longitudinal joint factor, taken from the relevant ASME code;

T = Temperature derating factor

Design

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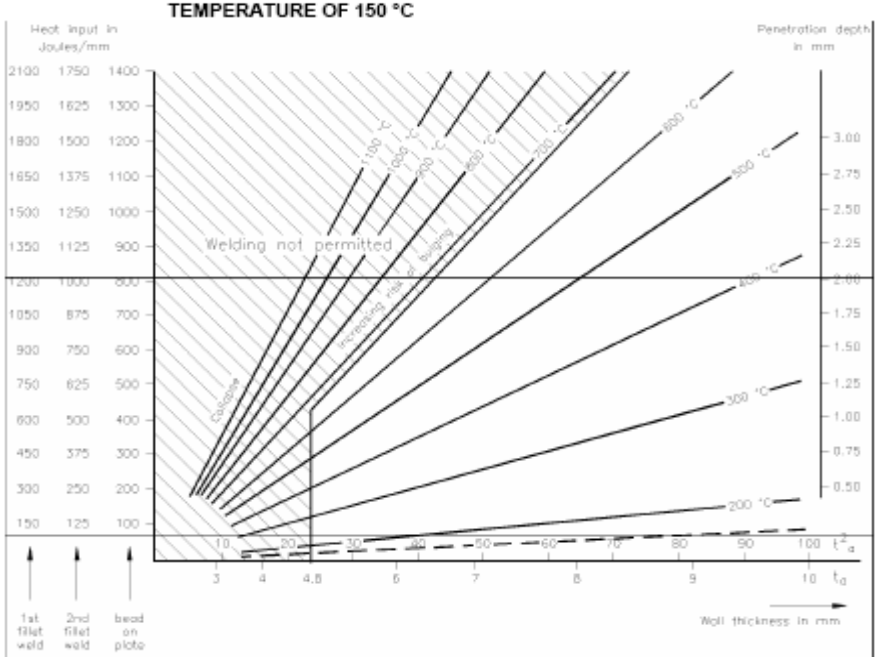
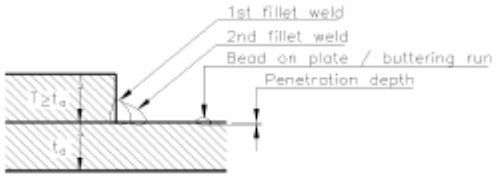
APPENDIX 3 **WELDING TEMPERATURE OF PIPEWALL - INITIAL PIPE WALL TEMPERATURE OF 150 °C**

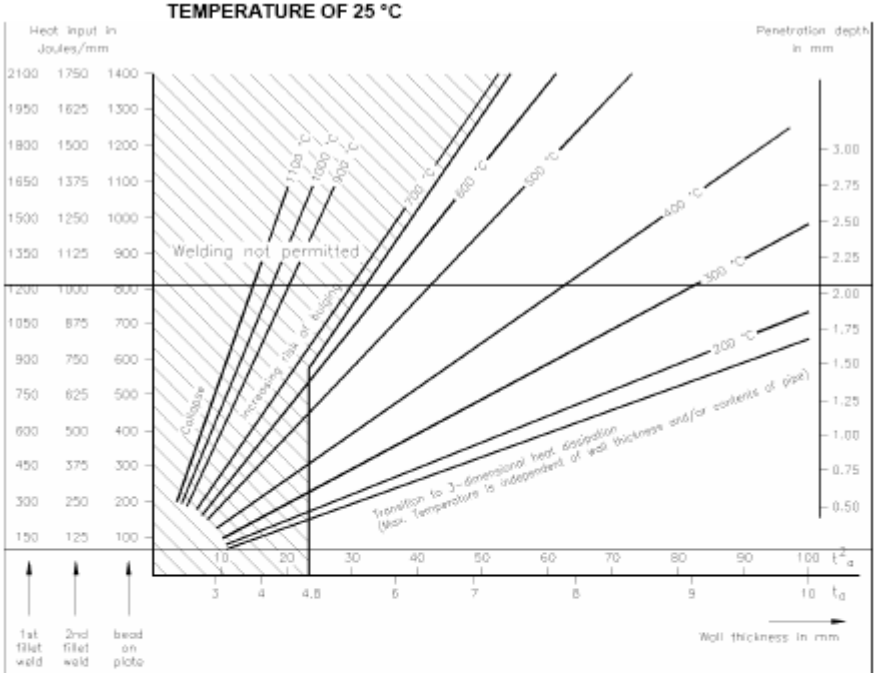


Maximum prevailing temperature at the inner side of carbon or carbon–manganese steel pipe when welding with covered electrode, starting from an initial pipe wall temperature of 150°C

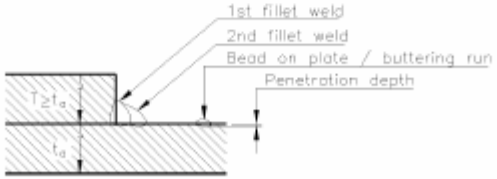


NOTE: 1. t_a = actual measured wall thickness.
2. Lines may be extrapolated horizontally for increased wall thickness.

APPENDIX 2 **WELDING TEMPERATURE OF PIPEWALL - INITIAL PIPE WALL TEMPERATURE OF 25 °C**



Maximum prevailing temperature at the inner side of carbon or carbon–manganese steel pipe when welding with covered electrode, starting from an initial pipe wall temperature of 25°C



NOTE: 1. t_a = actual measured wall thickness.
2. Lines may be extrapolated horizontally for increased wall thickness.

Design

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Predicted maximum pipe temperature (°C) during welding	Derating factor
More than 675	0.00
675	0.20
600	0.35
500	0.57
400	0.65
300	0.75
200	0.85
120	0.91
Less than 120	1.00

Table : Pipe Temperature Derating Factor

Design

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- **STRENGTH TEST PRESSURES**

Strength test pressure is the highest practical pressure depending on the internal pressure and ability to withstand external pressure against buckling

- **Split tees**

the maximum allowable pressure differential (ΔP) with respect to buckling of the run- pipe should be calculated as defined in ASME VIII, Division 1, Section UG-28.

PT=PA + ΔP

where

PT=test pressure

PA=actual operating pressure in the run-pipe

- **Set-on branch connections**

PT=test pressure in the branch pipe;

PA=actual operating pressure in the run-pipe;

F =ratio of actual operating stress level in the wall of the run-pipe to the material specified minimum yield strength (SMYS);

D=outside diameter of the run pipe;

d=outside diameter of the branch pipe;

t=actual wall thickness of the run-pipe.

Design

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- **STRENGTH TEST PRESSURES**

- **Split tees**

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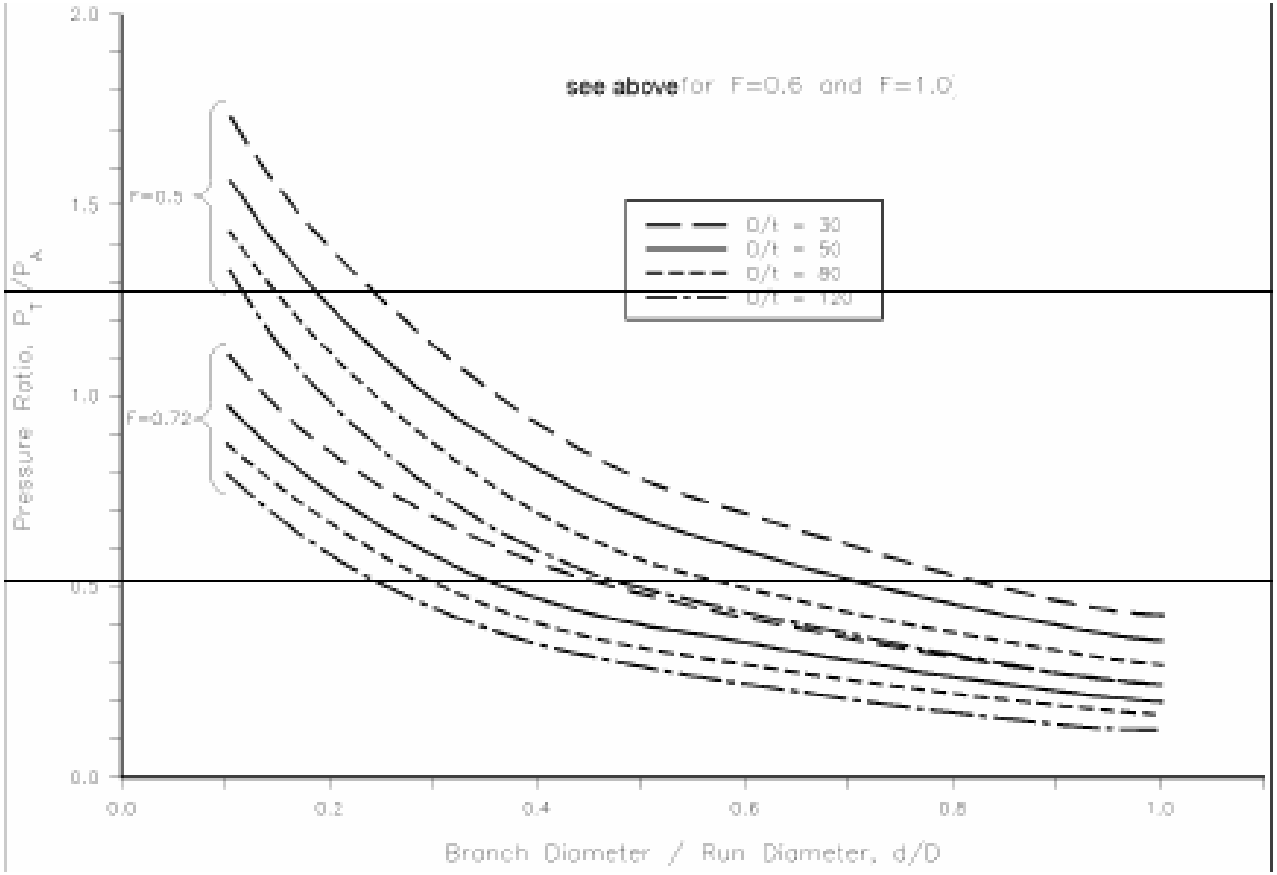
D=outside diameter of the run pipe;

d=outside diameter of the branch pipe;

t=actual wall thickness of the run-pipe.

Design


see above for F=0.6 and F=1.0
F=0.5
F=0.72
D/t = 30
D/t = 50
D/t = 80
D/t = 120
Pressure Ratio, P_T/P_A
Branch Diameter / Run Diameter, d/D
2.0
1.5
1.0
0.5
0.0
0.0
0.2
0.4
0.6
0.8
1.0

SAP and Hot Tap

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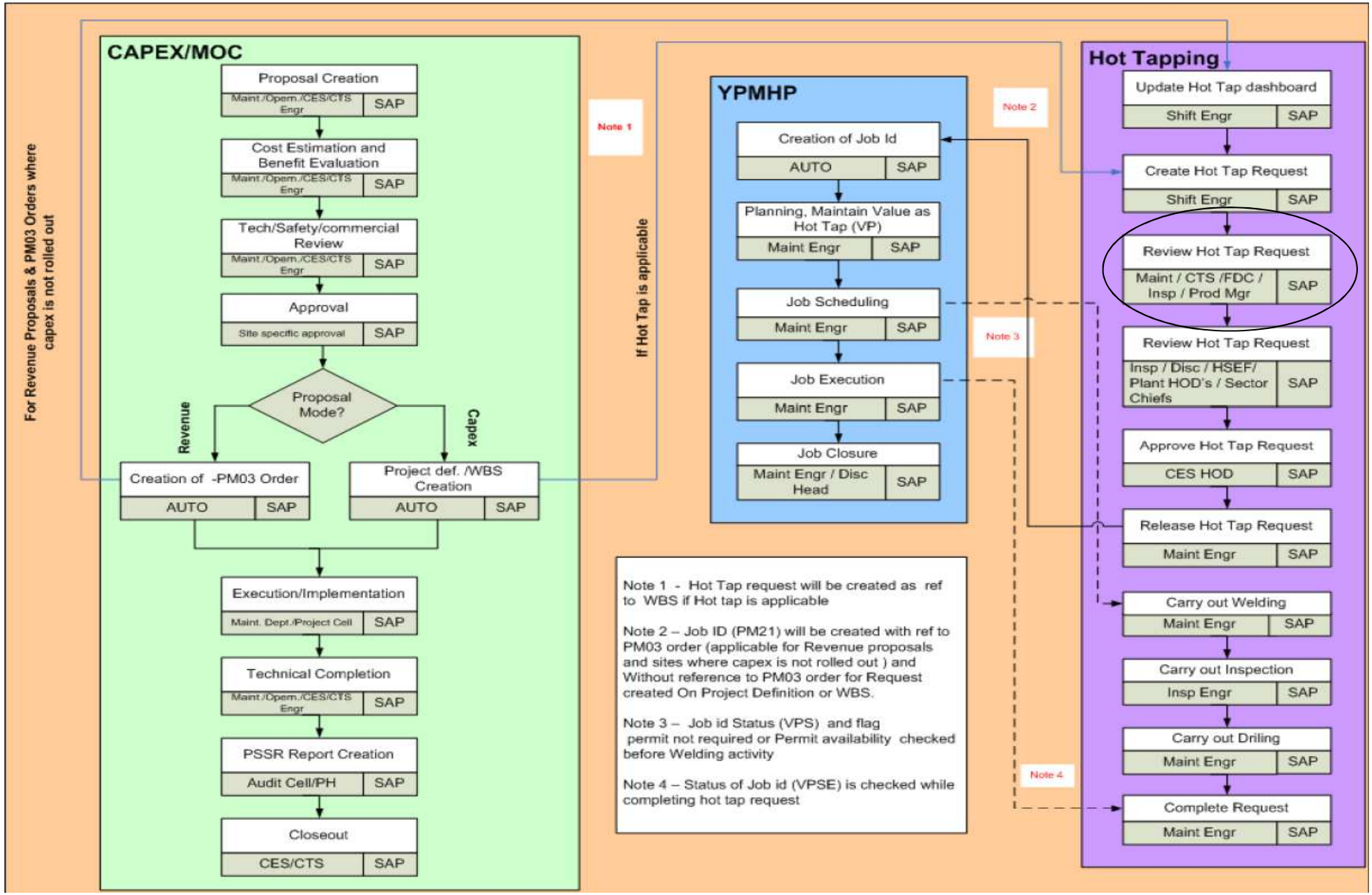

SAP and Hot Tap

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Operation Data

Display Hot Tapping Request

Documents | Change Document | Print

Header Data

Hot Tapping Request No	JRTFH00005	Approval Status	APCH Approved by CES HOD
Reference PCO/FCO No	PROJECT NO. 10044	Date Created	04.10.2011
Maintenance Plant	JRTF	Run Pipe Line no	641-PA-341-10"EC32EA
Equipment	MP-RW631-P010A		
Functional Location	1JR-RW-631		
Job Location	West of Gasoline blending station in pi…	PM Order	
P & ID Number	G42-RW641-038	Tie In No	641-718
Execution Status	Request Approved / Ready for Welding Activity		

Operation Data | Maintenance Details | CTS Details | Inspection Details | FDC Details | Precautions

Design Pressure	40.00	Kg/Sq.Cm	Run Pipe MOC	Carb…
Operating Pressure	8.00	Kg/Sq.Cm	Run Pipe Size	10.00 Inches
Operating Temperature	40.00	Deg C	Run Pipe Rating	300 #
Operating Fluid	GASOLENE		Branch Pipe ISO No	N1681-641-PA341
Fluid State	Liquid		Branch Pipe Rating	300 #
Fluid Velocity	1.35	m/s	Branch Pipe Size	6.00 Inches
Fluid Type	Hydrocarbon		Branch Pipe MOC	Carb…

Comments

Safety/Contingency Plan	ATTACHED
Comments By Operation Engr	FLUID VELOCITY AS PER LDT IS 1.92 M/S.

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Maintenance Data

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Header Data

Hot Tapping Request No	JRTFH00005	Approval Status	APCH Approved by CES HOD
Reference PCO/FCO No	PROJECT NO. 10044	Date Created	04.10.2011
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P & ID Number	G42-RW641-038	Tie In No	641-718
Execution Status	Request Approved / Ready for Welding Activity		

Operation Data | Maintenance Details | CTS Details | Inspection Details | FDC Details | Precautions

Head Room Clearance Available from Face of Isolation Valve Flange	20.00	Meters
Support adequacy Checked, Temporary Support Required	Yes	
Note: Confirm with FDC		
Branch Orientation	Vertical	
Type of Branch Connection	Full encircled Sleeve	
Note: Ensure removal of insulation / painting for carrying out inspection activities at weld location		

Comments

Comments By Maintenance Engr. Location seen along with operaions and …

Welding | **Drilling**

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CTS Data

System Help

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Header Data

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Operation Data | Maintenance Details | CTS Details | Inspection Details | FDC Details | Precautions

Fluid Hydrogen Content	NIL	PPM Weight %	H2S Content	NIL	PPM Weight %
Fluid Oxygen Content	NIL	PPM Weight %			
Purging Required	No		Other Chemicals	Qty	Unit
If Required Give Purging Rate	0.00		Chemical 1		
Purging Medium			Chemical 2		

Fluid Can't form Explosive/Flammable Mixture During Welding	Confirmed
Fluid Doesn't Contain Hazardous Substances	Confirmed
There is no possibility of Vacuum occuring during Welding	Confirmed
Fluid Stability during Welding/Drilling	Confirmed

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Inspection Data

System Help

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Header Data

Hot Tapping Request No	JRTFH00005	Approval Status	APCH Approved by CES HOD
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P & ID Number	G42-RW641-038	Tie In No	641-718
Execution Status	Request Approved / Ready for Welding Activity		

Operation Data | Maintenance Details | CTS Details | Inspection Details | FDC Details | Precautions

Checks Before Welding

Nominal Thickness	9.27	mm
Run Pipe is not Internally Cladded/Coated/Internally Lined	Confirmed	
Lamination Checks Done,None Observed	Confirmed	
Magnetic Particle Examination	Confirmed	
Note:Magnetic Particle Examination should be done over a band of 150 mm over proposed location.		
Actual Measured Hardness at Hot Tapping Location (Max)	148	BHN
Min. Thickness of Run Pipe Measured at Hot Tapping Location	8.9	mm
Type of Run Pipe	Seamless	
If ERW, Confirm that Stub Weld/RF Pad Weld is not interfering with Run pipe Seam		
If EFW, Check Whether Stub Weld/RF Pad Weld is not interfering with Run pipe Seam		
Note: If interfering, Change the Hot Tapping Location.		
If Stub/RF Pad weld is interfering with Run Pipe Seam		

SAP and Hot Tap

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JEC Data

System Help

Display Hot Tapping Request

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Header Data

Hot Tapping Request No	JRTFH00005	Approval Status	APCH Approved by CES HOD
Reference PCO/FCO No	PROJECT NO. 10044	Date Created	04.10.2011
Maintenance Plant	JRTF	Run Pipe Line no	641-PA-341-10"EC32EA
Equipment	MP-RW631-P010A		
Functional Location	1JR-RW-631		
Job Location	West of Gasoline blending station in pi…	PM Order	
P & ID Number	G42-RW641-038	Tie In No	641-718
Execution Status	Request Approved / Ready for Welding Activity		

Operation Data | Maintenance Details | CTS Details | Inspection Details | FDC Details | Precautions

Maximum Predicted Pipe Temp at inner side of Pipe wall during welding	418.49	Deg C
Maximum Fluid Operating Temp recommended during Welding	40.00	Deg C
Maximum recommended Operating Pressure during Welding	8.00	Kg/Sq.Cm
Support adequacy Checked, Temporary Support Required	Yes	
Branch Orientation	Vertical	
Maximum Actual Fluid velocity allowed during Welding	0.40 to 1.75	m/s
Flow Rate	-	Cu. M/Hr
Maximum Heat input allowed during Welding	1,900.00	J/mm
Type of Branch Connection	Full encircled Sleeve	
Strength Test Pressure [1.1 Times of Operating Pressure]	5.60	kg/Sq.Cm
Testing Medium	water	

Comments

Online welding and cold tapping

Reliance Industries Limited

Justification for WOLE jobs

Parent Line:
231-WC-952-20" and 231-WC-954-8"

Branch Line:
231-WC-909-8"

WOLE Job Location:
Tail Gas Compressor Interstage cooler ME-AY234-S02 cooling water return line bellow area.

WOLE Job details:
Encasing of the Cooling Water line at leak location.

Reason for WOLE:

Subject line is Cooling Return line from ME-AY234-S02 for Tail Gas Compressor inter stage Cooler, which is single line equipment. Pinhole leaks are observed in bellows of Cooling Water Return line. Replacement of the pipe section calls for shutdown of the compressor thereby New PSA unit & Recovery Plus unit shutdown and subsequent production loss.

WOLE job will allow online leak arresting of this line.

Subject line section will be replaced in the next available shutdown opportunity on priority.

This is as recommended by CES.

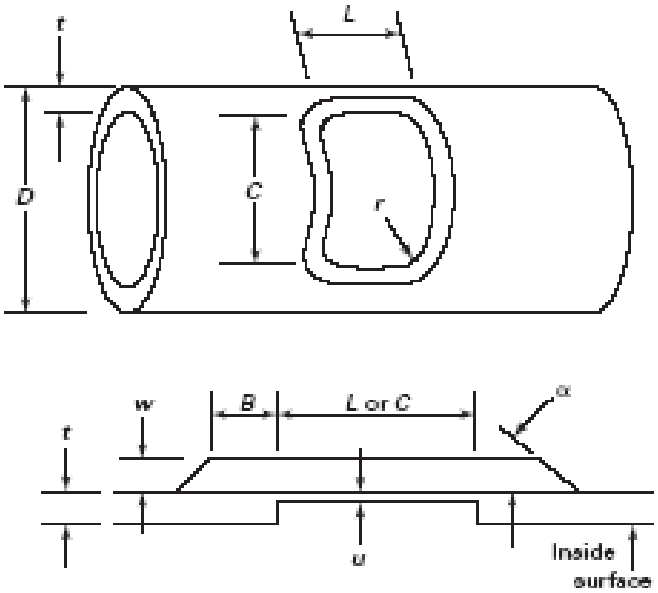
Prepared	Kapil Yeole	Sign:	Date: 24/04/12
Checked	Deep	Sign:	Date: 24/04/12
Approved	Suraj Nihar	Sign:	Date: 24/4/12

Online welding and cold tapping

Where online welding is done.

1. To add supports
2. For structural reinforcement
3. To arrest the leakages

Fig. 1 Weld Overlay Profile



Online welding and cold tapping

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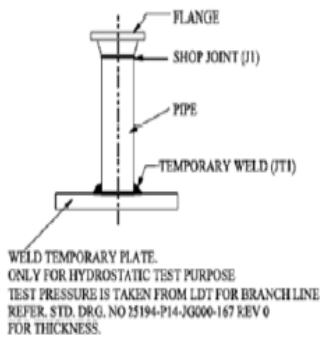
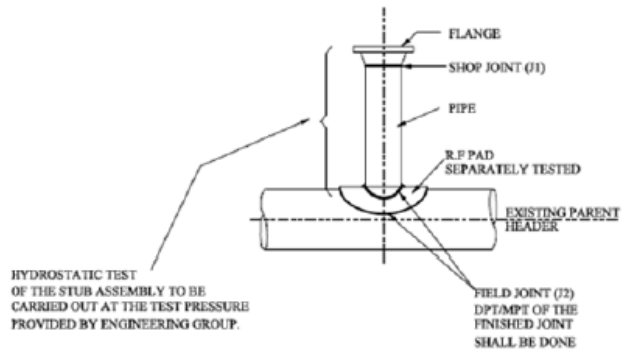
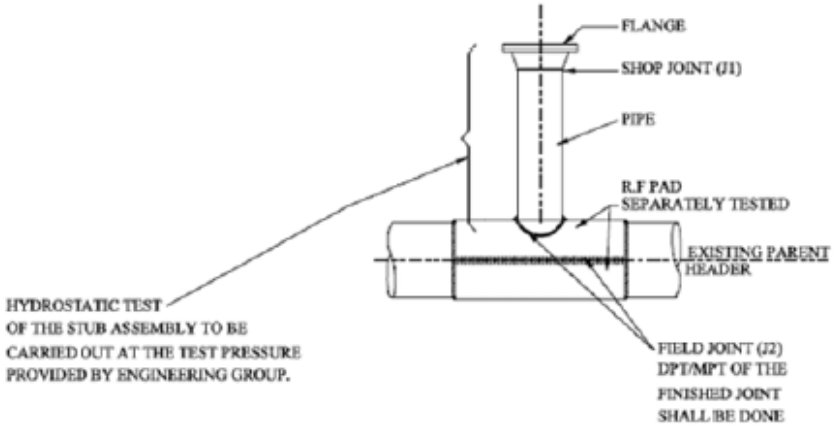
STEP.1



STEP 2 - CASE-1 (STUB WITH RF PAD)



CASE-2 (STUB WITH FULL SLEEVE RF-PAD)



NOTE:
1. APPLICABLE FOR BOTH 90° & 45° BRANCHES.
2. FABRICATION SHALL BE AS PER 3PS-PL-006 LATEST REV.

Latest development in hot tapping

Reliance Industries Limited

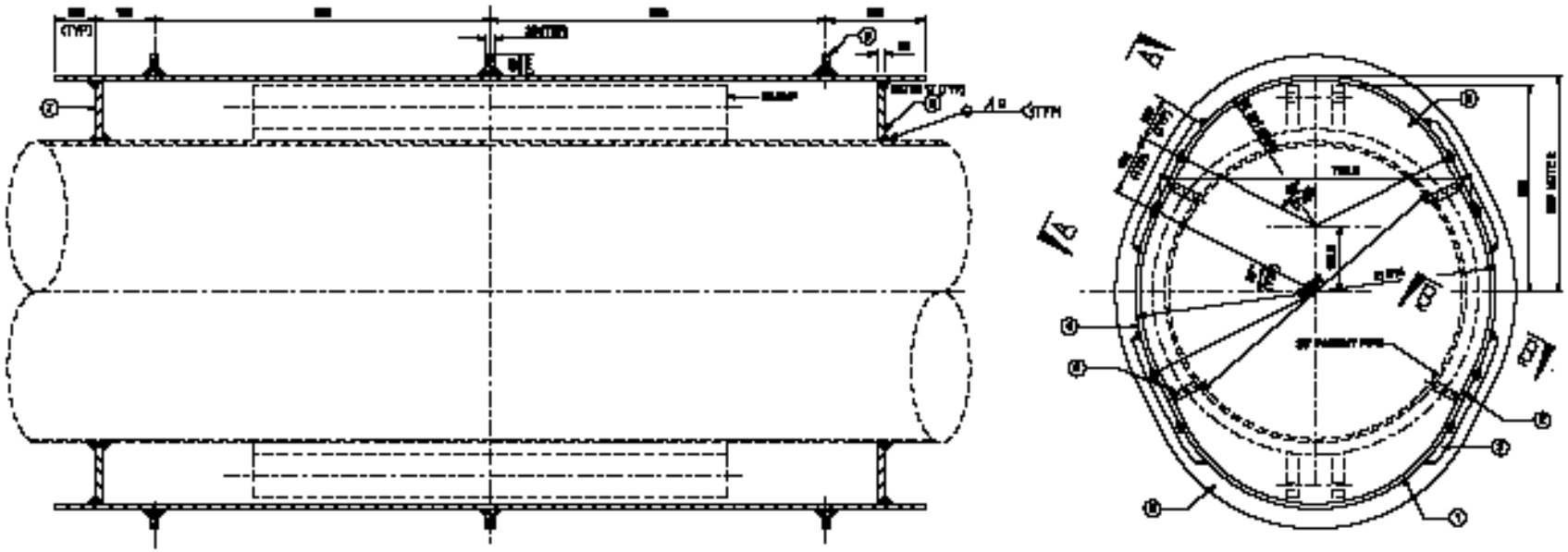
Line plugging and stopping

Pipe freezing

Piping Isolation

Stopple

Case study



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Thank you